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LIFE SCIENCE

Secondary Education

Third Year

Sections: Literature and Humanities
Sociology and Economics

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Republic of Lebanon

Ministry of National Education, Youth and Sports

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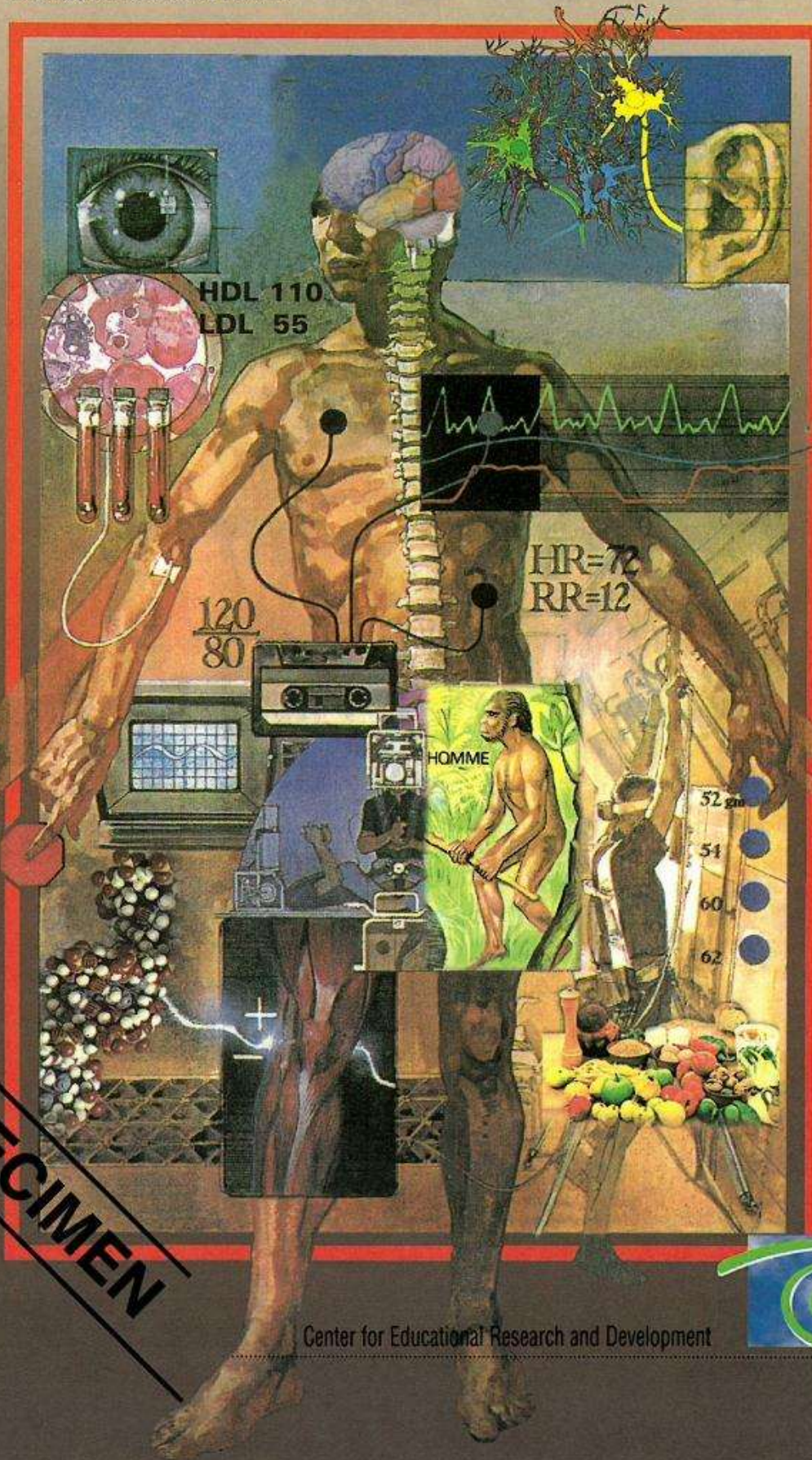
National
Textbook

New Curricula

Life

Secondary Education
Third Year
Literature and Humanities
Sociology and Economics Sections

Science



SPECIMEN

Center for Educational Research and Development



National
Textbook

New Curricula

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Forword

Dear students,

The phenomenon that marked our era the most, has been the extraordinary acceleration of the evolution of sciences and technologies. We have tried to provide you with the necessary basics and learning tools to help you stay focused and acquire the knowledge required for your continuing education.

- In last year's curriculum you studied several subjects covering the fields of reproduction, heredity, immunology and health.

- This year, the following themes are covered :

- health and nutrition,
- neurobiology, human behavior and health,
- theories of evolution,
- biotechnology (Sociology and Economics section only).

All the above themes will contribute to your knowledge of the functions of the human body and its surrounding environment as well as the interaction between them.

- You learned in the second year to exploit themes in the form of documents with a global approach but without going into details. You should be familiar with our method and we hope that it was easy to grasp.

- The presentation of this year's manual has not changed, but we have on the other hand enriched each of its documents with supplementary exploitation questions and we have added a summary of the essential points at the end of each chapter.

You should always be aware, alert and willing to understand what goes on inside and outside your body. Be curious, have a critical mind, try to understand how science is processing. Learn how to preserve your health and do not ignore how things evolve around you.

Good luck !

The authors

The National Textbook Project

By issuing the textbooks for the third year of each educational cycle, the Educational Center for Research and Development will have completed the third and last installment of books called for by the New Curricula. We are placing these books in the hands of students with the great hope that we are moving, step by step, toward the goal of acquiring sound learning, using sophisticated educational means and up-to-date methodology that encourage and reinforce individual thinking and research, acquisition of skills, development of ethical and national attitudes, the feeling of national belonging as well as the feeling of belonging to humanity at large.

The on-going revolution in information, communication and educational-aids technology has undoubtedly limited the role of the textbook and lowered the rank it used so recently to occupy. However, in our society and in many other societies, the textbook remains the basic means of education, and it is our duty to exert our utmost effort and care to come up with the best product as to form and content. Yet we should not lose sight of the fact that the textbook is not sufficient by itself but should rather be used as a stepping stone to access other sources of information. What is important is to keep a clear vision and maintain the right course toward our objective. The means should not turn into the end and the student should always remain the focus of the learning/teaching process.

No one ignores or denies the fact that textbook writing requires very high academic and educational qualifications and very wide field experience. The authors committees undeniably possess such qualities. Yet the textbooks of the last two years contained some negative aspects. Such is the nature of human work, no matter how good the intentions or how great the effort extended. Here constructive criticism constitutes a real contribution to raising the standard of authorship, minimizing errors and filling gaps. We say that, with all due appreciation and respect to all those who have contributed to the success of this project.

The Educational Center for Research and Development is embarking this year on a process of evaluating the New Curricula and related textbooks, teacher training courses and student achievement. This is a natural and necessary step now that the new system has been put into effect. This process aims at identifying the curricular objectives that have been achieved as well as those that have not been achieved, with a view to proceeding with the positive aspects and correcting the negative ones.

As part of this correction process, we plan to review the versions that have been issued in order to secure good textbooks for our students, who always deserve the best.

March 13, 2000

**President, Educational Center for
Research and Development**

Nemer FRAYHA



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Chapter 1

Diversity of food habits

A food habit regroups all activities of food processing, starting from the farm to the plate. It also includes the social ritual of the meal itself, the way to prepare and to serve it, as well as the utensils used in eating it. Food habits perpetuate traditions and characterize the evolution of the life style of a population.

We do not eat what our ancestors ate and we do not essentially eat what our contemporaries in other countries eat.

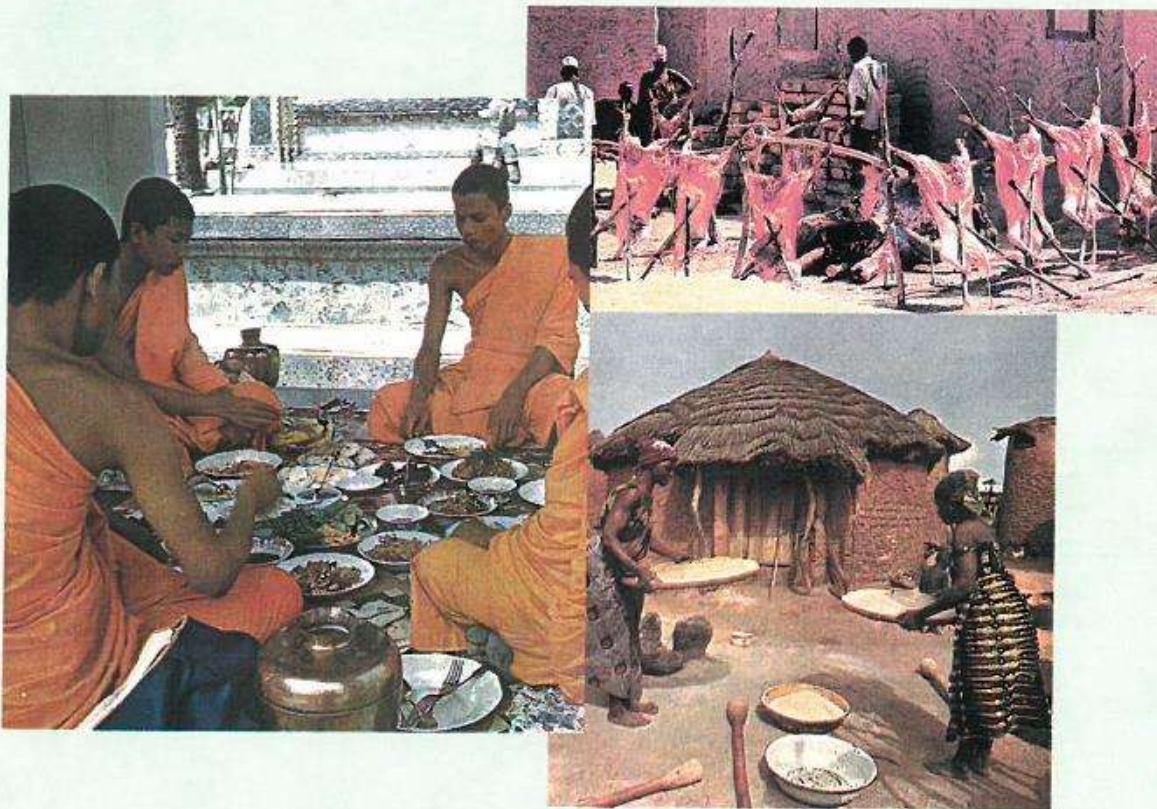


Fig. 1.2 Food diversity across the world

In this chapter we shall answer the following question:

 How do our food habits vary?

Unit I

Nutrition and Health

Nutrition is a complex process by which the living organisms consume food in order to fulfill their needs in energy and vital matter. Those needs are indispensable to the growth and to the repair of their tissues.

For humans, a meal represents a social and cultural ritual that varies from one society to the other. A meal is usually composed of many different types of food.

The quantity and the quality of food may vary in a meal according to the effort made to provide a balanced diet. This will depend on the scarcity or the accessibility of certain types of food.

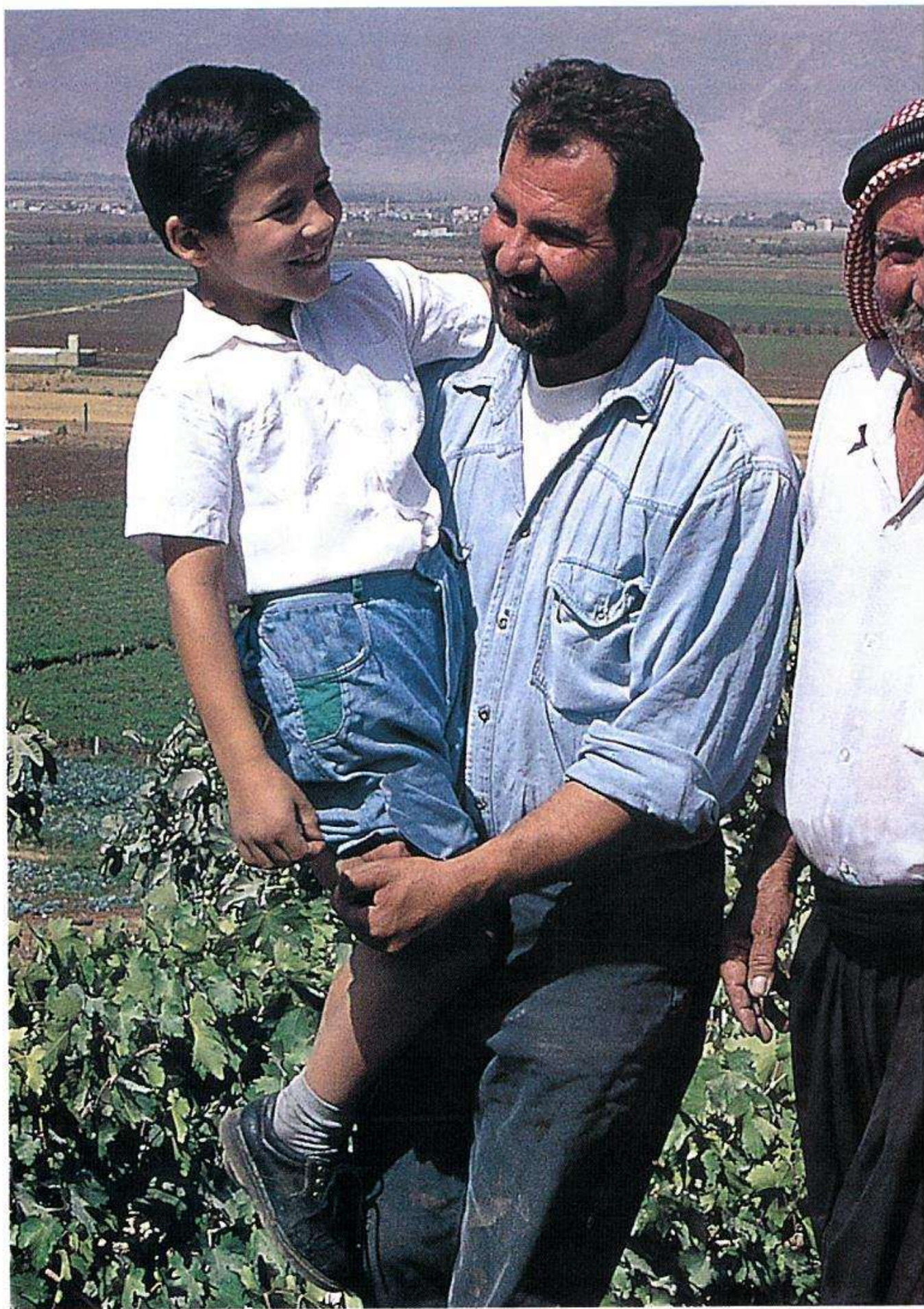
When a meal is not well balanced in nutrients it will not fulfill the individuals requirements, which may lead to a deficit in one nutrient or an excess in another, and in both cases the health of the individual is affected.

The entire world is concerned with these nutritional problems. Of all children who die each year 25% of them die from malnutrition and many adults develop health problems that are directly related to their nutritional habits.

Chapters:

- ① Diversity of food habits
- ② Principles of a balanced diet
- ③ Nutritional diseases

Fig. 1.1: Nutrition, road to health



Prosperity

The living standards in a country influence the budget spent on nutrition, as well as the nature of the consumed food products. This influence is not always positive; the technical progress has substituted the consumption of processed food for the consumption of natural food.

	% of budget value	% of food consumed from industrial source
Wealthy countries	20	70
Developing countries	70	10

Fig. I. 13 Diversity of food habits between countries with different living standards.

In wealthy countries, food products from all over the world, are consumed all the year long. In poorer countries, certain local food products are not consumed locally but are exported abroad to bring money to ensure the necessary funds to buy more essential food items for the population.

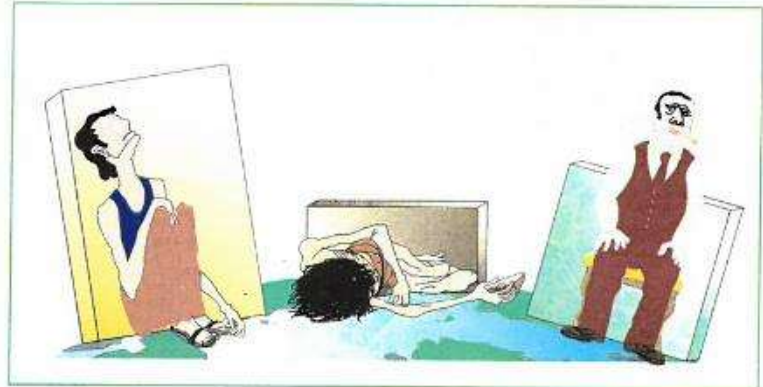


Fig. I. 14 Disparity in food distribution: certain populations are overfed; others suffer from undernourishment.

Exploitation of the document

1. Make an inquiry to note certain variations in food habits among different Lebanese regions?
2. How does the wealth of a country affect the food habits of its population?

Essentials

- Food concerns the society as a whole, whereas nutrition concerns only the individual and his needs.
- Different societies have different food habits, but the goal remains to ensure the needs of the organism. The observation of these habits show their historic as well as their geographic evolution.
- The technological progress in food industry and in agriculture during the 20th century has revolutionized our food habits.
- The ease of communication throughout the world has encouraged different countries to adopt each other's food habits.
- Universal food habits do not hide the traditional habits that continue to characterize each ethnic group.



Document 2 Geographic diversity of food habits

Despite the ease of communication between different countries that contributed to the development of universal food habits, nutrition remains probably what many define a social condition.

Influenced by climate, sociological and economical factors, food habits reflect the image of local resources as well as the technical and agricultural progress of a country.

How do food habits vary?

1. Diversity within a single country

Even in a small country like Lebanon, a difference of food habits may be observed between small and large villages, between the coast and the mountains, and between the North and the South.

Supermarkets changed our habits. We buy what we see on the shelves and we eat what we store.

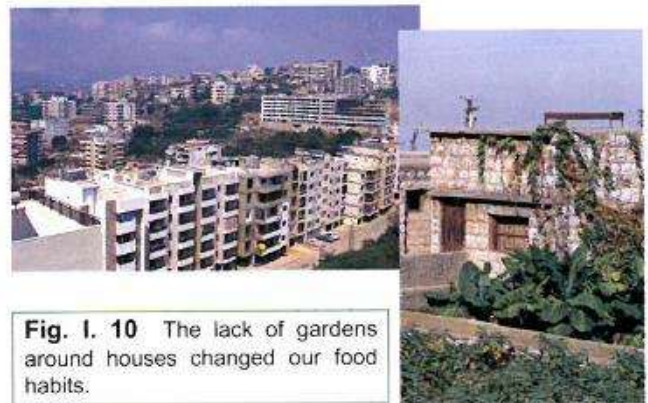


Fig. I. 10 The lack of gardens around houses changed our food habits.

2. Diversity among different countries

The climate

The climatic conditions of a country determine its local resources, and its food habits. For example, some of the exotic fruits that are imported and sold at high prices in Europe are considered as very ordinary in a country like Africa or South America where they originate.

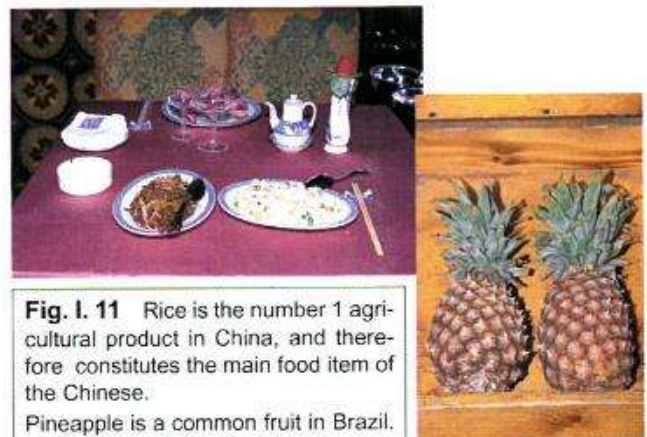


Fig. I. 11 Rice is the number 1 agricultural product in China, and therefore constitutes the main food item of the Chinese.
Pineapple is a common fruit in Brazil.

The culture

The traditions, the culture and the beliefs of a community affect their food habits. Certain products became synonymous with the country in which they are widely produced i.e., camembert with France, spaghetti with Italy; and Kebbe with Lebanon...

In one country people eat snakes meat while in others they don't eat beef.

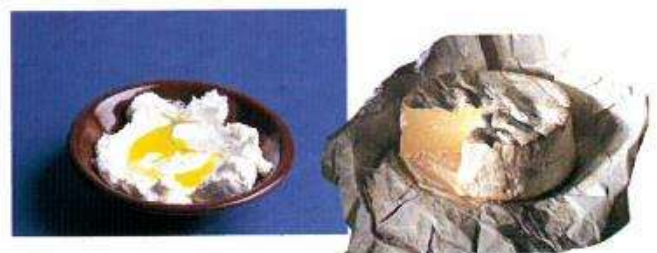


Fig. I. 12 In France the typical dairy product is cheese, in Lebanon it is "labneh".

Publicity plays an important role in our food choices . The worldwide expansion of foreign restaurant or restaurant chains, introduces the food habits of one country into another.



Fig. I. 6 Universality of new food habits

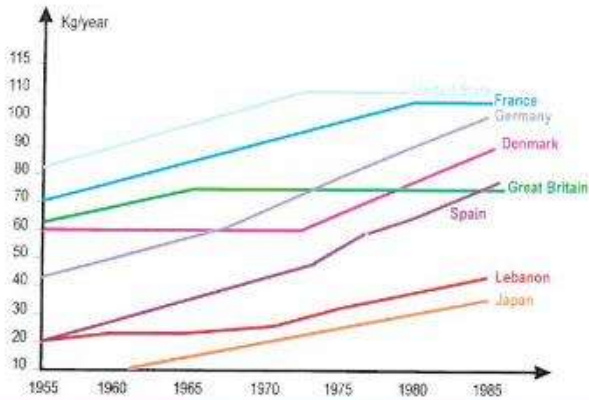


Fig. I. 7 Evolution of meat consumption in Lebanon in comparison to that of a few industrial countries at the end of the 20th century. In most Lebanese mountain villages, "Kawarma" cooked meat conserved in fat, constituted the only source of meat during most of the winter. Today, it is used only during festivities and celebrations.

The conservation techniques that are adopted in food processing introduce chemical substances and colorants that favor preservation.

In the past natural products such as salt, vinegar or spices were used to preserve foods.

Food additives can be natural or artificial substances meant to conserve foods and to enhance their flavors , smell or consistency. Their use is often criticized.

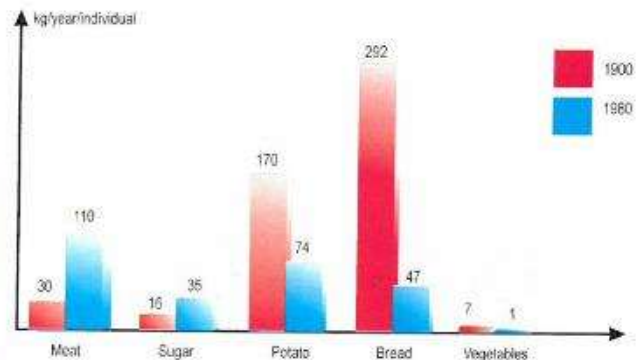


Fig. I. 8 Evolution of consumption of few essential foods in the world. According to FAO. Industrial farming has brought about a wide evolution in food consumption.



Fig. I. 9 Preservatives: The consumption of processed products introduces these substances into our body.

Exploitation of the document

1. Make an inquiry into the variations of food habits in your family or village.
2. How can we explain the increase of animal protein consumption in respect to the total intake of protein? Compare meat consumption in Lebanon to that in industrial countries.
3. Cite a few food products that have increased the quantity of sugar consumed by the average individual.
4. Notice the codes of the preservatives or colorant that are indicated on the labels of some of the foods that you eat.

Document 1 Evolution of food habits

Food habits follow very closely the evolution of a society. During the second half of the twentieth century there has been a revolution rather than an evolution in food habits. Forty years ago, a major change in our food habits occurred due to new techniques of food culture (green-houses), of food conservation (freezing) and of the food industry as a whole.

In what ways have food habits changed with time?

Modernization modifies food habits

Traditionally, the preparation of markouk (bread) used to be made at home. With new technologies there are now several industrial bakeries where any type of bread can be found ready made. However there is a shift to old traditions in making markouk but they remain commercial.



Fig. I. 3 We do not make bread like we used to.



Fig. I. 4 Industrial conservation techniques have replaced traditional ways of food storage



The tradition of preparing winter's provisions lost its meaning in the face of new food industry: we can find any type of food that we want all the year round.

Industrialization has brought about a change in the utensils used in preparing meals and in the way they are served. Clay, copper or steel utensils have been replaced by aluminum and stainless steel utensils that do not rust.



Fig. I. 5 Serving food is a habit that evolved with time



EVALUATION EXERCISE

- 1- What does this table represent?
- 2- Compare the consumption of sea fish among the above countries?
How can you explain the numbers obtained for Lebanon, even though like the other two, it is a maritime country?
- 3- Using histograms, represent the evolution

of poultry and egg consumption in Lebanon, and justify this evolution ?

- 4- What do you think of the diversity in fruit consumption?
- 5- Which food product characterizes each country?

consumed product (kg /person/year)	1961	1961	1961	1996	1996	1996
	Lebanon	France	Egypt	Lebanon	France	Egypt
Butter	1	7.65	1.50	1.52	8.69	1.93
Milk and dairy products	64.04	220.32	30.83	85.96	256.15	38.42
Eggs and derivatives	2.67	10.55	1	16.94	15.93	2.26
Fish (fresh water)	0	0.11	1.93	0.06	3.01	2.88
Fish (sea water)	4.54	18.15	3.91	0.71	27.93	6.61
Meat	20.94	76.08	10.80	29.61	97.95	12.68
Poultry	3.44	11.05	2.43	14.27	24.60	6.65
Cereals and products	133.13	126.42	39.22	124.82	96.09	144.49
Fruits	60.36	19.67	32.04	106.63	35.51	37.41
Vegetable oils	7.16	7.81	4.73	16.24	16.48	7.56
Potato and derivatives	11.66	115,18	7.72	57.20	72.46	21.12
Sugar	20.24	30.08	13.37	55.23	36.16	29.65

Chapter 2

Nutrition and health

Principles of a balanced diet

Food provides all the basic constituents of the cells and the necessary energy for their activity.

The manner in which we eat has always followed empirical rules. Now dietetics helps establish more scientific rules of hygiene, based on the knowledge of the organism's needs and the food supplies.

One can choose between several nutrients in order to maintain an equilibrium in one's body, i.e., to preserve and maintain good health.

Nutritional balance revolves around 3 pivotal questions:

What to eat? When to eat? How much to eat?



Fig. 1-15: A balanced food ration : staying healthy

In this chapter we shall answer the following questions:

- What are our food supplies?
- What are the needs of the organism?
- How do we ensure a balanced diet?

2- The requirements of the organism: matter and energy

Structural molecules

The analysis of the chemical constituents of different tissues of the body (blood, bone, skin) reveals the presence of minerals (water, ions) and organic components (carbohydrates, fat and proteins).

constituents	g.l ⁻¹
Water	900
Mineral salts	9
Glucose	1
Lipids (triglycerides)	8
Proteins	80
Nitrogenous wastes	0.3

Fig. I. 19 Major components of plasma. Blood forms an internal medium from which cells draw their needs.

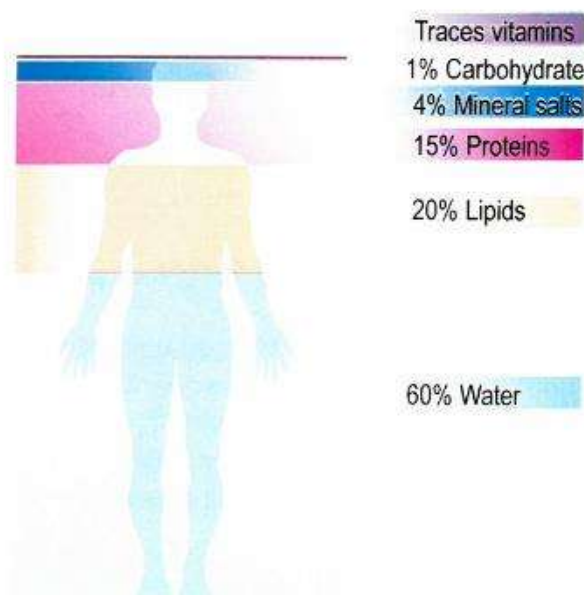


Fig. I. 18 We are what we eat

Functional molecules

They are responsible for cellular activity:

- Enzymes and hormones

The cytoplasm is the stage for the many chemical reactions that constitute the cellular metabolism. These reactions require the presence of enzymes. Other protein molecules, the hormones, ensure communication

between organs.

- ATP

The oxidation of organic nutrients in the cell liberates energy that will be stored in highly energetic molecules, ATPs or adenosine triphosphate. The tertiary phosphate group is easily detached, thus releases energy.

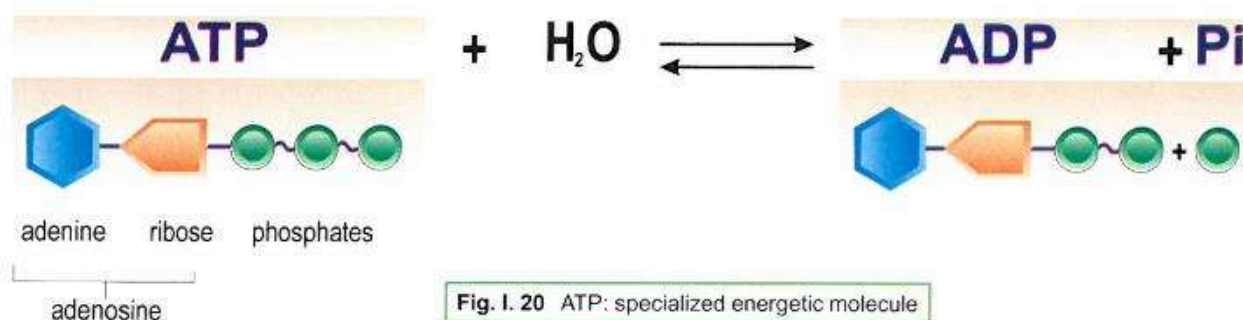


Fig. I. 20 ATP: specialized energetic molecule

Exploitation of the document

1. Group the nutrients of Fig I 16 according to their richness in carbohydrates, lipids or proteins ?
2. What happens to the nutrient constituents before they get utilized by the cell?
3. Why does the organism need a permanent nutrient intake?

Document 1 Food supplies

Whether of animal or vegetable origin, the complex food that we consume is a mixture of the same basic components but in variable proportions. By digestion, they are broken down into nutrients that are used by the cells.

What are the basic components of food? What are the needs of the organism?

1- Foods, an apparent diversity

simple food complex food (100g)	proteins %	lipids %	carbohydrates %	water %	kJ
butter	1	80-82	0.5-1	15-17	3177
breast milk	2-2.1	3.6-3.74	6.3-6.4	87-88	293
cow's milk	3-3.5	2.5-4	4.7-5	86-88	372
eggs	12.5-14	12-16	0.5-1.8	70-75	334
beef	17-22	10-20	1-2	60-70	836
lamb	16-22	20-30	1-2	60-70	1380
fish	20-22	3-9	0-1	65-75	400
poultry	20-21	10-20	1-2	60-70	627
wheat	26-27	1-1.5	70-75	10-14	950
green beans	2-3	-	7-8	85-90	167
oil	-	99	-	-	3762
lentils	24-25	1.9-2	60	10-14	1413
margarine	1-1.5	82-83	1-2	14-15	3156
honey	0.4-0.5	0.1-0.15	70-75	19-21	1333
white bread	7-8	0.7-0.8	55-60	25-30	1066
whole wheat bread	8	1-1.2	50-55	30-35	960
potato	2-2.5	0.8-1	21-25	70-75	402
raisins	2-3	1.2-1.5	70-75	15-18	1354
soya bean	32-35	1-1.1	40-42	25-30	700
tomato	0.5-1	-	4-5	94-95	85

Fig. I. 16 A complex nutriment is composed of many simple foods.

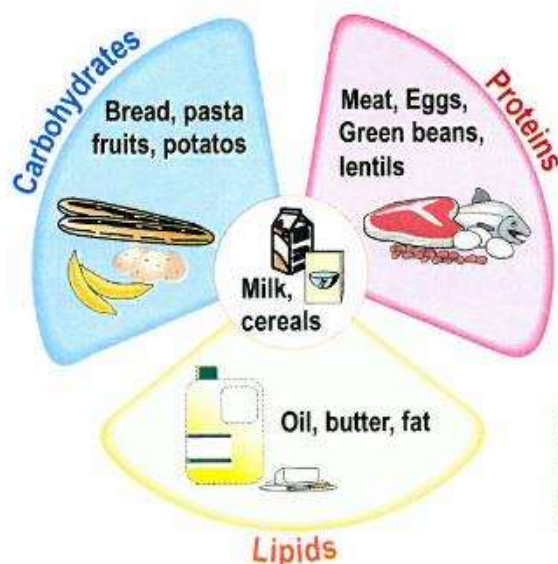


Fig. I. 17 Complex foods are usually rich in carbohydrates, proteins and fats. A complete nutriment contains all the mineral and organic components.

Additional Information

"Bourghol" Production

"Bourghol" is a typical Lebanese food product. It constitutes the principle ingredient of many of the plates of Lebanese cuisine, such as "kebbe" and tabbouli.

"Bourghoul" is not cornstarch or "couscous"; it is cooked ground wheat.

Traditionally, its preparation in villages took place only in the summer, but now it has become industrialized and "bourghol" is being produced all the year round.

How is "bourghol" prepared?

- 1- Wheat is boiled until it is well cooked.
- 2- It is then spread in the sun to dry.
- 3- It is then ground into small grains a few mm in diameter.
- 4- It is then separated from the bran, and it is divided into larger or smaller sized "bourghol".

Quelques additifs dangereux dans notre alimentation

code	additive nomenclature	found in	additive role	toxicity
E102	tartrazine (yellow)	candies, pastries	coloring agents	risk of
E110	yellow-orange	cured fish	synthetic products	allergic reactions
E120	cochenille carminic acid	cidre, drinks vinegar, red fruits	red coloring agent natural products	risk of allergic reactions
E123	amarante	candies, biscuits	red coloring agent	risk of allergy
E124	cochenille red	canned redish fruits	synthetic products	carcinogenic
E127	erythrosine			
E211	sodium benzoate	canned fish	preserving antiseptic agent	carcinogenic
E220	sulfuric anhydride	beer, cidre, fruit juices, wines	preserving antiseptic agent	deficiency in Ca^{2+} and vitamin B_2
E230	diphenyl	citrus fruits	preserving antiseptic agent	risk of allergic reactions
E239	hexamethylene tetramine	canned caviar	preserving antiseptic agent	carcinogenic
E250	sodium potassium nitrites	cured meats	stabilising and salting agent	water detention cardiovascular risk
E251				
E252				
E311	gallate	industrials fats (many foods)	antioxidant agents	risk of skin reactions increase in blood lipids
E312				
E330	citirc acid	soda drinks	emulsifiers and acidifying agents	risk of aphthous
E407	carraghenates	sorbet, icecreams	thikening agents	intestinal disturbances
E450	Sodium phosphates	gruyere, salted pretzets	homogenisers	intestinal disturbances

- Energetic expenditures =
 $V.O_2 \text{ l.kg}^{-1} \cdot \text{h}^{-1} \times 20 \text{ kJ.l}^{-1} = \dots \text{ kJ. kg}^{-1} \cdot \text{h}^{-1}$
- The KiloJoule has replaced the calorie as the energetic unit of expenditure.
 It is a common unit for all forms of energy. 1 kcal= 4.18 kJ...

	Daily expenditures of energy (kJ)
Child	7000-9000
Adolescent boy	12500
Adolescent girl	10000
Man	11300
Woman	8400
Pregnant woman	10000
Nursing woman	10750

Fig. I. 23 Different individuals have different energy expenditures.

2- Basal metabolism

It is the minimal expenditure in energy that covers the basic functions of an organism: cardiac, respiratory, nervous, excretive...

It is evaluated by the volume of O_2 that is consumed by a resting individual who has been fasting for 12 hours and placed in a neutral thermal temperature ($18^\circ\text{C} - 20^\circ\text{C}$).

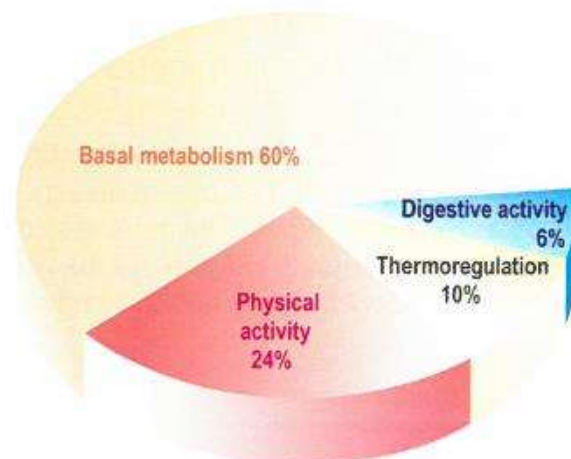


Fig. I. 24 The basal metabolism covers 60% of the total energetic expenditure of an adult individual with moderate muscular activities.

Exploitation of the document

1. How can we explain the difference in energetic values of proteins obtained from either the calorimetric bomb or the organism?
2. Calculate the expenditure in energy of an individual who weighs 50 Kg when resting if the respiratory intensity is $0.25 \text{ l.kg}^{-1} \cdot \text{h}^{-1}$.
3. How do energetic expenditures vary?
4. Make a list of your daily meals, evaluate approximately their energetic value and compare it to your energy expenditure.
5. Justify the conditions for measuring basal metabolism.

Nutrients (100 g)	Lipids (g)	Cholesterol (mg)	polyunsaturated fatty acids (g)	monounsaturated fatty acids (g)	saturated fatty acids (g)
Chicken Grease	100	65	17.6	45.3	32.5
Vegetable	100			1.5	98.5
Arachidonic oil	100		18.5	61	1.6
Olive oil	100		9.3	69.5	16.8
Corn oil	100		61	24	-
Butter	81	250	2	24.5	51.3
Gruyère cheese	32	106	1.7	9.9	18.7
White cheese	7	25	0.2	1.9	4.5
Yogurt	1.5	8	0.04	0.41	1
Beef	10.6	65	0.45	5.4	4.8
Egg	11.6	504	1.4	4.5	3.4

Fig. I. 27 Lipid composition of certain nutrients.

Essential fatty acids

Not all acids can be synthesized by the organism. Certain unsaturated fatty acids must be provided by food and then are called essential fatty acids.

Experiment of Evans-Burr:

It has been proven experimentally that a diet totally free of fat provokes a decrease in the overall growth of young rats and a weakening of their ability to reproduce. The addition of 3 unsaturated fatty acids, linoleic, linolenic and arachidonic, was enough to eliminate these disabilities.

nutriments	essential fatty acids %
butter	7
olive oil	10
arachidonic oil	23
sunflower	60

Fig. I. 28 Composition of certain lipids in essential fatty acids.

Exploitation of the document

1. Why does a carbohydrate excess in foods lead to an increase in body weight ?
2. According to Fig. I. 26 can the lipid reserves produce all the fatty acids ?
3. Cholesterol is produced in the body from saturated fatty acids. Referring to Fig. I. 27, determine the origin of the cholesterol-containing lipids ?
4. Explain the symptoms observed in the mice that lacked lipids in their diet ?

Document 3 Qualitative requirements, energetic needs

In addition to the quantitative, a diet must cover the qualitative needs and bring essential substances that cannot be synthesized by the body. The nutritional value of a nutriment depends on its intake of essential substances.

What is the nutritional value of every food category?

1- Carbohydrates

When carbohydrates are digested they produce glucose, it is the combustible of choice for most cells.

If the glucose is present in excess, it will be stored in the liver and in the muscles in the form of glycogen or transformed into lipids and stored in the adipose tissue under the skin. Slow sugars, that is sugars that are absorbed slowly like starch, are not instantly available, and are stored as glycogen. Fast sugars such as sucrose, release their energy immediately. If this energy is not used, the fast sugars will be stored as lipids.

In the absence of stored energetic elements (glycogen), cells utilize lipids as a source of energy. Once the reserves in lipids expire, the muscles start to degrade their own proteins to generate the required energy for their activity. This leads to a reduction in the muscular mass.

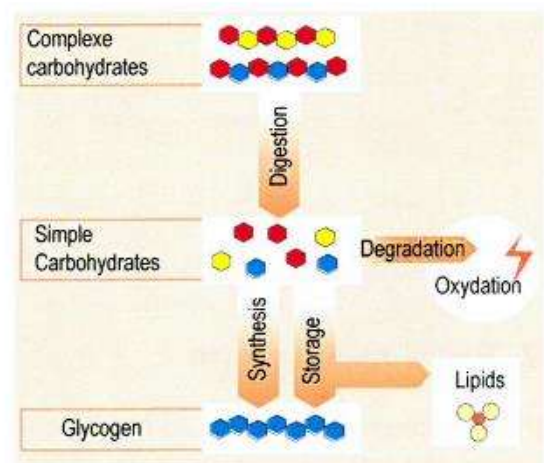


Fig. I. 25 Carbohydrates metabolism.

2- Lipids

When digested, lipids are hydrolyzed to fatty acids and to alcohol (glycerol or sterol). They are a source of energy for certain cells and for muscles during endurance sports or physical exercises. In addition to their energetic values they also have nutritional values.

They provide essential substances that enter in the structure of nerve cells, cell membranes and the composition of certain hormones and vitamins.

A fatty acid is an organic acid formed of a long carbon chain. The presence of simple bonds around the carbon radicals indicates a saturated fatty acid.

The presence of one or several double bonds indicates mono- or poly-unsaturated fatty acid respectively.

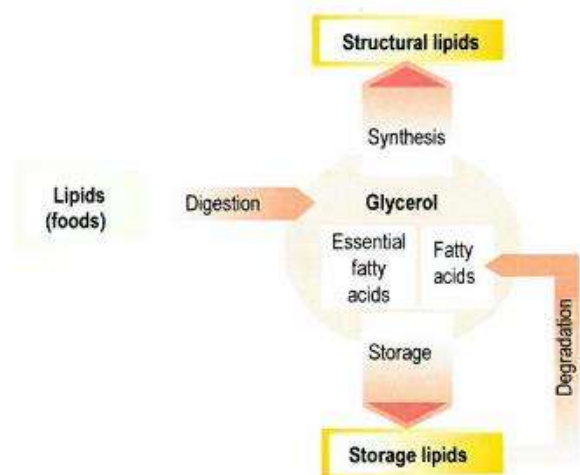


Fig. I. 26 Flux of lipids throughout the organism.

Document 2 Quantitative needs, energetic needs

The body's requirements in energy are satisfied by organic substances, such as carbohydrates, lipids and proteins. These requirements cover the various activities of the organism. Balancing between the intake of organic substances and the expenditure of energy by the organism is the basis of a rational diet.

How are energy expenditures evaluated ? Are they constant ? Can energy expenditure be zero?

1- Evaluation of energy expenditures

The energy released by oxidation of nutrients is proportional to the quantity of the oxidized food and to the volume of oxygen necessary for the reaction. This energy can be measured by burning a nutrient in the calorimetric bomb.

food	liberated energy k J.g ⁻¹	
	calorimetric bomb	organism
carbohydrates	17	17
proteins	23	17
lipids	38	38

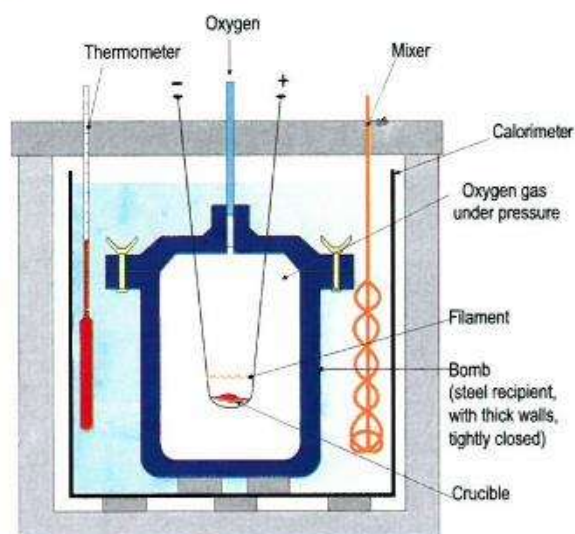


Fig. I. 21 Average energetic value of foods

In the body, oxygen is used to oxidize different nutrients. Therefore, we use an average value of 20 KJ.l⁻¹ for the thermal coefficient that corresponds to the energy released by using 1 liter of oxygen in the degradation of a mixed and varied meal. All the activities of the body require energy and increase the consumption of oxygen. The respiratory intensity allows the measurement of the volume of consumed O₂. It is expressed in liter of O₂.Kg⁻¹.h⁻¹. The respiratory rhythm varies according to the activities of the body in order to ensure the varying needs of energy at all times.

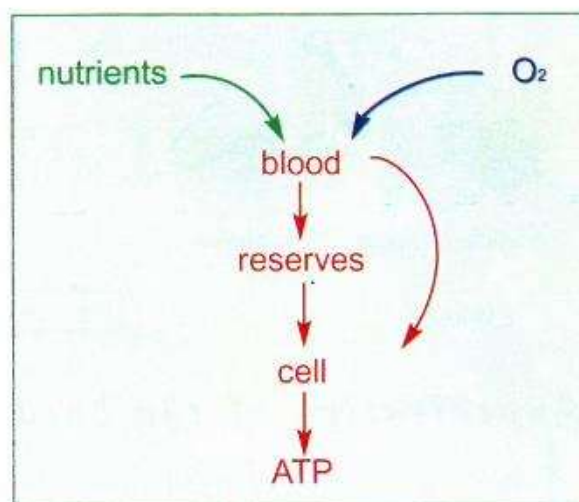


Fig. I. 22 Release of energy in the cell.

Document 4 Qualitative needs, requirements in proteins

Proteins are the structural molecules that are found in muscles, bones, skin, internal organs, blood, hormones and enzymes. They are the builders of cells. Are food proteins similar to the protein of the body? Is protein intake necessary?

1- Proteins, a particular structure

Proteins are complex molecules formed of long chains of amino acids that are the building blocks of a protein molecule. Proteins of animals (including human beings) and plants are made up of twenty different amino acids. These are arranged in different combinations specific for each species. Each combination marks the genetic identity of individuals.

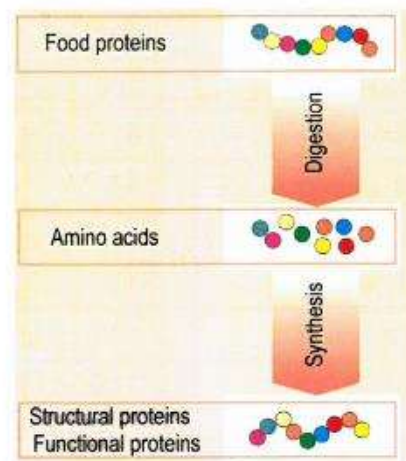


Fig. I. 29
Protein metabolism

2- Daily need in proteins

The daily needs of an individual vary with the age, sex and physiological state of the individual. A minimum intake of $1 \text{ g.Kg}^{-1}.\text{day}$ is reasonable.

(20 g of proteins are obtained from 100g of meat or 100 g of peas or 2 eggs or a half a liter of milk).

Intake deficiency : this causes a destruction of structural proteins without their replacement, which leads to loss of muscle tissue.

Excess intake: this contributes to the formation of reserve lipids and to insufficient elimination of the excess products (urea, ammoniac) by the liver and kidneys.

3- Does every protein contain all the amino acids?

Experiment of Osborne and Mendel

Two identical groups of young rats received meals that differed in their composition but were quantitatively sufficient. The first group received a balanced intake of proteins; the second received zeine (corn proteins) as the only source of proteins, and were later supplemented with tryptophane and then with lysine. The results are shown in the graph.

From this experiment, OSBORN & MENDEL have identified certain essential amino acids that cannot be synthesized by the organism. A protein that does not contain a certain essential amino acid has a diminished nutritional value; this amino acid is considered to be the limiting factor of this protein.

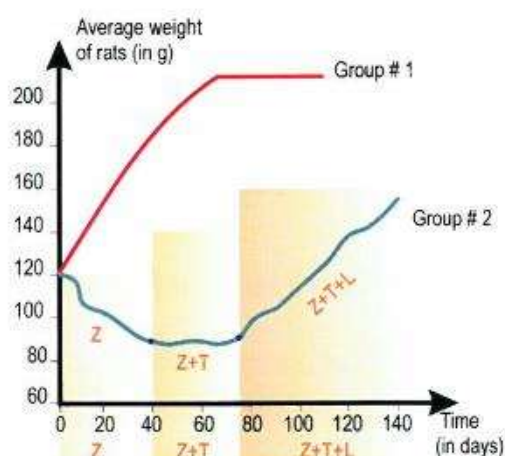


Fig. I. 30 Results of the experiment of Osborn & Mendel.

Where are the essential amino acids found ?

	egg	beef	fish	milk	soybean	daily needs g.kg ⁻¹
Leucine	9	8,2	7,2	8,5	6,7	1,1
Isoleucine	6,9	5	8,1	4	4,2	0,7
Lysine	7,2	8,9	8,1	7,3	6,1	0,8
Methionine	5,8	4,2	3,8	3,7	3,4	0,65
Phenylalanine	5,9	4,3	3,5	5,3	6,1	0,65
Threonine	5	4,7	4,9	4,3	4	0,50
Tryptophan	1,6	1,3	1	1,4	1,2	0,25
Valine	7,4	5,3	5,4	5,5	4,8	0,80

Fig. I. 31 Composition of essential amino acids in 100g of food : soy is the only vegetable food that contains all essential amino acids.

The nutritional or biological value of a protein is its capacity to provide on its own all the needs in essential amino acids. This value is

limited by the lack of one amino acid or by its presence at low concentrations.

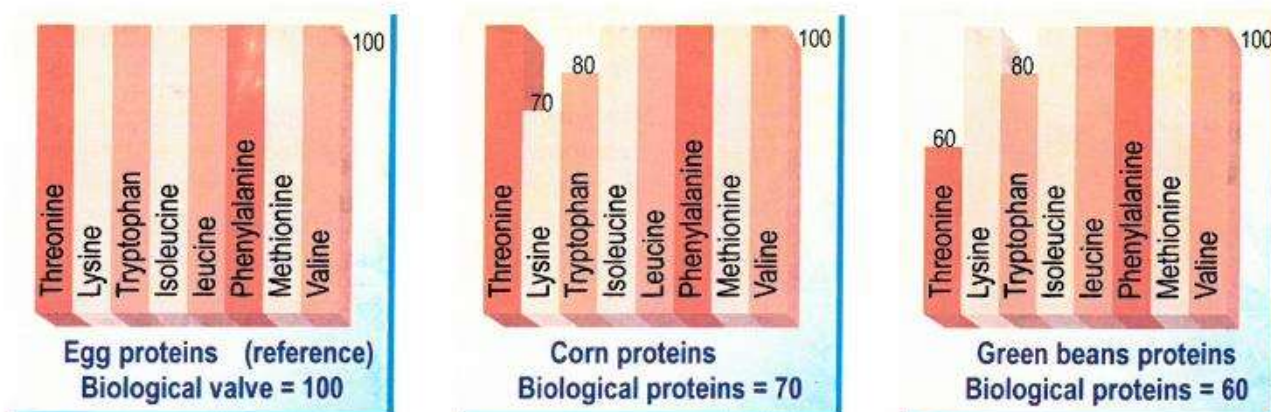


Fig. I. 32 One essential amino acid that is present in insufficient quantity limits the nutritional value of a protein.

Exploitation of the document

1. Arrange the following three amino acids (Lysine-Methionine-Valine) to build a hypothetic chain of proteins made of 5 amino acids each. It is not necessary to use all three amino acids at the same time. Deduce the number of possible combinations that will produce a protein chain.
2. Analyze the graph of Fig. I. 30. Deduce the nutritional value of the zeine.
3. How does soy culture improve nutrition in poor countries ?
4. What is the nutritional value of a meal that contains a mixture of beans and corn? What are the benefits of such a meal for vegetarians (who consume only vegetable products) ?

minerals	principle dietary sources	roles	Needs/J
calcium	cheeses, especially hard cheeses, dairy products, almonds, hazelnuts, parsley, fava beans, fish, eggs.	essential constituent of bones, and teeth. It is involved in the integrity of cellular membranes and in muscular contraction. A deficiency causes osteoporosis.	1 µg
phosphorus	meats, poultry, fish.	essential constituent of bones; important role in nerve conduction and muscular contraction.	0,8 mg
iron	meats, fish, sea shells, vegetables, and dried fruits, cereals. Daily needs increase and are not always satisfied in women during menstruation, pregnancy and breast-feeding, and in teenagers.	essential constituent of hemoglobin (it transports oxygen inside the red blood cells to the tissues); deficiency causes anemia.	15 mg
iodine	iodized salts, fish liver oil.	indispensable to the thyroid gland for the synthesis of thyroid hormones; its absence causes goiters.	0,2 mg
copper	eggs, whole flour, beets, beans, asparagus, spinach, liver, fish.	participates along with iron in the manufacturing of hemoglobin. Intervenes in the synthesis of melanin pigment (responsible for the skin coloration and tanning) helps to fight against infections.	0,02 µg
sodium	table salt, salty foods, canned foods, smoked foods, generally well covered by the diet	indispensable for the proper cellular functioning, at the nervous, cardiac, and muscular level. Intervenes in the water balance of the organism.	2 mg
potassium	dry yeast, vegetables and dried fruits, smoked ham, oily fish, caviar, chocolate, bananas, mushrooms, spinach, potatoes.	essential in nerves, muscular and cardiac activity. Intervenes in many enzymatic reactions.	4 µg
chloride	in most foods, table salt (sodium chloride).	important role in the acid-base and water balance of the organism.	
magnesium	chocolate and powdered cocoa, wheat germ and dried legumes, oats, corn, barley, brown rice, lentils, whole bread;	essential in the cellular functioning, cell membrane permeability, enzymatic activity and energetic metabolism. Intervenes in muscular contraction.	0,3 mg
sulphur	poultry, beef, lamb, eggs, fish, dairy products.	intervenes in the composition of numerous hormones (as insulin), participates in the regulation of numerous metabolic reactions.	
zinc	especially foods from animal origin; meats, giblets, fish, shellfish, seashells, egg yolk.	essential constituent of many enzymes and proteins. anti-inflammatory action in acne lesions.	15 mg
fluorine	water, fluoride table salts.	hardener of bones and teeth, anti-decaying action.	1,5 mg

Fig. I. 38 Principle mineral elements that are essential to the body.

Exploitation of the document

1. If we burn mineral substances in the calorimetric bomb, do they produce heat ?
2. Why do we use iodized salt instead of sodium chloride ?
3. Why do we have a goiter in the case of hypothyroidism ?

Document 5 Qualitative needs, requirements in vitamins

Certain micronutrients are substances that are needed by the body but in small quantities. These include vitamins and minerals. Essential to the organism, vitamins cannot be synthesized; they should be provided by food.

Where do we find vitamins? What are they essential for?

1- History of their discovery

Working in Java in 1889, Eijkman identified in Java the cause of the paralysis of the lower limbs in some prisoners. He noticed that chickens raised in the yard of the prison and fed with the same husked (polished) rice as the prisoners developed the same symptoms. He concluded that the disorder was caused by a food deficiency. He rapidly cured the prisoners by supplementing their diet with the bran of rice. He called this particular substance, which is essential to life and present in rice husk and cereals, a vitamin.



Fig. I. 33 The slightest excitation provoke convulsive crisis in a pigeon that is only fed with polished rice for one month. An injection of vitamin B make those convulsions disappear within few hours.

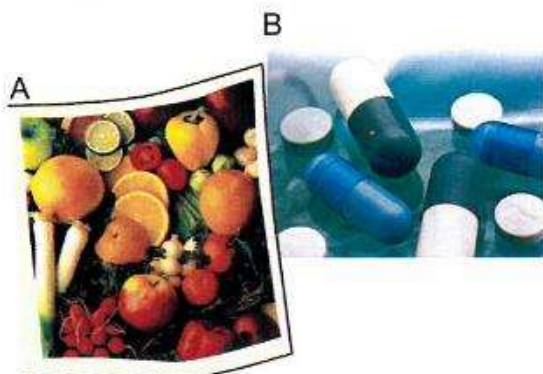


Fig. I. 34 A) : Vitamin rich foods; we consume vitamins without knowing it. B) : It is also possible to consume synthetic vitamins through capsules.

Vitamins constitute an original type of food; Even though they are organic substances, they do not have energetic value. The suppression of their supply or their lack contributes to the development of serious disorders such as scurvy and beriberi ...

2- Properties of vitamins

Vitamins are usually classified in two groups: hydrosoluble or water soluble (Vitamin C and vitamins of group B), and liposoluble or fat soluble (Vitamin A, D, E, K).

Hydrosoluble vitamins are eliminated from the body in the urine, whereas liposolubles are stored in the body, in the case of excess consumption, there can be intoxications.

Notice !

- Vitamins are fragile: a nutriment that is cooked easily loses 50% of its vitamins.
- Certain vitamins are oxidized by light (A, B₆, B₉, C and D), others are destroyed by heat, humidity or low pH (B, B₂, B₅ and D).
- On composition labels of some conserved foods, the name of certain vitamins is marked. This means that these vitamins have been added after the food was treated.

Document 6 Qualitative needs, mineral requirements

Even though some elements are not a source of energy, they are indispensable to the proper functioning of the body. These mineral elements can be found free in the body or fixed in certain tissues. They are found in the food we eat and are often taken as supplements.

What are mineral substances? Why do we need them?

1- Water requirement

Water is the main component of the human body (60%-70%). It is present in all the structures of the body and intervenes in the dissolution of several nutrients.

nature of losses	volume of water L per day
urination (urine)	1,4 - 1,7
defecation (feces)	0,1
respiration (water vapor)	0,4 - 0,9
transpiration (sweat)	0,4 - 0,7
total	2,3 - 3,4

Fig. I. 36 Water losses should be recuperated in the forms of beverages (1.2L) or food content.

A person can fast for several days, but cannot go more than three days without water.



Fig. I. 37 The water intake must be increased in hot temperatures.

2- Mineral ions requirements

Certain mineral salts such as sodium chloride dissolved in water ensure electrolytic equilibrium which is an essential condition for the proper functioning of the organism.

Others are components of different parts of the body, and the lack of certain minerals such as calcium, phosphorus, and iron, leads to severe disorders.

Osteoporosis

This is a progressive decrease in the protein mass of the bone. The bone becomes less dense and more fragile. The causes are diverse (age, hormonal problems ...) as are the ways of treatment. Prevention is indispensable and it is recommended :

- to have a diet rich in calcium and proteins, starting from a very early age
- to limit the consumption of alcohol and tobacco
- to practice sports such as walking regularly

Bearing children and nursing may have a negative effect on bone mass.

Tooth decay: an epidemic

An effective way to prevent tooth decay is to use fluoride; its incorporation into the tooth enamel renders it resistant to the attacks of acids.

Its action is beneficial in small doses (1.5 mg- 4mg). An excess intake can cause the appearance of spots on the tooth enamel).

Thyroidism

This is an increase in the volume of the thyroid gland. It is a frequent disease, with more than 800 million people affected (people living in the Lebanese mountains are mostly affected). A deficiency in iodine, which is an obligatory component of thyroid hormones, can be the cause of this disease.

3- Various vitamins and their different roles

Vitamins	principal dietary sources	roles	quantity needs
A retinol	fish liver oil, fish and animal livers, egg yolk, parsley, spinach, green salads and certain fruits.	Important role in vision especially in dim light; defense against infections, maintenance of the integrity of the skin and the mucus membranes, proper growth in children.	800 µg
B ₁ thiamine	yeasts and cereals, and in lesser quantity: vegetables, eggs, meats fish and dairy products.	essential vitamin for the breakdown of carbohydrates in the organism: necessary to the proper functioning of the cells and the transmission of the nerve impulses. Its deficiency causes the beriberi disease which symptoms are paralysis, serious psychic troubles, and in severe forms, cardiac failure and edemas.	1,5 mg
B ₂ riboflavine	cereal sprouts, liver, milk, eggs, greens, yeasts.	major role in the metabolism of carbohydrates, lipids and proteins. Its deficiency causes lesions of the skin of the mouth and of the ocular conjunctiva.	1,8 mg
vitamine PP or vitamine B ₃ or nicotinamide	yeasts, meats, fish, dried fruits, cereals, (except corn, millet and sorghum), liver, mushrooms, and dried vegetables.	necessary for growth. Its deficiency causes pellagra characterized by an inflammation of the skin, the mouth, and the digestive tract with diarrhea, insomnia, depression, delirium and hallucinations.	20 mg
B ₁₂ cyanocobalamine	food from animal origin essentially; goblets, fish, meats, eggs, cheese and dairy products.	intervenes in the metabolism of neurons, blood cells, intestine and uterus. Its deficiency causes anemia and different neurological troubles	3 µg
C ascorbic acide	fresh fruits (especially kiwis, red currants, lemons, oranges, grape-fruits, strawberries), raw fruits and vegetables salads, and greens (mainly parsley green peppers, cauliflower) important loss during cooking.	favors iron absorption and therefore fights against anemia; stimulates the anti-infectious defenses of the organism; its deficiency causes the appearance of scurvy (fatigue, hemorrhages of gums and skin, anemia, bone pain.)	80 mg
D calciferol	fat fish, liver, egg yolks, meats, milk-fat: exposure to sun favors the synthesis of vitamin D from a precursor located in the skin.	favors the intestinal absorption of calcium and its fixation in the bony tissue (bone mineralisation), ensures the normal development of the skeleton. Is essential to the newborns and the osteomalacia in elders.	15 µg
E tocopherol	vegetable oils, liver, cereal germs, green vegetables, egg, milk, butter.	natural antioxidant; extends the life of red blood cells, favors the synthesis of the good cholesterol (HDL cholesterol).	12 mg
K phytoménadione	green vegetables (spinach, cabbage), liver, meats. It is also manufactured by intestinal bacteria.	essential vitamin for blood clotting: hemorrhagic disease occurring in premature newborn babies, or children born to mothers treated during their pregnancy by certain antibiotics, rare deficiencies in the adults are observed.	1,5 mg

Fig. I. 35 Principal vitamins.

Exploitation of the document

1. Find in a CDI the symptoms of certain vitamin deficiencies.
2. How do we explain the fact that, even though we consume polished rice, we don't suffer from beriberi ?
3. Why do vitamin deficiencies occur even when there is sufficient food intake? Does malnutrition result only from undernourishment ?
4. Vitamin C is very fragile. How can we consume citrus fruits to satisfy our needs in vitamin C ?
5. Why do we say that the cases of hypervitaminosis are observed especially with liposoluble vitamins ?

The fate of molecules

Knowing that the total mass of proteins in the body is constant (1/7 of the total mass), we can deduce that proteins are constantly degraded and replaced with proteins of similar nature and quantity. Amino acids are not stored, hence the need of a daily intake of protein in order to maintain the equilibrium of the body. An equilibrium is established between the degradation and the synthesis of the same molecules.

The reserve molecules(carbohydrates and lipids) are constantly renewed. We cannot estimate their life span because they can be stored. When taken in excess the glucose will be stored. When there is a deficiency in the intake of glucose, it will be regenerated from the body reserves (glycogen and lipids).

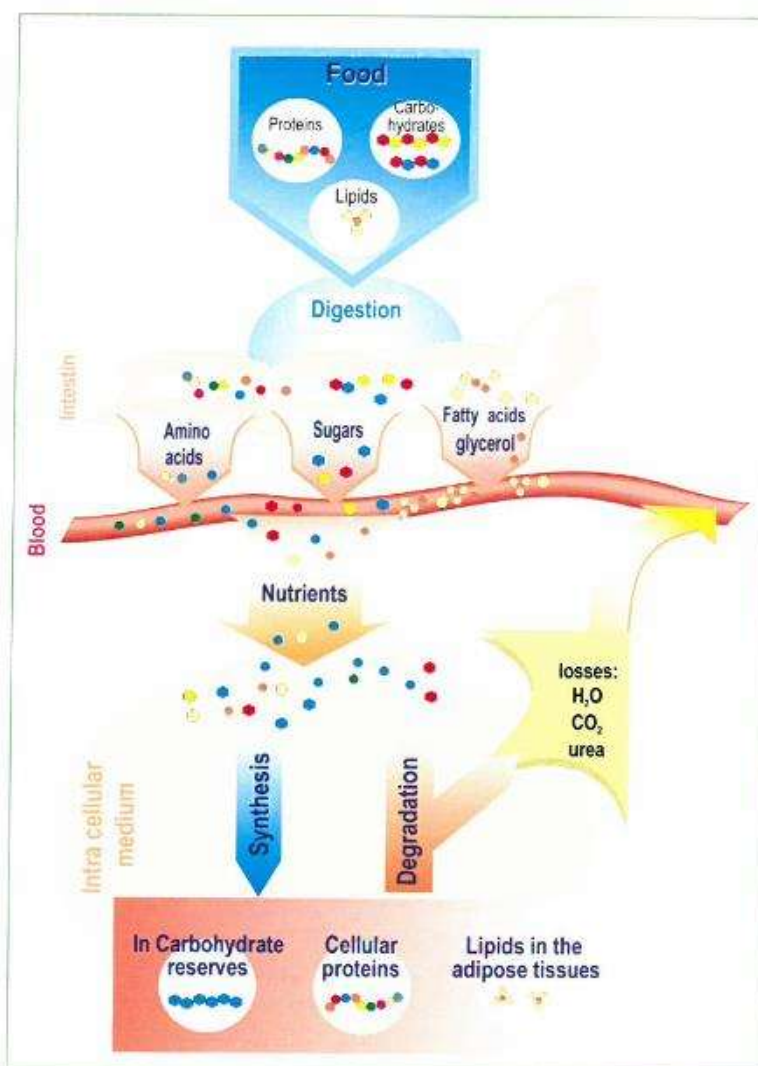


Fig. I. 45 Flux of matter throughout the body.

Exploitation of the document

1. Why does cellular renewal require an intake in matter ?
2. Is cellular renewal limited to the growth period only ?
3. Through which processes are body losses produced ?
4. How does the body maintain the same cellular and molecular characters despite biological renewal ?
5. Analyze the graph of Fig. 44 and deduce the life span of the studied protein.
6. Can we obtain the same graphs of Fig. 44 with carbohydrates and lipids? Why ?

Balance animal and vegetable proteins

In a rational meal the ratio

$$\frac{\text{animal proteins}}{\text{vegetable proteins}} = 1$$

Balance the lipid intake

$$\frac{\text{lipids from animal source}}{\text{lipids from vegetable source}} = 1/2$$

Balance the carbohydrate intake

$$\frac{\text{sugar with rapid assimilation}}{\text{sugar with slow assimilation}} = 1/3$$

Do not forget fiber-rich foods

Carbohydrates also include indigestible fibers (cellulose, pectin...). Fiber-foods do not have energetic value, but they improve intestinal transit and give a sensation of satiety . They thus favor a slimming diet.

What is a portion ?

- bread and cereals : 1/4 of a loaf of bread or 1 slice, 1/2 bowl of cereals
- legumes : 1 tomato
- fruits : 1 apple or 1 banana or 1 orange
- meats, eggs, fish : 90 g of meat or fish or 1 egg
- milk and dairy : 1 glass of milk or 60g of cheese.

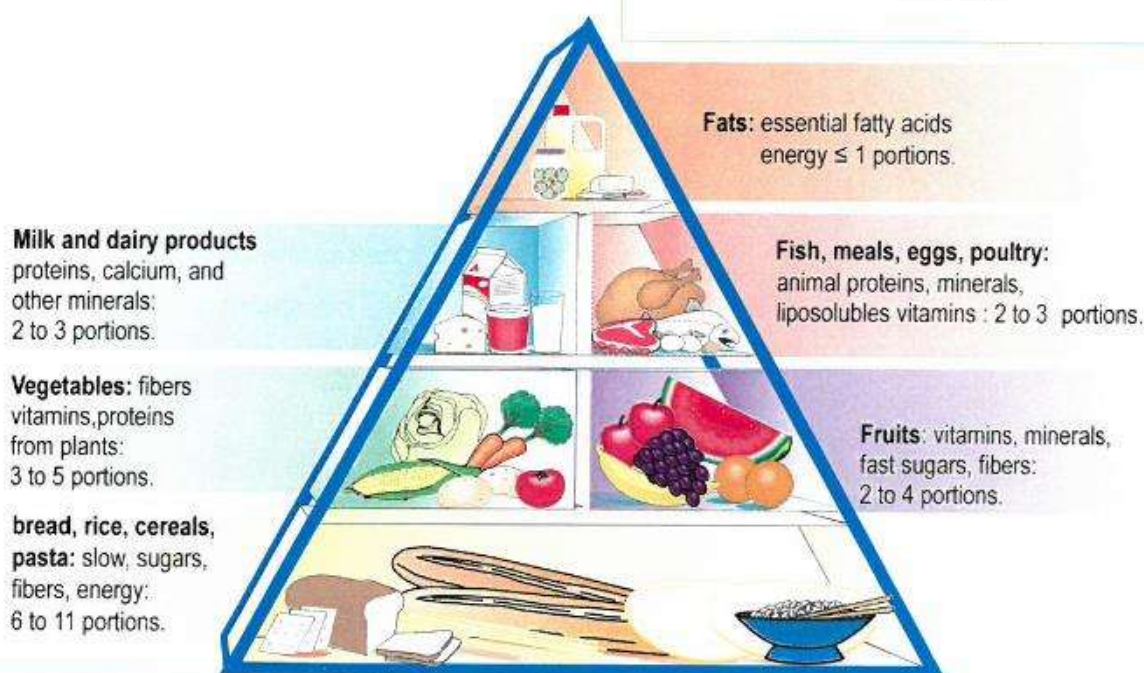


Fig. I. 41 Example of a food pyramid

Exploitation of the document

1. Name the qualitative needs of the organism . Why are they necessary ?

2. Here is an example of a diet :

milk + sugar + bread + butter + cold cuts + fruit juices + fruit.

Study the global composition and deduce if it follows the formula 4,2,1

3. Try to find out if a meal that you consume corresponds to the formula 4,2,1.

Document 8 The fate of nutrients

The dietary requirements of the organism are permanent and are ensured by the daily food intake. During growth the organism undergoes noticeable changes. Outside the growth period, its appearance remains unchanged even though it receives the same intake. What is the fate of nutrients in the body? Why are the dietary needs permanent?

1- Cellular renewal

Certain cells have a limited life span, and they need to be replaced by new ones. The life span of leukocytes and platelets is a few days. That of the red blood cells is of three months. Every day we replace 2,500,000 red blood cells.

2- Molecular renewal

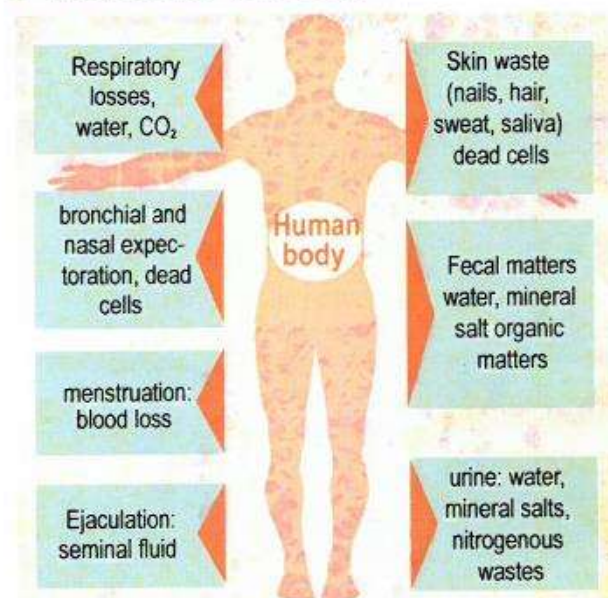


Fig. I. 43 Forms and compositions of losses observed in the body

Evaluation of a protein's life span

Labeling an amino acid with radioactivity allows the monitoring of a protein (which is marked by the radioactive amino acid) from its synthesis until its degradation. The presence and then disappearance of radioactivity shows that proteins have a determined life span.

This graph shows the variation of the radioactivity in proteins of the plasma in a rat that was injected with ¹⁴C (radioactive) labeled glycine. The 100% radioactive value is an arbitrary one.

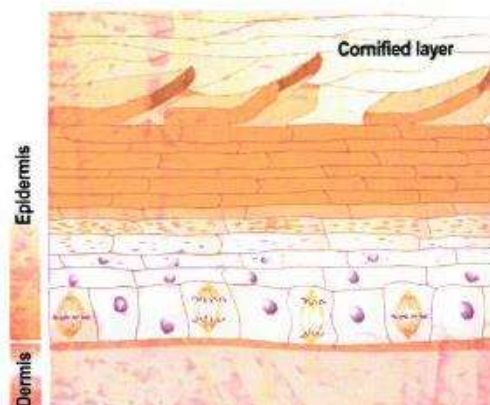


Fig. I. 42 Skin cells are renewed. We change skin every week.

The chemical analysis of the constitution of cells reveals the presence of water carbohydrates, lipids and proteins.

The molecules, constituting the cells have a very limited life span. They are degraded and renewed constantly.

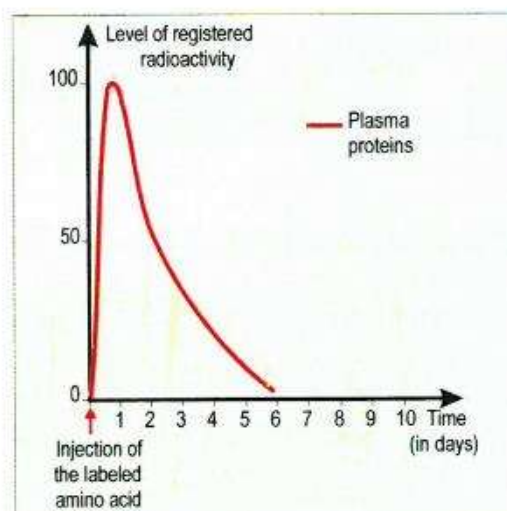


Fig. I. 44 Amino acid labeling allows to trace the fate of the radioactive protein.

Document 7 Constructing a balanced food ration

A food ration is called balanced if it responds to all the needs of the body, i.e., it ensures the body's proper functioning, its proper structure and maintain its good health. A balanced food diet can only be attained from varied nutriment. Is there one balanced ration? What principles should be followed to construct such a ration ?

1- To ensure the energetic needs

The expenditures of energy vary with age, gender, and the physiological state.

type of activity	energy spent
office work (sitting, writing...)	340 - 420
march (4 km/h), manual work, carpenter...	460 - 670
march (6 km/ h), biking - tennis	710 - 1000
race - football	1050 - 1450
athletes exercises	> 1450

Fig. I. 39 For each physical activity there is a given energetic expenditure.

age (years)	% of decrease relative to the standard
30 - 40	3
40 - 50	6
50 - 60	13,5
60 - 70	21
>70	31

Fig. I. 40 Adjustment of energetic needs according to age.

The rule to follow

The first rule to follow is that to evaluate the needs and to bring a sufficient quantity in organic nutrient to cover them. This equivalence permits to keep the same body mass. All supplementary physical activities require an additional energy intake.

A few formulas to apply

All dieticians agree on the benefits of various nutrients but use different formulas to ensure the balance.

- Formula 4, 2, 1 = CPL by Dr A. Creff. Energetic expenditures are ensured by 4 portions of carbohydrates equivalent to 55% of the total energetic intake (TEI), 2 portions of proteins equivalent to 15%, and 1 portion of lipids equivalent to 30%. This is not an ideal formula but it is easy to apply.
- Distribute the energetic intake throughout the day. Breakfast should provide at least 25% of the total energetic intake of the day.

2. To ensure all qualitative needs

A nutrient cannot be replaced by another. Each has its own nutritional value. Hence the need to adopt a varied diet. To avoid an

excess in certain nutrients, nutritionists propose that food balance should to be respected :

Principles of a balanced diet

The liberty that we have in choosing the quantity and the quality of food plays a very important role in what we consume. Dietary science is considered the true science of food. It proposes the hygienic rules of diet to follow in order to conserve good health and to recuperate in case of sickness. In summary, it looks to balance the diet and to adapt it to everyone's need.

1- Food diversity

Foods are formed of a limited number of constituents: Water – minerals – carbohydrates – proteins and vitamins. They provide the organism with the needed matter:

- to build cells
- to enable cells to function
- to liberate the energy needed for cellular activities

Organic foods are the only ones that produce energy.

2- Balanced diet

This means a diet that ensures to the living organism all what is required without deficits or excesses. The requirements are not always the same, they vary from one individual to the other, they vary with age, gender and the individual's physiological status.

It is imperative to know the body's needs in order to balance the diet :

Quantitative needs

They represent the quantity of organic nutrients that ensures the energy that is spent by the body.

Energy is spent to perform different activities: muscular, digestive, thermo-regulation, basal metabolism.

Spent energy is evaluated by the O_2 volume that is consumed. This volume reflects the respiratory intensity

Energy expenditure =

Volume of consumed O_2 X 20kJ.l^{-1}

Qualitative needs

These represent the essential substances that the body cannot synthesize but must obtain from nutriment.

• Requirements in proteins

Proteins are made up of simple molecules of amino acids that are not stored. So, a daily intake of protein is needed. Among the 20 amino acids that constitute the proteins, some are essential. Eight amino acids cannot be made in the body and must be supplied in the diet (Food). They are all present in proteins of animal origin but not all are present in vegetable proteins.

• Requirements in essential fatty acids

Certain essential unsaturated fatty acids are only present in lipids of plant origins and must be supplied to the body by diet.

• Carbohydrates requirements

Carbohydrates mostly under the form of slow sugars must cover more than half of the energetic needs. In addition they must bring nutritious fibers that are the most filling and facilitate the intestinal transit.

Summary diagram

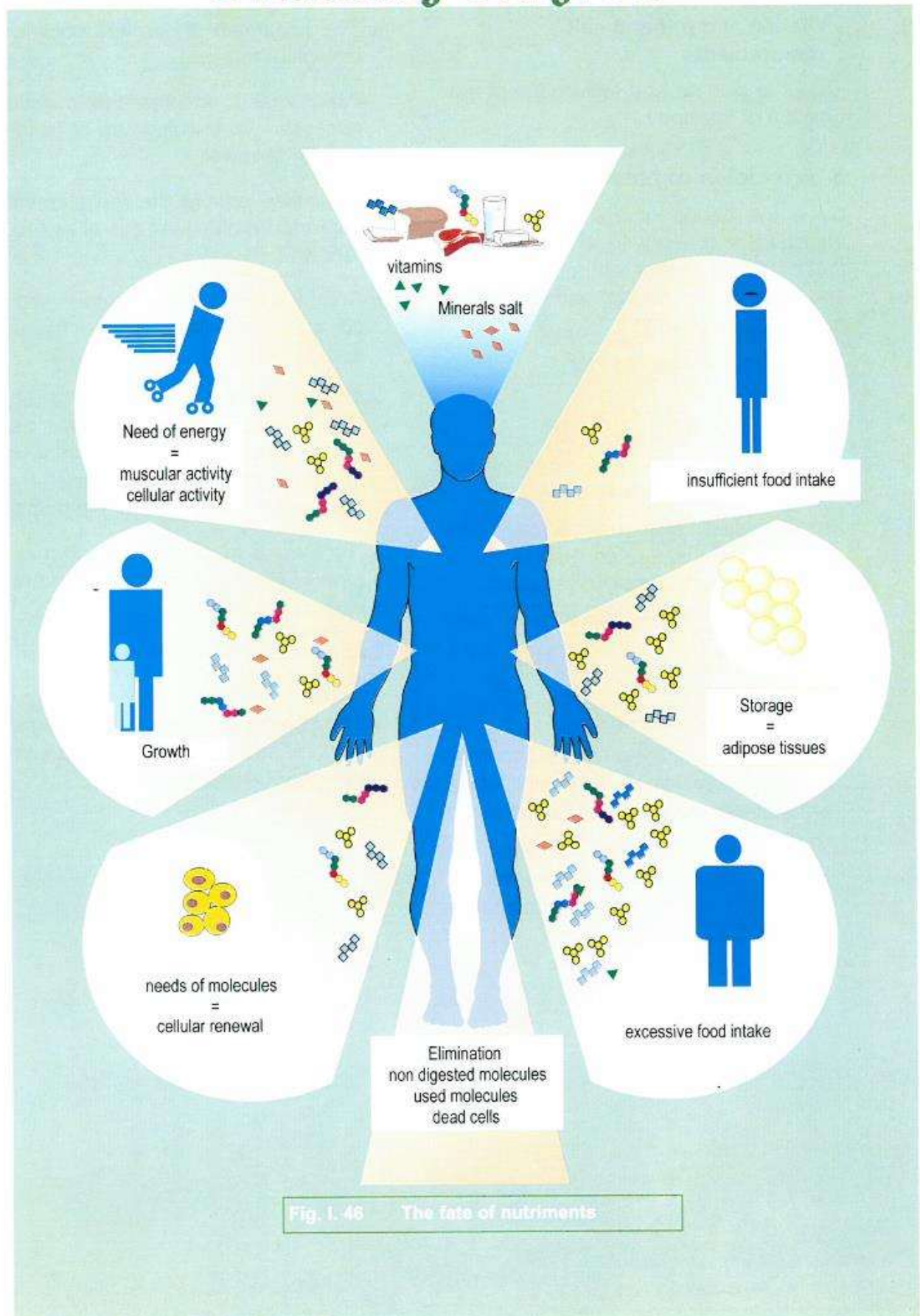


Fig. 1.46 The fate of nutrients

- **Vitamin and mineral salt requirements**

They are diverse and ensure the proper function of the body.

3- Principles of balanced diet

Once the needs of the organism are identified, it is easy to elaborate rules that permit to ensure all the quantitative and qualitative needs without either excess or deficiency:

- Adopting a diversified diet.
- Balance
 - a) Energetic nutrients: 4 portions of carbohydrates: 2 portions of proteins:1 portion of lipids (4:2:1),
 - b) Saturated and unsaturated fatty acids: 10% saturated, 10% monounsaturated, 10% polyunsaturated,
 - c) Proteins from animal and plant origin.

4- The fate of nutrients

- The body is in a dynamic equilibrium, it sustains daily losses in matter which are replaced by intakes of nutrients.
- All the molecules that constitute the cells or ensure their activities have a limited life span. They are constantly degraded and then replaced.

- The organism undergoes continuous biological renewal :

Food intake ensures the molecules necessary for building new cells: this is cellular renewal.

They also ensure the replacement of degraded molecules: this is molecular renewal

- Complex foods are consumed, digested and transformed in the digestive tract to nutrients.
- The nutrients are molecules of simple carbohydrates, amino acids, fatty acids, glycerol, water, vitamins and minerals. They can be assimilated and distributed in the cells by the blood.
- The excess in glucose is stored in the form of glycogen in the liver and muscles.
- The excess in lipids is stored in the adipose tissue.
- The excess in proteins is degraded in the liver and cleared by the kidneys as nitrogenous waste.

Chapter 3

Nutrition and health

Nutritional diseases: characteristics, causes and prevention

Nutrition is necessary for our health. Any nutritional unbalance can be responsible for diseases which may lead to some serious consequences. Malnutrition and undernourishment are the two faces of hunger. Undernourishment is a quantitative deficiency in the diet. When the food quantity is sufficient, but the diet is unbalanced, this may lead to malnutrition. In poor countries (Africa, South America) undernourishment and malnutrition often coexist and they particularly affect children. On the other hand, in developed countries, diseases that are associated with diet are due to excessive food intake.

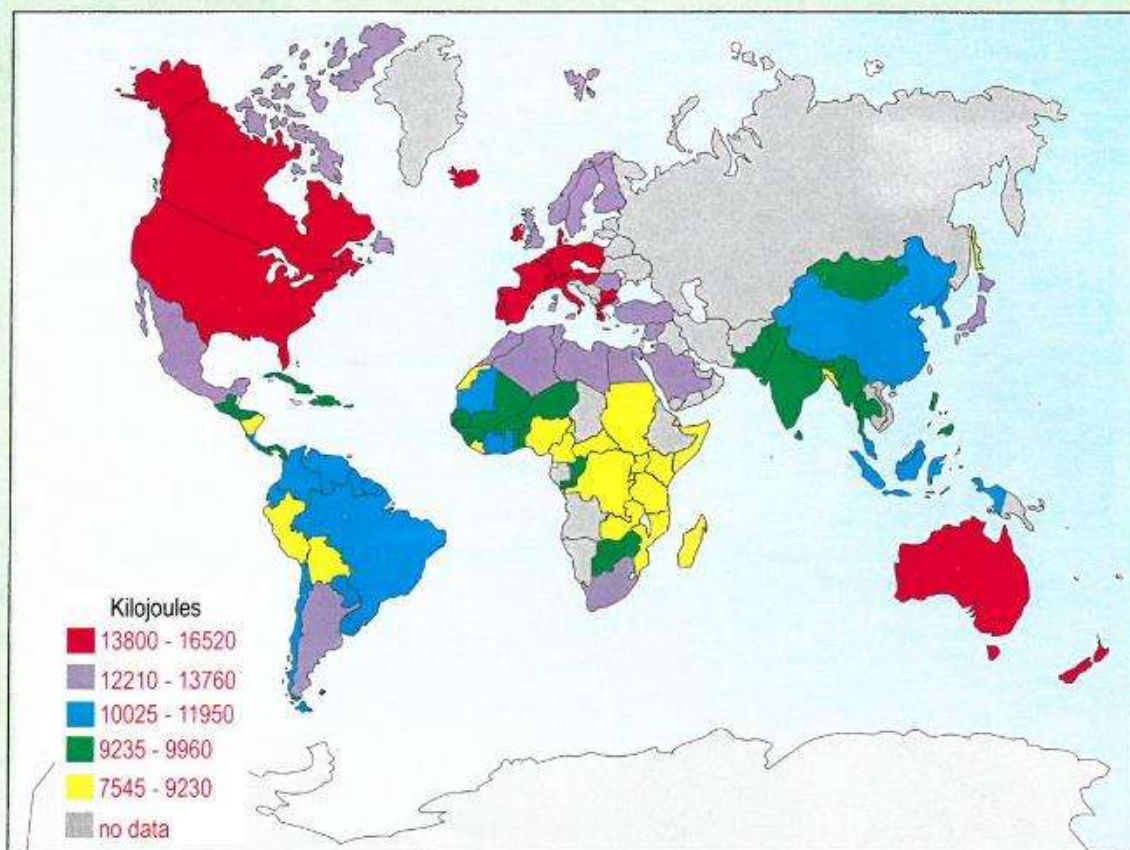


Fig. 1.47: Inequality of food distribution in the world.

In this chapter we shall answer the following questions :

- What are the consequences of hunger ?
- How does an excessive food intake affect health ?

Additional information

We consume sugar without knowing !

In developed countries an average meal provides one tea spoon of refined sugar per hour. In addition to the presence of sugar in cakes and pastries, hidden sugars are found in almost every precooked plate (conserved food, sausages, soups)

Can we heat all oils ?

Heat can destroy the oil.

- Arachidonic oil resists when it is fried for long periods.
- Corn oil is less resistant and can only be used when frying for a short period.
- Olive oil is usually used as seasoning and sometimes for quick frying.
- Sunflower oil, soy oil: should never be heated.

Hidden Fats

Our food is already rich in fat without the consumption of lots of butter and oil. It is because we consume large quantities of meat and cheese, which contain certain lipids in their composition.

The importance of soybean

The introduction of soybean with its vegetable proteins radically improved the diet. In fact soy is present in many foods that are consumed daily, such as pastries, precooked food, cold cuts, sauces, etc. In poor countries where food of animal origin is rare and very expensive the introduction of soybean made a startling improvement to diet.



Eating between meals

It has become a common habit to eat sweet or salty snacks between meals. Extras have become a habit. This is partly because breakfast is neglected, unbalanced, insufficient, prepared and swallowed hastily. The result is that people feel the need of light refreshment during the morning. Commercial interest profits from this demand, and responds with easily available chocolate bars, pastries and biscuits. These can be carried in the pocket, easily unwrapped and eaten at any time.

These are not eaten because of actual hunger, more often they are eaten because habit has created a physiological demand for sweets (candies). Eating food with a pleasant taste makes the body excrete endorphin molecules that act against stress and pain.

To avoid the habit of eating between meals we should understand what drives us to do it. First we should take enough time over our meals. If we feel a little hungry between meals, we should choose well balanced foods, like fruits, a piece of whole wheat bread, dried biscuits, small tomatoes, radishes or small vegetables for those who prefer something crispy.



EVALUATION EXERCISE

A- Choose the correct answers(s)

- 1- Nutrients are molecules of:
 - a- glycogen,
 - b- lipids,
 - c- glucose,
 - d- proteins,
 - e- amino acids.
- 2- Basal metabolism involves the energetic expenditures relative to :
 - a- muscular activity,
 - b- cardiac activity,
 - c- nervous activity,
 - d- thermoregulation,
 - e- metabolic activity.
- 3- An essential substance is a substance that:
 - a- is synthesized by the organism
 - b- is stored in the organism
 - c- cannot be synthesized by the organism
 - d- is provided by food intake .
- 4- The biological renewal takes place in the organism :
 - a- in the growth period
 - b- during food intake
 - c- during muscular activity
 - d- during one's life time.
- 5- A balanced energetic meal brings :
 - a- a sufficient quantity of carbohydrates
 - b- a quantity of vitamins
 - c- a sufficient quantity of lipids
 - d- a sufficient quantity of organic nutriment.

B- Choose the right statement(s) and correct the false one(s)

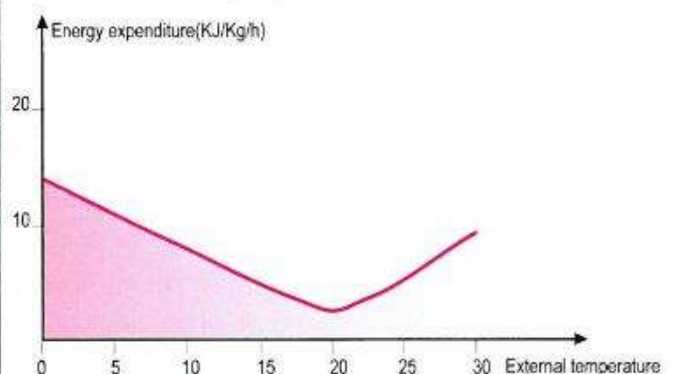
- 1- Carbohydrates, lipids and proteins are complex nutriment.
- 2- The body utilizes glucose from its reserves.
- 3- Supplementary muscular activity of an organism provokes a decrease in the respiratory intensity.
- 4- Essential fatty acids are present in lipids coming from an animal source.
- 5- Vitamin deficiencies are observed with diets based on diverse fresh products.

C- Think about it

- 1- During exercise, an athlete consumes three liters of oxygen per minute.
 - a- How much energy does he spend if his exercise lasts 10 minutes.
 - b- Following this exercise, he eats 100g of bread (8g of proteins, 1g of lipids, 54g of carbohydrates) with 50g of white cheese (10g of proteins, 1g of lipids, 0.5g of carbohydrates) and one apple (0.4g of proteins, 0.4g of lipids and 14g of carbohydrates).

Evaluate the energy provided by this meal.

- c- What advice will you give him regarding the energetic intake?
- 2- An individual follows a food diet, lacking proteins. Chemical analysis of his losses indicate the presence of nitrogenous wastes.
 - a- What is the source of those wastes ?
After a few weeks, he loses weight. to compensate, he is to eat food rich in proteins, like cereals (wheat and corn).
 - b- Why did he loose weight ?
 - c- Can this new diet stop the weight loss ? Justify your answer.
- 3- We evaluate the energetic expenditures of an individual at rest, normally dressed, by exposing him to different temperatures. The values are recorded on a graph.



- a- How do the energetic expenditures vary with the temperature of exposure ?
- b- What does the 20°C temperature correspond to.
- c- Suggest a simple method to evaluate the energy expenditure.

Document 3 Obesity. Symptom of the rich countries

When the food intake exceeds what the organism needs, the mass of the latter increases. This is due to the accumulation of the synthesized products. It has been estimated that 50 km of supplementary blood vessels develop for every 500 g of fats in the obese person. Thus, the heart must work vigorously in order to distribute the blood into all those vessels.

Is obesity a disease ?

1- What are the criteria for obesity ?

Obesity is defined as a 20% increase over the ideal weight. This increase is the consequence of the accumulation of adipose tissue. Nutritionists propose several formulas for determining the ideal weight of a person. An athlete may be considered overweight because of an increase in his (her) mass, but is not considered obese.

Calculation of the ideal weight

A widely used formula which measures the degree of obesity, is the body mass index (BMI) :

$$\text{BMI} = \text{weight kg} / (\text{height m})^2$$

Even moderate obesity may be dangerous for health. This is because obesity is considered to be one of the three main risk factors of cardiovascular diseases.

Fatal obesity

obesity may become fatal when the weight of the body is two to three times the ideal weight. In this case, it is often accompanied with hypertension, diabetes, and atherosclerosis.

excess weight%	% increase of mortality
20-25 %	16%
25-30%	30%
35-40%	54%
50-74%	130 to 182 %

Fig. I. 65 Obesity: fatal risk factor.

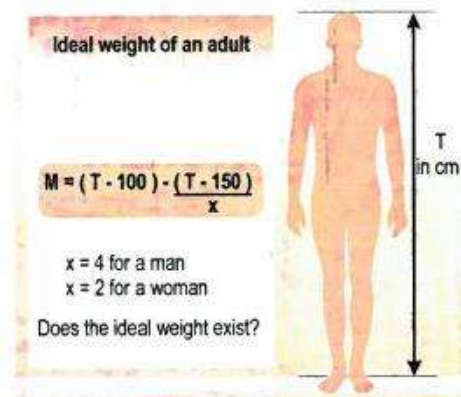


Fig. I. 62 Does the ideal weight exist ?

BMI	category	risk of weight-related diseases
<18	very thin	much below average
18-20	thin	below average
20-25	normal	average
25-29	moderate	above average
> 29	obese	much above average

Fig. I. 63 Obesity often increases the risk of atheroma. Sedentary life style, diabetes, increase of cardiac activity, hypertension and hypercholesterolemia are also risk factors.

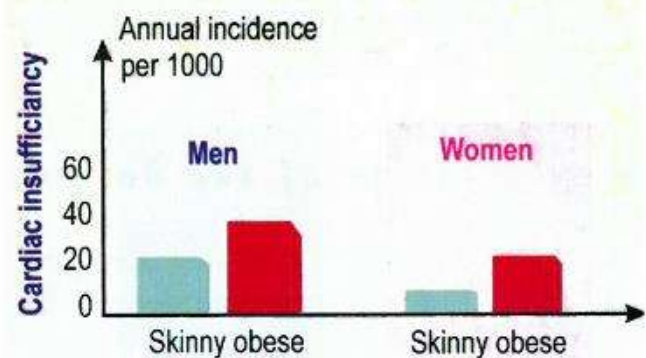


Fig. I. 64 Obesity: risk factor of cardiovascular diseases.

According to their density, we can divide lipoproteins into two categories:

HDL (high density lipoproteins) which contains 13% cholesterol

LDL (low density lipoproteins) which contains 55% cholesterol.

LDL provides the cholesterol for the body cells. However, under abnormal conditions, LDL deposits cholesterol on the internal walls

of arteries, where it accumulates, causing the formation of a plaque called atheroma: this is the "bad cholesterol".

HDL is called "good cholesterol". It plays a protective role since it carries cholesterol from tissues to the liver to prevent its accumulation in the blood. Thus a high level of HDL is associated with a low risk of cardiovascular disease.

Amount of cholesterol in adults' blood

	normal (mg/dl)	high risk (mg/dl)
Total cholesterol	<200	>239
LDL	< 130	> 159
HDL	> 40	< 40
Total / HDL	< 5	> 5

Fig. I. 58 Good and bad cholesterol, and risk of cardiovascular diseases.

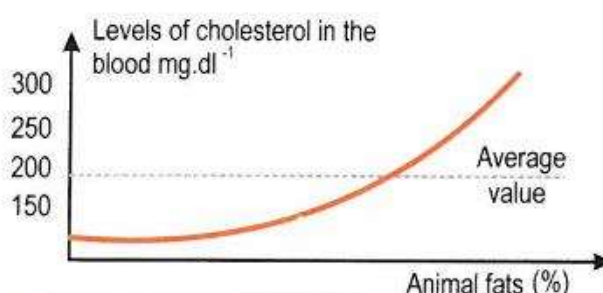


Fig. I. 59 The rate of blood cholesterol increases with the increase in the intake of animal fats in food.

3- Risk factors and coronary morbidity

Framingham has studied many factors believed to be related to coronary diseases :

- hypercholesterolemia (more than 250 mg/dl),
- arterial hypertension (a systolic tension more

than 16 mm Hg),

- smoking.

He observed that the presence of more than one factor increases the risk of death by coronary accidents.

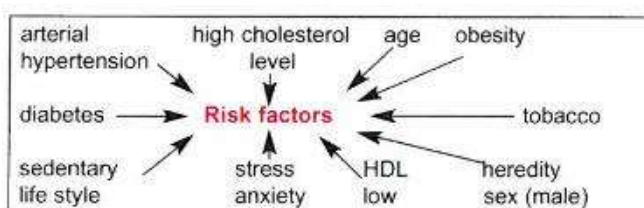


Fig. I. 60 Risk factors are related to our lifestyle. Can they all be controlled ?

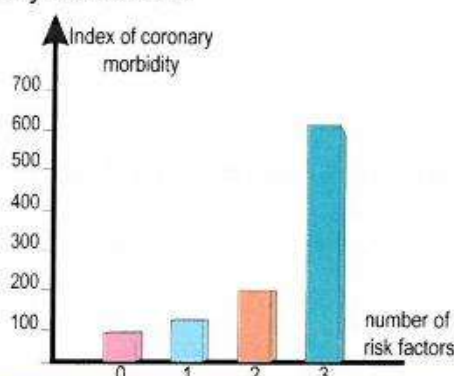


Fig. I. 61 Addition of risk factors.

Exploitation of the document

1. What is the relationship between plasmatic fats and cardiovascular diseases ?
2. Why is atheroma formation dangerous to health ?
3. Try to compare a blood test from a relative with the results of the table Fig. I.58.
4. What do you observe in the Framingham graph ?
5. What type of lipoproteins contributes to the development of atherosclerosis ?

Document 1 Food deficiency diseases

A person's diet must provide enough matter and energy. If the diet is quantitatively and qualitatively insufficient, many diseases may appear. These diseases are often serious and threaten the life of undernourished or malnourished children (6 months to 5 years) living in poor countries.

Other diseases appear when the diet is lacking certain vitamins.

1- Marasmus and kwashiorkor: two starvation diseases

Marasmus : If the diet is lacking energy (less than 4500 KJ/day), then the child's growth becomes limited. In children suffering from this condition, the brain is smaller and its cell composition is altered. Marasmus is an irreversible disease and may be life threatening.



Fig. I. 49 Kwashiorkor syndrome is due to a qualitative and quantitative protein deficiency.



Fig. I. 48 Marasmus is a total deficiency in food.

Kwashiorkor : This disease is due to a qualitative food defect. Food quantity may be sufficient but the diet is unbalanced. In this case, the child who has just been weaned is placed on a cereal diet or manioc porridges which are poor in proteins and lack some essential amino acids. The child becomes apathetic and suffers from serious edemas (water infiltrations in the dermis), dermatoses (skin diseases) and gastro-intestinal disorders; 30% of the children who are affected by kwashiorkor before they are five years of age will certainly die.

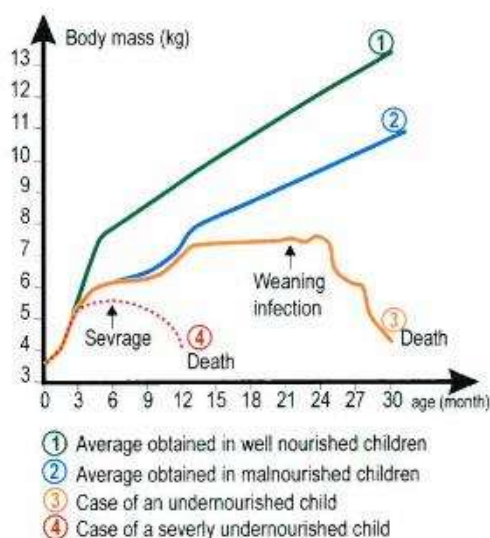


Fig. I. 51 Comparison of weight gain in well fed and malnourished children.

average value for 100g of food	proteins	absorbed carbohydrates	lipids
maternel milk (dry extract)	11	55	30
manioc	1 to 2	86	0,2

Fig. I. 50 Comparison table between maternal milk and manioc (or semolina).

2- Treatment and prevention

To treat the children who are affected by marasmus or kwashiorkor, we must :

- watch their growth by taking their weight regularly.
- feed them with a quantitatively (quantity of kilojoules) and qualitatively (quantity of proteins notably) balanced diet.
- treat the children for possible infections and parasites.

Concerning prevention, many world organizations have plans to improve the standards of living of the populations concerned, and to educate the mothers about the elementary principles of hygiene and nutrition.



Fig. I. 53 Rachitis, a deformation of the skeleton.

Rachitis : It is a bone mineralization disorder that occurs during growth, and is characterized by arching of the legs. It is due to a deficiency in calcium or vitamin D that is necessary for the fixation of calcium in the bone. It can also be due to the lack of exposure to the sun. The skin has pro-vitamin D that upon exposure to sunlight is transformed into vitamin D. Rachitis affects fragile children or children who live in unfavorable social conditions.



Fig. I. 52 All the shame of our time is in a child's look. Note that every year, 800 million children still die of hunger. The programs that are implemented by the FAO remain insufficient.

Anorexia : It is an auto-deprivation food deficiency due to a psychological disorder. This disease is more common among young girls who unconsciously refuse to become adults. They follow a very strict diet that has serious consequences and requires emergency hospitalization and psychiatric help. Severe loss of weight is not the only indicator of auto-deprivation. There are other symptoms such as cessation of the menstrual periods, diarrhea, constipation, infection, and refusal to speak or take part in social life, as well as visual troubles..



Fig. I. 54 Severe food deficiencies caused by self-starvation

Exploitation of the document

1. What are the causes of marasmus and kwashiorkor ?
2. Why is milk necessary for children under five years of age ?
3. What can you deduce from the graph of Fig. I.51 ?
4. How can rich countries help poor ones ?

Document 2 Diseases of excessive food intake : cardiovascular diseases

Clinical investigations have shown that there is a relationship between the cholesterol level in the blood and cardiovascular diseases. Excess lipids (cholesterol) tend to be deposited on the internal walls of arteries, causing atherosclerosis, which can lead to myocardial infarctions.

These disorders are a major cause of deaths (40% of total deaths) in developed countries.

1. Atherosclerosis

This disease causes narrowing of the coronary arteries. After a 75% obstruction, the blood flow to the cardiac muscle is severely diminished. Symptoms vary from a mild pain in the chest to complete heart failure.

It is believed that the first step in atherosclerosis

is due to a lesion of the interior wall of the artery. Cholesterol and triglycerides will deposit on the walls of arteries, causing the formation of a plaque called atheroma, which produces a narrowing of the artery.

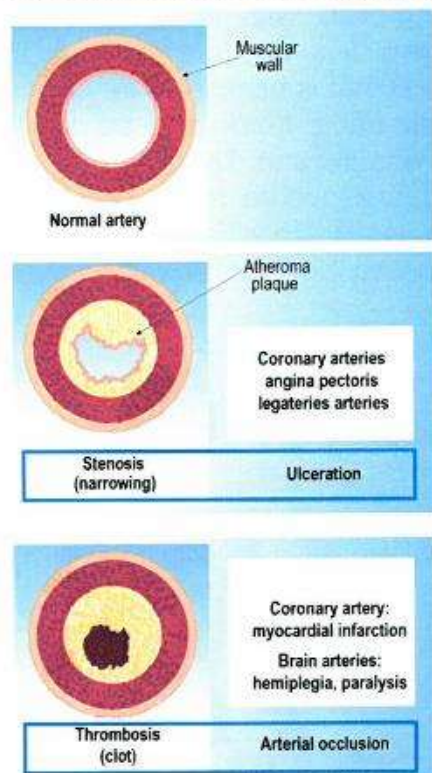


Fig. I. 55 Transversal section of an artery partially obstructed by an atheroma

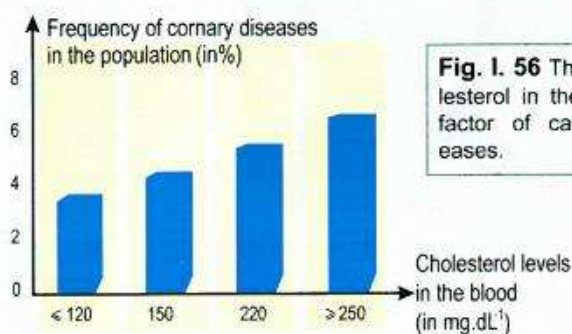


Fig. I. 56 The amount of cholesterol in the blood is a risk factor of cardiovascular diseases.

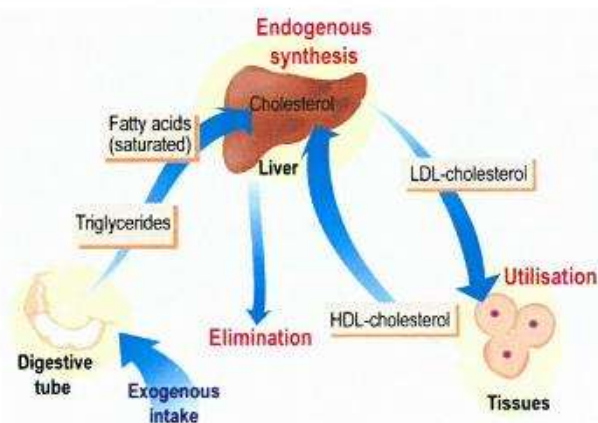


Fig. I. 57 Cholesterol cycle in the organism.

2- Good and bad cholesterol

Part of the cholesterol present in the body comes from food (eggs, some meats, dairy products, etc...). But most of it (2/3) is synthesized in the liver.

Cholesterol is one of the constituents of cell membranes. It is also a precursor of sexual hormones, corticoids, vitamin D and biliary

salts useful to the digestive process.

Cholesterol is insoluble in plasma. That is why it has to bind to huge protein molecules forming complexes called lipoproteins. These either provide the needed lipids to the cells or eliminate them from the circulation when they are in excess.

- a- Draw the graphs that represent the limits of the superior and inferior variations for weight and height.
- b- A 15-months-old boy weighs 8Kg and has a height of 79cm. Is he in the normal zone of growth? Calculate his BMI. Are the results concordant ?

E- Comment the following text

One of the major principles of dietetics consists of the adaptation of consumption to needs. The food ration must be established according to the particular physiological situation of each individual. One should be aware that "besides the problem of quantity, the quality of nutriment and their balance have a determined importance, particularly during the period of organic buildup of the infants" (M.Bour)

F- Justify the advice of the ministerial circulation

Ministerial circulation of June 9, 1971 in France concerning the meals at a lunch canteen:

"Every day, lunch must be composed of a main dish of meat, fish or eggs, in addition to milk or cheese as sufficient source of animal proteins and calcium. This dish is supplemented with a vegetable dish on the side, or a soup, depending on the season and the local habits, and complemented with one or two desserts of different composition.

The substances used in dressing or in cooking, as well as bread and cereals, provide an additional caloric supply.

As a general rule, a meal must always contain :

- a raw vegetable (salad or fruit).
- animal proteins, a part of which is provided by milk or cheese.
- cooked fresh vegetable (twice a week).
- potatoes, pasta, rice or vegetables the other days."

EVALUATION EXERCISE

A- Choose the correct answer(s)

- 1- Atheroma plaque :
 - a- can obstruct an artery.
 - b- builds up inside an artery.
 - c- favors blood circulation.
 - d- forms spontaneously.
- 2- Cholesterol is :
 - a- an unsaturated fatty acid.
 - b- manufactured by the liver.
 - c- a cardiac disease.
 - d- a saturated fatty acid.
- 3- To prevent nutritional diseases, one should:
 - a- go on a diet.
 - b- decrease unsaturated fat intake.
 - c- increase saturated fat intake.
 - d- have regular physical activity.

B- Choose the right statement(s), and correct the false one(s)

- 1- Kwashiorkor indicates a total food deficiency
- 2- LDL carries the cholesterol from the liver to the cells
- 3- Obesity is a multifactorial symptom

C- Answer the following questions

- 1- Is " losing weight" equivalent to staying hungry?
- 2- Is an adequate energetic intake the only main aspect of a balanced diet ?

3- What are the two main questions an adolescent should ask himself (herself) before going on a diet?

4- How does physical activity prevent cardiovascular diseases?

D- Think about it

1- The two tables below show the normal evolution of weight and height of a well nourished child.

age	average weight in kg (usual variations ± 1 kg)	
	girl	boy
birth	3.3	3.4
3 months	5.4	5.8
6 months	7.4	7.6
9 months	8.4	8.9
12 months	9.3	9.8
15 months	9.9	10.6
24 months	11.6	12.2

age	average height in cm (usual variations ± 3 cm)	
	girl	boy
birth	49	50
3 months	58.7	59.9
6 months	64.8	66.5
9 months	69	70.7
12 months	73	75
15 months	75.9	77.6
24 months	84.3	85.6

2- What is the origin of excess weight ?

The liver maintains a constant level of glucose in the blood (0.8 to 1.2 g / l), and stores the excess of glucose as glycogen. When glucose is abundant, and can no longer be converted to glycogen by the liver, it is transformed into fats which will be stored inside fat cells called adipocytes. Thus, the body mass increases. Excess food without exercise equals fat.

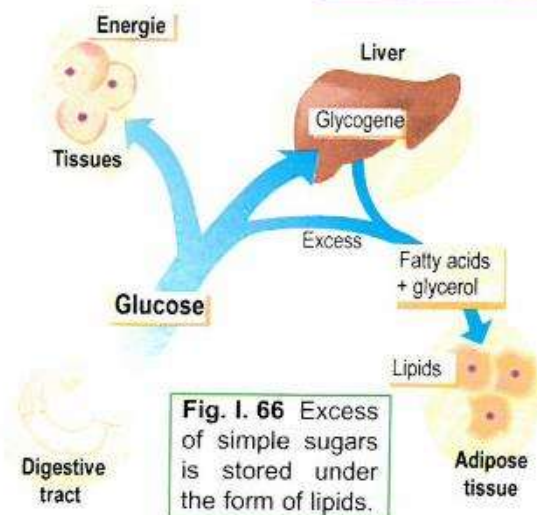


Fig. I. 66 Excess of simple sugars is stored under the form of lipids.

3- What are the factors that may cause obesity

- Bad feeding habits acquired in early life
- overfeeding in order to appease anxiety
- social habits such as meals rich in simple sugars and fat
- and sedentarism are all factors that cause obesity.
- In addition, certain situations such as pregnancy, menopause, quitting smoking, and some medical treatments (some antidepressants and antiepileptics) can also lead to obesity.
- Recently it has been discovered that obesity may be linked to a genetic factor. Thus, some persons may be more predisposed than others to becoming obese.

- Obesity as an epidemic.

The WHO (World Health Organization) qualifies obesity as an epidemic in rich and

well developed countries. 10 to 25% of the population in west Europe are considered obese, whereas in America it is between 20 - 25%. In France, this rate is situated between 6 and 11%, with a great variability depending on the region, age, sex, and social class. The highest proportion is encountered in the Pacific island of Samoa, where 60% of men and 75% of women are obese. The statistics for the number of obese persons in England and the United States will double by the year 2025. Moreover, Americans claim that the expenses associated with obesity amount to 70 billion dollars per year. These are direct expenses and indirect expenses caused by a decrease in the productivity of obese people.



Fig. I. 67 Obese family.

Exploitation of the document

1. Calculate your BMI.
2. Is obesity dangerous ?
3. Is food quantity the only reason for being overweight?

Nutrition disorders: characteristics, causes and prevention

1- Deficiency diseases

- Famine, which still prevails in certain regions of the world (Africa, southeast Asia..), primarily affects children under 2 years of age. The high mortality rate (one child out of four) is due to a qualitatively and quantitatively insufficient food intake.
- Two diseases are characteristics of hunger : Marasmus, which is due to total food deficiency, and kwashiorkor which is due to protein deficiency. Children do not receive adequate food. The situation is aggravated by infections that a weak individual cannot fight.
- Deficiency diseases can be prevented by improving the standard of living of the populations concerned and by educating mothers about elementary principles of hygiene and nutrition.

2- Diseases of excessive food intake

A sedentary life style, accompanied by the intake of energy-rich food such as sugars and lipids, contribute to the appearance of certain diseases in countries where food is abundant.

- Cardiovascular diseases are the main cause of death in industrialized coun-

tries. Atherosclerosis causes the formation of atheroma plaques in the lumen of the blood vessels, slowing down the blood circulation and putting the individual's life in danger.

- Cholesterol is an important risk indicator. High levels of cholesterol in the form of LDL in the blood is associated with atheroma plaque deposits.
- Saturated fatty acids are more hypercholesterolemic than unsaturated fatty acids. This is why it is desirable to control the intake of animal fats, which are rich in saturated fatty acids.
- Obesity, frequent in rich countries, is not a disease.
- Obesity is defined as a 20% increase over the ideal weight as determined according to the age and the sex of the individual.



Fig. I. 68 Hidden fats, added sugars and salts are abundant in these meals, making them unhealthy. Certain nutritionists advise people to eat red meat and pastry only once a week, and to eat mainly food rich in fibers.

Additional information

Bulimia

When the state of hunger is not appeased by food consumption, there is a sign of deep perturbation of an emotional nature. Obesity is one of the first manifestations of bulimia.

In industrialized countries, being either obese or thin is often the result of a genetic predisposition associated with bad dietary habits

Monotony makes the fat

In France, they consume an average of many different kind of meals each week against fewer meals in the United States. This monotony in nutrition may partly explain the obesity that is characteristic of a large number of people in the United States.



Diet

Diet is not synonymous with exaggerated constraints. Many teenagers influenced by fashion consider themselves obese while they are well below their ideal weight. They go on rigorous diets without medical advice in order to loose weight rapidly. They deprive themselves of balanced meals, which often leads to dramatic consequences. In addition, repetitive and unsuitable diets have an adverse effect on the health of the individual; the lost weight is rapidly regained afterwards. On the other hand a progressive and well balanced diet gives long-term results.



How to prevent cardiovascular diseases?

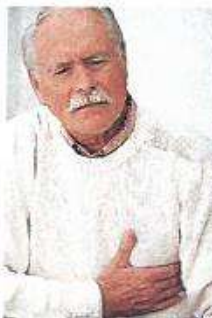
- 1- Watch your diet:
 - * Consume chicken, meat and fish
 - * Grill instead of frying
 - * Avoid cream
 - * Consume polyunsaturated fats
 - * Use corn and sun flower oil instead of animal fat for frying
 - * Consume a lot of vegetables and fruits
- 2- Check your blood pressure regularly :

Normal blood pressure:

diastolic less than 85 mm Hg

systolic less than 140 mm Hg
- 3- Maintain your ideal weight; any decrease in the body mass of an obese person may lead to an increased HDL, which prevents cardiovascular diseases.
- 4- Practice regular physical activity. Physical activity increases the efficacy of the heart and the cardiac output, increases the HDL concentration, decreases the triglycerides concentration, dissolves blood clots and enhances the pulmonary function.
- 5- Don't smoke; stop smoking right now. Better late than never!
- 6- Relax:

Take more free time for leisure. Breathe in fresh air deeply. Listen to music and dance. Do not overwork. When preventive or medical treatment is not effective, a surgical intervention may be necessary and can save the life of the patient
- 7- Fruits and vegetables. These protect the organism, prevent vascular disorders and facilitate drainage.





Unit II

Neurobiology, Human behavior and Health



Fig. II.1: Closing your eyes to avoid intense light, playing music, driving a car, sneezing, reading, yawning, all those behavioral reactions in our daily actions, are they similar? What is their meaning?

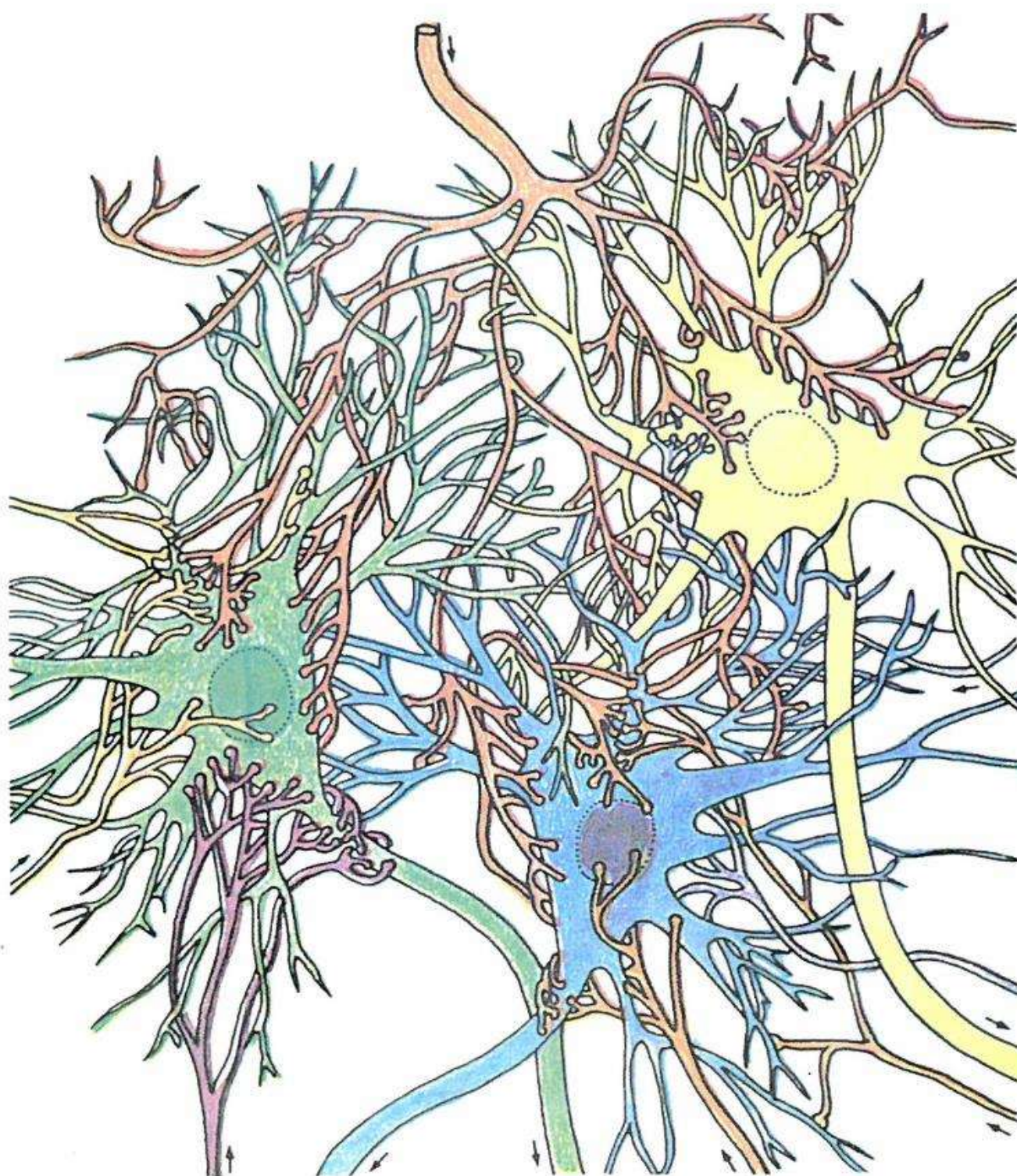
Behavior is the integrated summary of all our reactions when responding to the stimulation of the surrounding environment. Certain reactions are present from birth, and they are manifested in a similar fashion among all human beings; these are reflexes. Others are voluntary, chosen and decided for by the individual. Finally, there are the reactions that are acquired through training.

Sensory functions that are responsible for the reception and the analysis of outside information play a fundamental role in our behavior. All behavioral reactions necessitate an exchange of signals, i.e. circulating the message to ensure the communication between different cells and organs in order to establish a well adapted response.

In order to maintain the equilibrium of our body despite our changing environment, the nervous and endocrine systems together regulate and ensure the harmony of any behavioral reaction. Agents such as stress, drugs, bad hygiene, etc. may cause a failure in the system of communication, and the equilibrium of the organism will be disrupted, leading to dramatic consequences. Our health is very important and we should preserve it.

Chapters:

- 1 Neural communication
- 2 Cerebral activity and conditioned reflexes
- 3 Endocrine system and hormonal communication
- 4 Stress and emotional reactions
- 5 Drugs and drug-addiction
- 6 Biological rhythms



Document 1 *The nervous system : an organized network*

The nervous system commands all the actions of life: the most instinctive, like respiration, the most mundane, like walking, and the most complex like memory and reasoning. This system is everywhere, in the muscles, at the tips of the fingers, around the intestines, and most certainly in the skull where the cerebral mass occupies the dominant place and by its importance makes the human being the most intelligent of all creatures. Despite its complexity the nervous system is made of only two principal sorts of cells: neurons, specialized in the transmission of messages between different regions of the organism, and support cells called neuro-glial cells. For a better understanding of the functions of the nervous system as well as the malfunctions, one should know the structure that makes it.

1- Structure of the nervous system

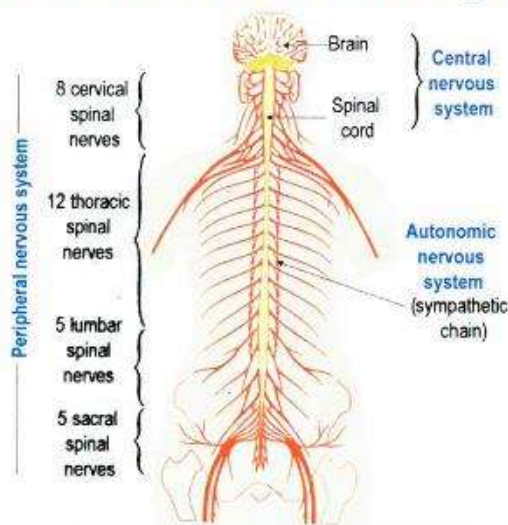


Fig. II. 3 General organization of the nervous system. All nerves are paired.

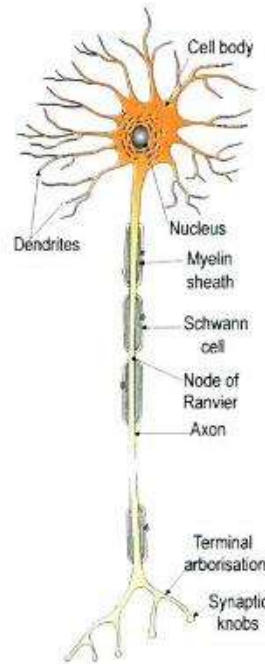


Fig. II. 4 Structure of the neuron: basic element of the nervous system (multipolar).

2- The neuron: a specialized cell

The neuron is the elementary structure of the nervous system. Neurons, under various forms are present in each part of the nervous system. A neuron is composed of a cellular body and several extensions: the dendrites that are finely branched and the axon, which is the longest and the largest part of the cell.

The axon often has a particular structure: the cylindrical axis is enveloped by an isolating myelin sheath made of lipids, which is in turn surrounded by the protective Schwann membrane.

The nervous fibers extended from neurons (axons) join together in fascicles to form the nerve.

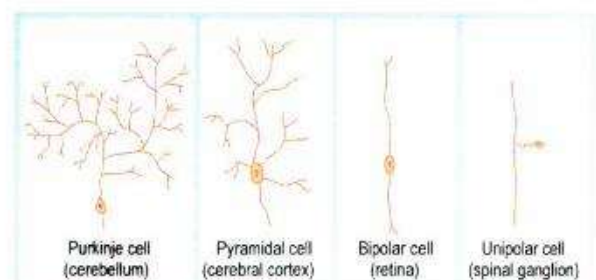


Fig. II. 5 Some type of neurons.

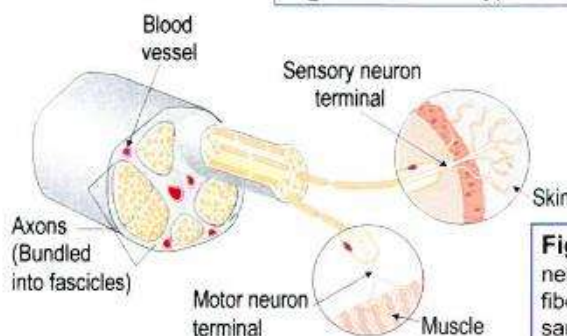


Fig. II. 6 Cross section of a nerve. Sensitive and motor fibers often circulate in the same nerve.

Chapter 1 Neural Communication

The nervous system fills three principal functions: sensorial, integrative and motor. With the sensorial function the nervous system detects the internal changes of the body as well as the external changes through specialized receptors or transducers. The integrative function permits the analysis of the sensorial information and undertakes appropriate decisions. With the motor function, it reacts to stimulation by muscular contraction, glandular secretion, etc...

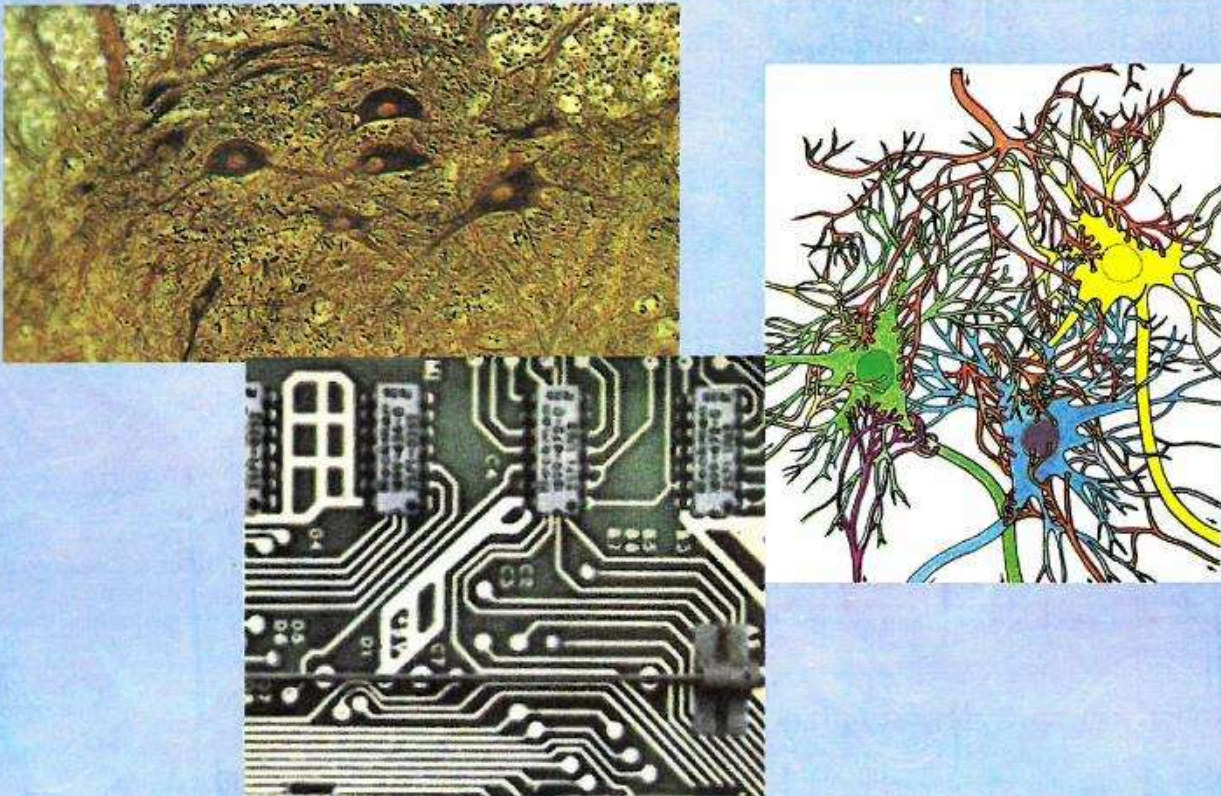


Fig.II. 2 The nervous system is probably the most complex structure in the body: 1 cm³ of human brain can contain several million nerve cells. Each cell can communicate with thousands of other cells and form a network that analyses the information received. Compared to this network, all of the integrated and perfectly constructed electronic circuits look very primitive.

In this chapter we shall answer the following questions :

- ☞ How is the nervous system organized ?
- ☞ What is the structure and the function of a neuron ?
- ☞ What are the factors that contribute to the generation of a neural influx ?
- ☞ How is the influx propagated along the nervous fiber ?
- ☞ How does the transmission between neurons occur ?
- ☞ What happens when neurons malfunction ?

Functionally there are three type of neurons :

- Sensitive neurons (afferent) that conduct the nervous influx from the sensorial receptors to the CNS.
- Motor neurons (efferent) which conduct the nervous influx from the CNS to the effectors (muscles and glands).
- Association neurons (inter-neurons).

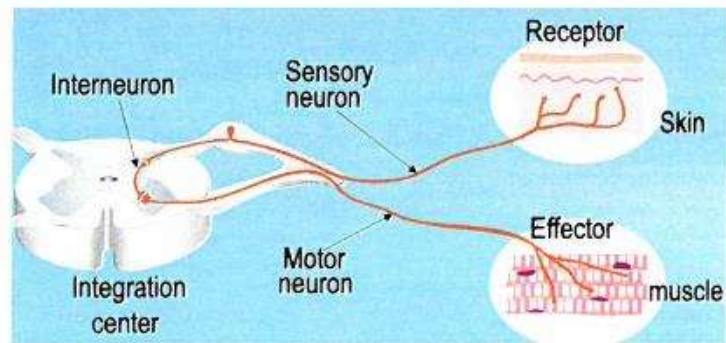


Fig. II. 7 A simple communication network; most of the networks are however more complex.

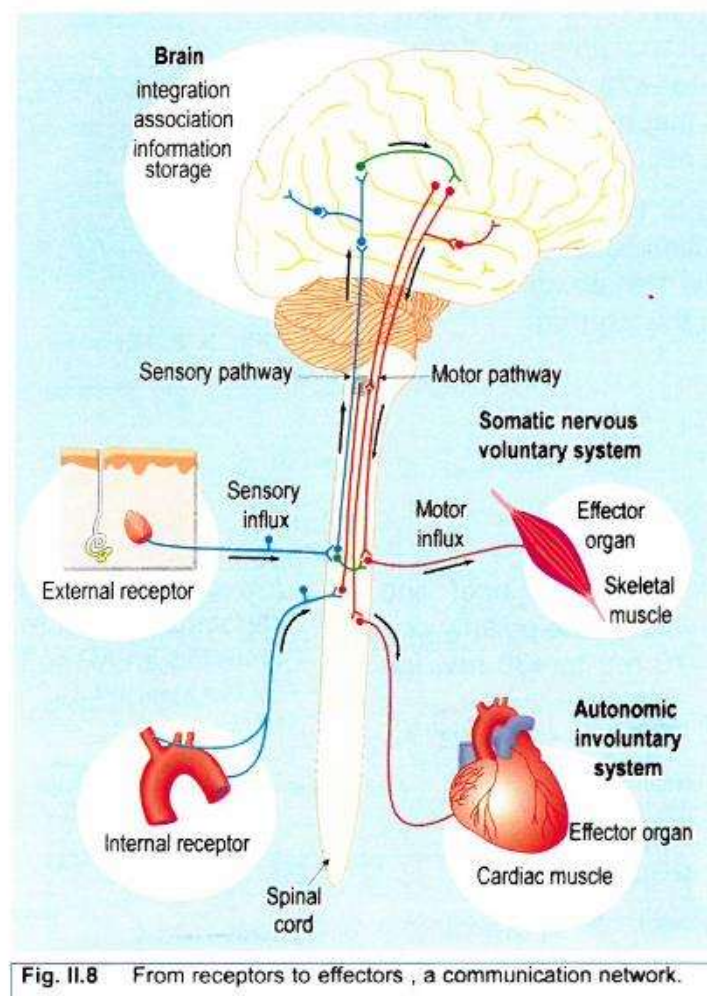


Fig. II.8 From receptors to effectors , a communication network.

Exploitation of the document

1. Is the efferent part of the peripheral nervous system always motor ?
2. Why do we talk about a network of neurons ?
3. Can we say that neurons allow the circulation of information inside an organ ?
4. What advantage does a neuron possess by having ramified dendrites ?
5. What are the consequences of the appearance of a stimulus at the level of a sensory receptor ?
6. Are the sensitive neurons voluntary ?

Document 2 *Nervous information : nature and propagation*

At rest, nervous cells are electrically polarized. A difference of potential exists between the inside and the outside. Membrane properties will be modified after stimulation, giving rise to an action potential (AP). What is an AP ?

1. Electrical phenomenon

The resting potential

A non-stimulated neuron possesses a membrane potential or resting potential that is approximately equal to -70 mV (the negative charge indicates that the interior is negatively charged in respect to the exterior).

The resting potential is the result of difference in ionic composition between intra- and extra-cellular fluid (the Na^+ predominates at the exterior and K^+ in the interior).

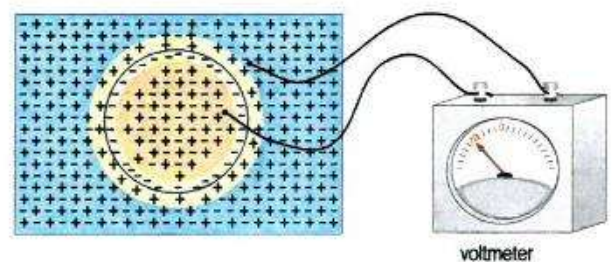


Fig. II. 9 Non-stimulated neuron in the state of polarization.

The action potential

An efficient stimulus triggers a brief and rapid inversion in the membrane polarity or a depolarization from -70 mV to $+30$ mV, fol-

lowed by a return to the resting potential or repolarization from $+30$ mV to -70 mV. This is called an AP of 1 m.sec. duration.

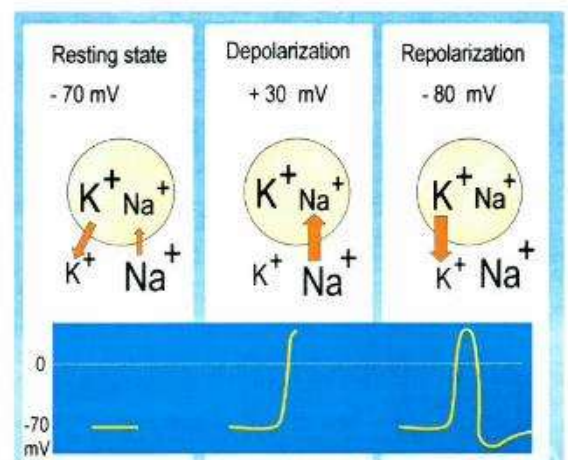
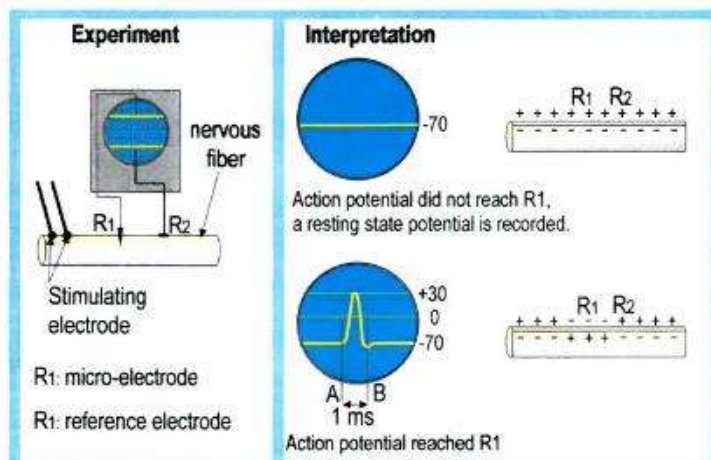


Fig. II. 10 Recording of action potentials on an isolated fiber: a) neuron is electrically stimulated and the response is recorded on the screen of an oscilloscope. b) ionic interpretation. Depolarization is due to a sudden permeability to Na^+ .

2- Neurotransmitters

It was thought for a long time that neurons release only one type of neurotransmitters at synaptic junctions. It is now clear that two or maybe three neurotransmitters are present in a single neuron.

There are about twenty molecules that act as neurotransmitters and another ten that

can be added to the list. They are present in the CNS and peripheral nervous system (PNS) and play a primary role in the function of the organism and its behavior.

Neurotransmitter	Role
Acetylcholine	An excitatory neurotransmitter in the nerve terminal of a motor axon in the junction with a skeletal cell muscle. However, it is inhibitory in the heart. In the brain it possesses several functions that are not well determined, especially when it comes to memory and learning.
Noradrenaline	Released in the autonomous nervous system (acceleration of heart beats). In the brain it plays a role in maintaining the general state of wakefulness, mood regulation...
Dopamine	Linked to the regulation of motion reactions and to the control of complex movement.
Serotonine	Regulator of the body temperature, sensorial perception, sleep...
Endorphine	A natural analgesic.

Fig. II. 14 Some important neurotransmitters.

Exploitation of document

1. What are the different types of synapses ?
2. Can we affirm that the appearance of the action potential leads to the appearance of an action potential in the post-synaptic neuron ?
3. Propose an hypothesis to show the link between the frequency of action potentials in the pre-synaptic neuron and the neurotransmitter concentration in the synapse ?
4. If a drug blocks the liberation of an excitatory neurotransmitter, what do you think are the consequences ?
5. Why is a neurotransmitter degraded after the response ?

Document 3 *Synaptic transmission*

In addition to transmitting information all along the axons, neurons must have a way to pass on this information from one cell across the entirety of the nervous system. This is done by means of the synapses. How does transmission operate at the level of synapses ?

1. Synapse

A synapse is a functional junction between one neuron and another or between a neuron and an effector cell, such as a muscular fiber or a glandular cell. The transmission of nervous messages is unidirectional. It goes from a pre-synaptic neuron towards a post-synaptic cell through a synaptic cleft.

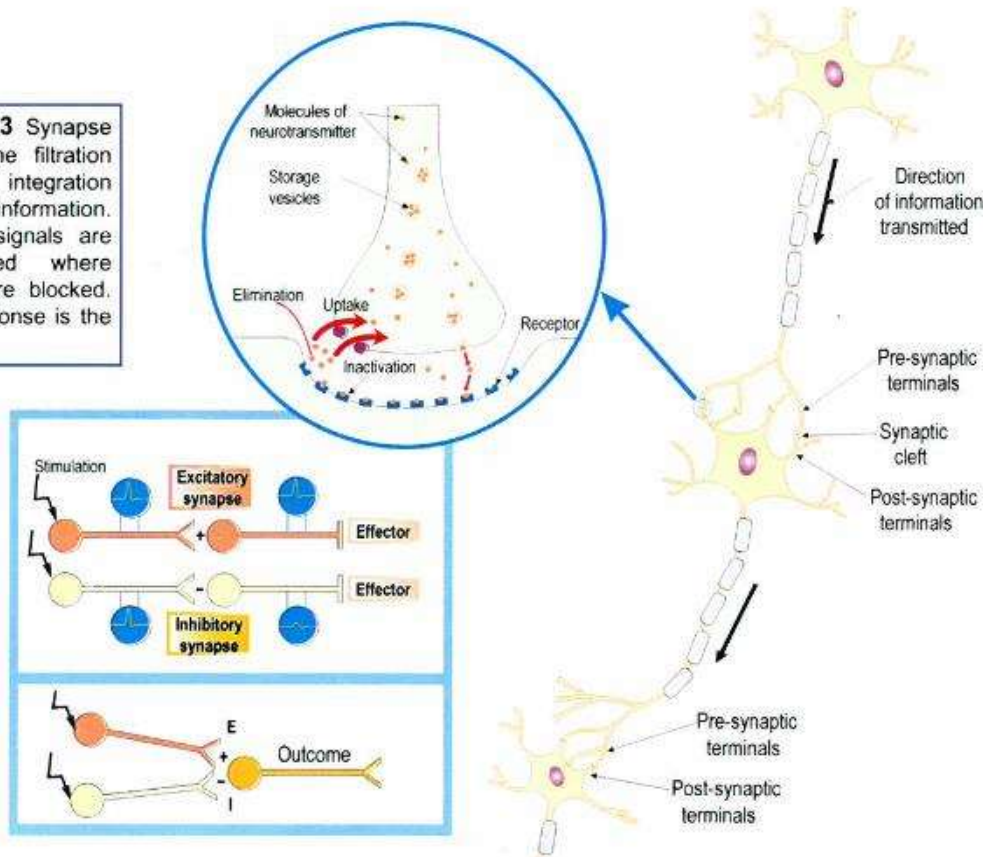
The transmission of the nervous influx occurs through chemical messengers or neurotransmitters. These are synthesized by the neuron and stored in vesicles at the end of the axon, where they are released. After stimulation the vesicles liberate the neurotransmitters in the synaptic cleft. Once liberated, they attach to specific receptors in the post-synaptic membrane.

An excitatory transmitter is the one that can depolarize or render less negative the potential of the post-synaptic membrane in order to facilitate transmission, i.e., it forms action potentials. On the other hand, an inhibitory neurotransmitter renders the potential of the post-synaptic membrane more negative, and thus it blocks transmission.

After its release the neurotransmitter is taken up by the axonal terminal and the fraction that is attached to the receptor is quickly degraded by enzymes.

Note that the same neurotransmitter can be excitatory in certain places and inhibitory in others, depending on the nature of the receptor.

Fig. II. 13 Synapse allows the filtration and the integration of information. Certain signals are transmitted where others are blocked. The response is the result.



2- Conduction of the message

No turning back!

When the polarity of the axonal membrane is reversed during an action potential, the potential difference between the depolarized zone and the neighboring zone creates a local current which will then depolarize it. This activity continues to happen all along the membrane and creates an action potential which is propagated in a unique sense.

Nervous influx speed

In general, the conduction speed increases with the size of the axon. In a myelinated axon, the conduction is faster (100m/s) than in a non-myelinated one (1m/s). This difference is due to the saltatory conduction where action potentials occur only in the nodes of Ranvier.

The law of all or non

An action potential has the same amplitude regardless of the stimulus intensity which created it as long as the intensity of the stimulus is above a value called the threshold of excitation. If the intensity of the stimulus is inferior to the threshold, no action potential is triggered. Hence, the intensity of the stimulus is coded by the frequency of the action potentials. This law is applicable to the nerve fiber but not to the nerve.

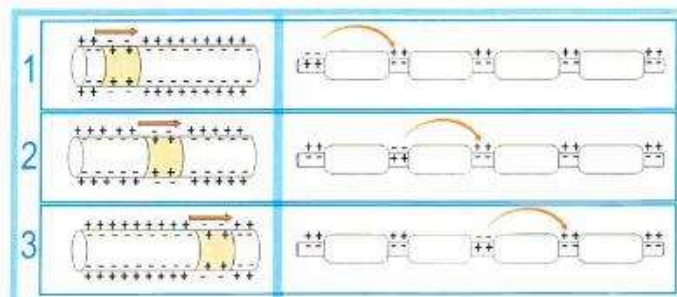


Fig. II. 11 Propagation of the action potential. A) In non-myelinated axon. B) In myelinated axon.

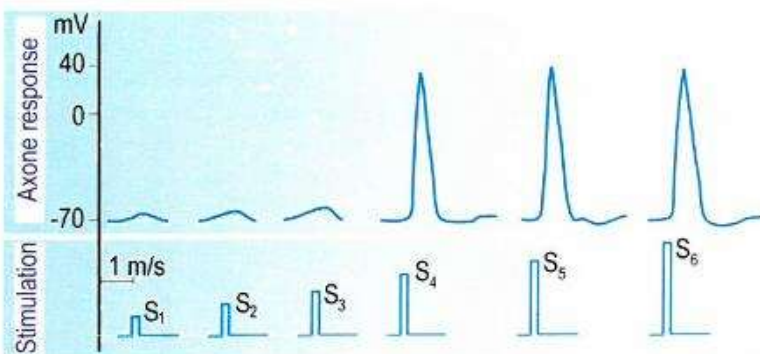
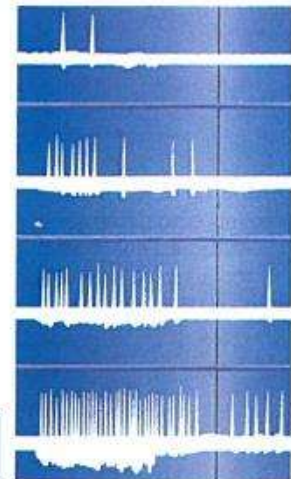


Fig. II. 12 The nervous message in a fiber is coded in serial successions of AP of which the frequency and number are functions of the stimulus intensity.



Exploitation of the document

1. What does an AP represent ?
2. Does the action potential move all along the axon ?
3. Why is the nervous message unidirectional ?
4. What is the advantage of myelination ?
5. Analyze the graph II. 12 (left). Analyse the intensities that gave the measurements in the right panel ?

Summary diagram

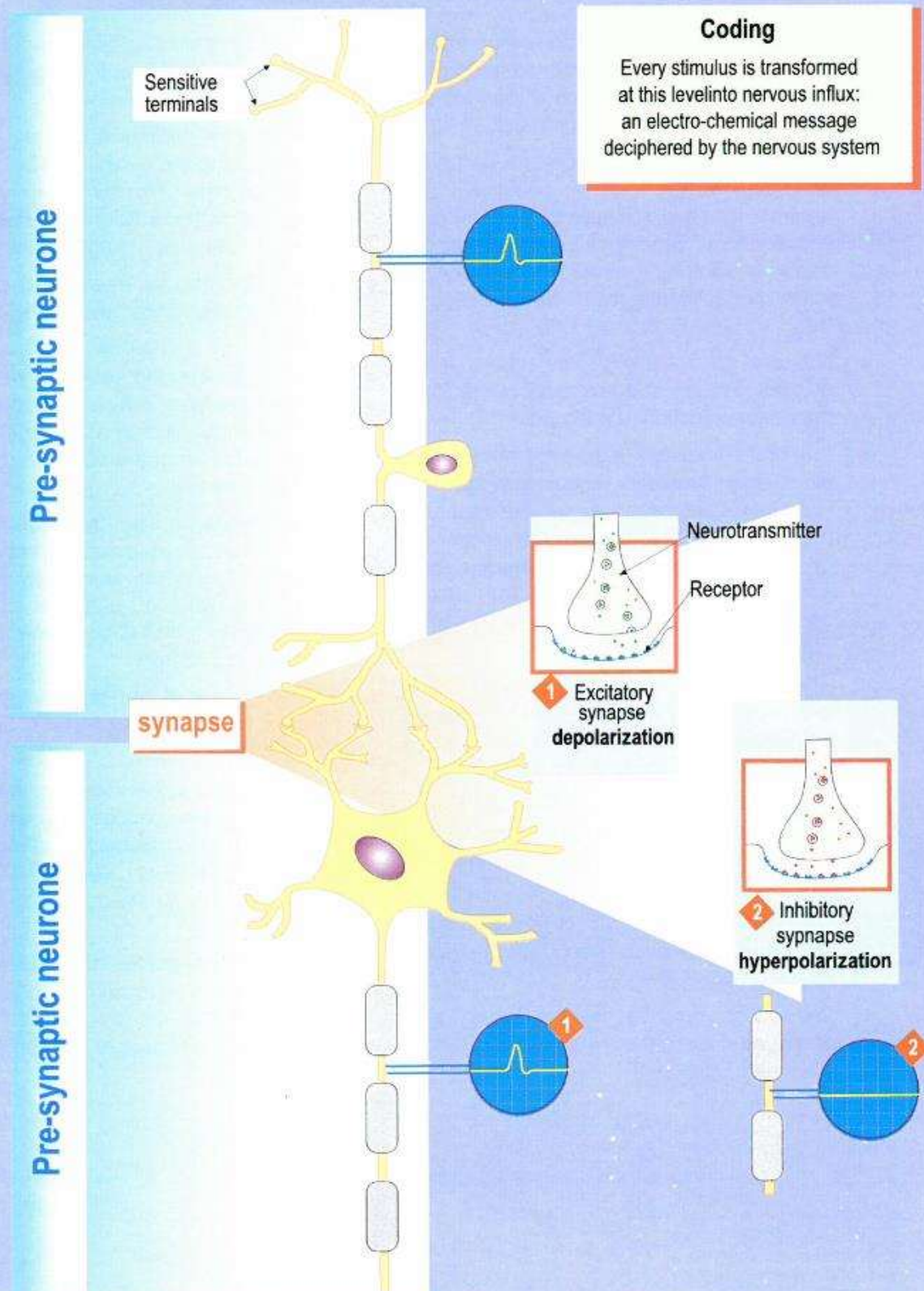


Fig. II. 16 Neuron, the support of information.

EVALUATION EXERCISE

A- Indicate the correct answer (s)

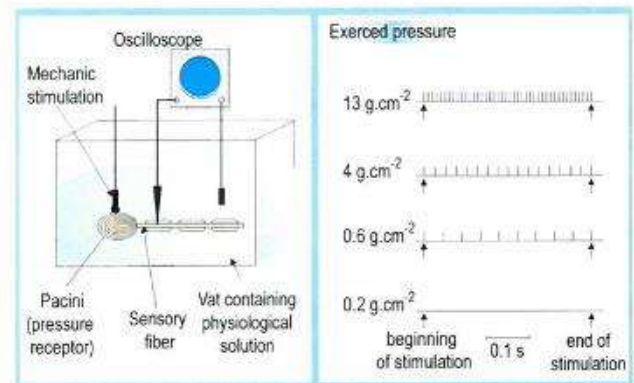
- The peripheral nervous system comprises :
 - the somatic Nervous System.
 - the autonomous Nervous System.
 - the central sensitive nerves.
- The nervous influx moves from :
 - the periphery to the CNS.
 - from the CNS to the periphery.
 - only towards the CNS.
- Efferent neurons that transmit the nervous influx to the effectors are :
 - motor neurons
 - sensitive neurons
 - associative neurons.
- Receptors of neurotransmitters are found :
 - at the nodules of Ranvier on the axonal membrane.
 - on the post-synaptic membrane.
 - at the axonal tip.
- In an excitatory synapse the binding between the neurotransmitter and the receptor :
 - depolarizes the post-synaptic membrane.
 - stabilizes the resting potential of the post-synaptic membrane.
 - increases the polarity of the post-synaptic membrane.
- The variation of the stimulus intensity can bring :
 - nervous influx with different amplitude.
 - nervous influx with different speed in the same neuron.
 - different frequencies of nervous influxes in the same neuron.

B- Choose the right statement(s) and correct the false one(s).

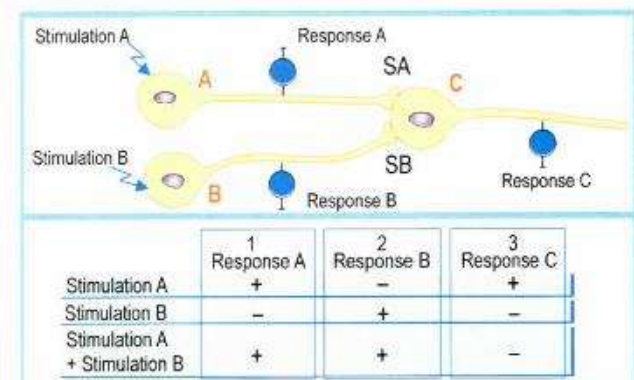
- There are two systems of communication between cells: the nervous system and the endocrine system.
- The amplitude of the action potential diminishes during propagation.
- Axons of different neurons transmit nervous influx at the same speed.
- Synapse functions only in one direction, from the pre-synaptic neuron towards the post-synaptic element.

C- Think about it

- Analyze these measurements



- We test a system of three neurons, A, B and C. The results are summarized in the table below. (Stimulations performed are efficient and have the same amplitude and frequency) Explain.



Additional information

Did you know ?

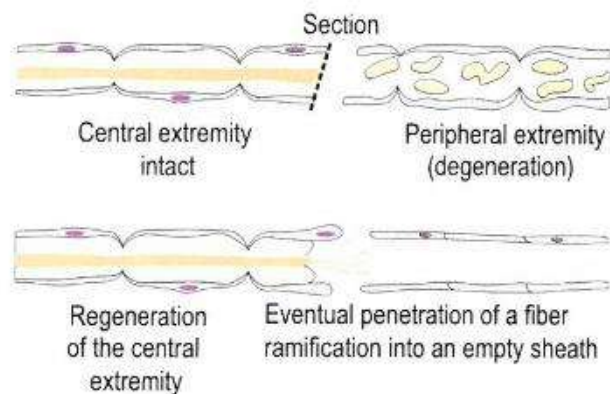
- Messages can circulate all along the neurons at a speed of 100m/s. It only takes few msec to connect the brain to the hand.
- The number of neuro-glial cells is 10 to 50 times more elevated than that of the neurons.
- The sciatic nerve possesses one axon that spreads from the inferior part of the

spinal cord to the leg muscle, a distance of about 1 m.

- In 1791 Luigi Galvani noticed that an isolated frog's leg conserved with it's nerves contracts when touched with a pair of pliers in which each branch is made of a different metal. This observation was behind the invention of the flash light by Volta.

Degeneration and regeneration of a nerve

When a peripheral nerve is severely damaged or sectioned, the unattached part to the cellular body undergoes degeneration, but the attached ones can develop new fibers. This property is used in modern surgery during grafting of a hand or foot lost in an accident. However, given the large number of nerve fibers, the connection of the central nerve end to the sectioned nerve is not always perfect. The grafted member can also lose a part of its motricity and sensibility. In all cases re-education is necessary.



Multiple sclerosis

It is an auto-immune disorder that is due to a loss of myelinization of nervous fibers which affects about one in every 1500 individuals. It is manifested by lesions that are initially inflamed and then become sclerotic (hard) and often form plaques. It can create severe disorders of movement and vision.



Local anesthesia

Products based on the principle of the arrest of the nervous influx are used during dental treatment or the suturing of a wound, in order to attenuate the pain. The arrested nervous influx will not cross the borders of the region being treated. A small dose of anesthesia prevents the opening of the sodium channels.

Document 4 *Chemical perturbation of the synapse*

At the level of synapses, the mediators responsible for neurotransmission may sometimes not operate properly. What are the consequences of such dysfunction ?

1- Parkinson's disease

Disorders of the CNS are frequent in people over 60 years of age but the cause is not known. It is unlikely to be a hereditary disorder because only 5% of patients with such disorders have a family history of this disease. Therefore toxic factors present in the environment are believed to be implicated.

Tremors are the most frequent symptoms of this disorder. It is due to involuntary skeletal muscle contraction which disrupts voluntary movement. Certain muscles may remain permanently contracted, which leads to the rigidity of the affect-

ed part. Muscular movement in affected patients becomes slow, their steps become short and they drag their feet. It takes more time and effort for them to do mundane things, like eating and getting dressed. Their handwriting deteriorates and their speech is affected. Parkinson's disease is linked to degeneration of dopamine-producing neurons in a certain area of the brain, which leads to the symptoms that characterize this disease.

Treatment consists of increasing the amount of dopamine in the brain.

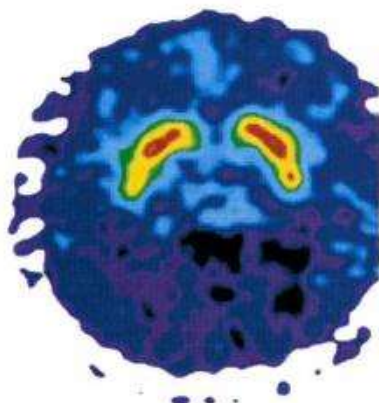


Fig. II. 15 On this tomography of the brain we can observe the activity of dopamine (in red) which is decreased in patients suffering from Parkinson's disease.



2- Alzheimer's disease

This is a disabling neurological problem that affects about 10% of the population above 65 years of age. The cause of this disease is still unknown and there is no treatment. Victims of this disorder have a hard time remembering recent events. Confusion is accentuated with a progressive loss of reasoning and patients become totally incapacitated. Unfortunately this disease cannot be diagnosed until the late

stages of the disorder. An autopsy of patients' brains reveal a massive loss of neurons in the encephalon region that is linked to memory and intelligence (cortex). Most destroyed neurons utilize acetylcholine as the neurotransmitter.

A virus, toxicity to aluminum, a genetic modification or an insufficient input of oxygen into the brain could be causes of this disease.

Exploitation of the document

1. What are the common points between the disease of Parkinson and Alzheimer's disease ?
2. Why is it so hard to treat these disorders ?

Essential Points

Neural communication

- We permanently receive information from the environment. We react in an adapted manner to this information through the coordinated work of many organs.
- Communication between organs is done according to the general scheme : a receptor which is sensitive to a modification of the environment sends information to a treatment center; the appropriate response is then realized by an effector.
- Neurons are the basic elements of the nervous system and are responsible for the communication of information.
- The non-stimulated neuron membrane is electrically polarized. There is a difference in the potential between the interior (negatively charged) and the exterior (positively charged) of the membrane as a result of the difference in the ionic composition of the two environments. This is the resting potential.
- The stimulation of a nervous cell inverses the polarity in such a way that the interior becomes relatively positive in respect to the exterior. This is action potential.
- A stimulus with sufficient intensity triggers an action potential with an immediate maximal amplitude. This is the law of all or none.
- The intensity of a stimulus is coded by the frequency of action potentials and not by their magnitude.
- The action potential lasts 1 msec, is propagated along the axon and produces a nervous influx.
- The speed of propagation is a function of the fiber diameter and the presence of myelin.
- Synapses are structures that are capable of transmitting the nervous influx from one cell to the other. This transmission is uni-directional; it goes from the pre-synaptic neuron to the post-synaptic element.
- Axonal terminals synthesize one or several neurotransmitters and store them in vesicles.
- After a stimulus, the vesicles release the neurotransmitters, which are diffused in the synaptic cleft and attach themselves to the receptors of the post-synaptic neurons or effectors.
- If the binding between neurotransmitters and receptors depolarizes the post-synaptic membrane and generates an action potential, the synapse is excitatory; if no action potential is generated then it is considered inhibitory.
- The response of the post-synaptic element is the resultant of the inhibitory or excitatory synapses.
- A perturbation in neurotransmitter liberation may have serious consequences; thus Parkinson's disease is caused by a dopamine deficiency, while Alzheimer's disease is due to an acetylcholine deficiency.
- The same transmitter may play several roles; thus acetylcholine can be an excitatory transmitter at the neuromuscular junction but inhibitory of cardiac contraction.

2. The implicated cerebral structures

- A specific zone of the motor area commands a precise group of muscles. Each cerebral hemisphere commands the muscles of the opposite moiety of the body.
- The various parts of the body are projected onto the motor area (primary area) in a manner that is proportional to the precision of the gesture performed by the organ and not by its size (fig. 26).
- The pre-motor zone situated in front of the motor zone ensures the coordination of movements.
- The cerebellum, as well as the other subcortical centers, intervenes in the control of the gesture.

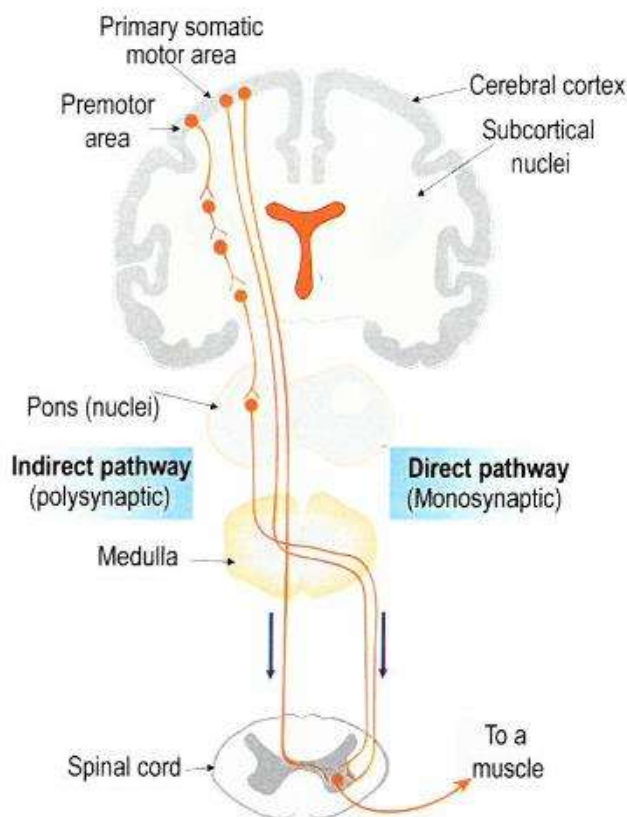


Fig. II. 29 The descending pathways of voluntary motricity.
Direct pathways : command precise and fine movement.
Indirect pathway : command general movements and control posture.

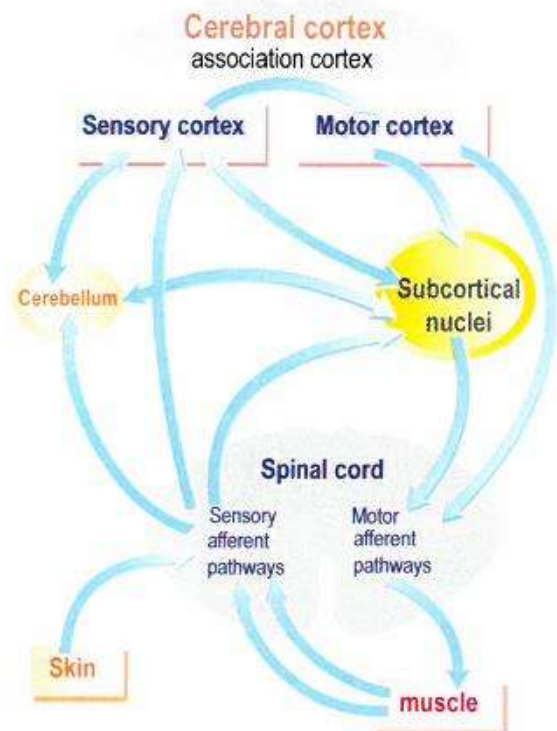


Fig. II. 30 Nervous pathways intervening in the control of a movement.

Exploitation of the document

1. Are observation techniques capable of giving all the necessary information concerning the control of a gesture?
2. Why are the organs on the homunculus represented in this manner?
3. You decide to throw a piece of paper in the garbage can. Can you describe all the steps involved in performing your movement?

Chapter 2

Cerebral Activity and Conditioned Reflexes

The brain is the center for registering all sensations. These are networked, and connected to information already present in order to make decisions, and cause action.

It is the site of intelligence, emotions, and memory. It dictates our behavior towards others. We decide voluntarily to take an object or to memorize a text. However, our heart beats involuntarily, we breathe, we walk, we move our head backwards if someone abruptly brings a hand close to our face. We salivate when we see an appetizing dish; we drive our cars mechanically. Many of these reactions liberate the brain from its enormous tasks.



Fig.II. 17 Kasparov is beaten at chess by Deep Blue, but it is man who invented computers. Where is the site of intelligence? Progress in the fields of neurophysiology as well as imagery techniques have contributed to the localization of the cerebral regions. We are only beginning to understand the functions of the brain.

In this chapter we will answer the following questions:

- ☛ How is the brain structured?
- ☛ How is received information treated?
- ☛ In what manner does the brain command gestures?
- ☛ What is the importance of conditioned reflexes?

Document 1 The brain, an organ being fully explored

The brain is formed of two cerebral hemispheres. Each is composed of an external gray matter substance (cortex) containing mainly neuron bodies, and an internal white substance containing nervous fibers in addition to central gray nuclei (basal ganglia). How is the brain explored?

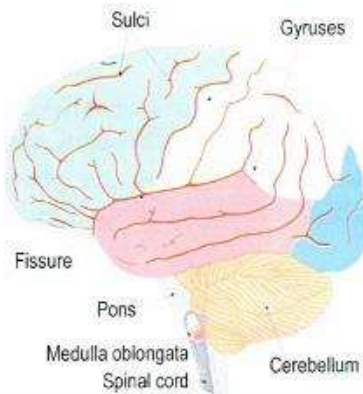


Fig. II. 18 Lateral view of the encephalon.

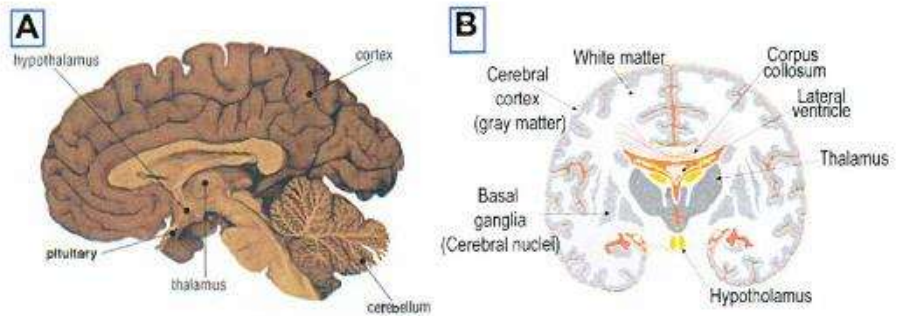


Fig. II. 19 A-Longitudinal cut of the encephalon; B-Transversal cuts of the brain.

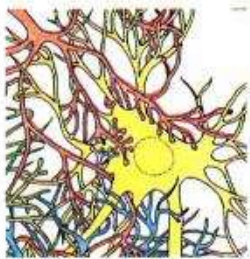


Fig. II. 20 Electroencephalogram.

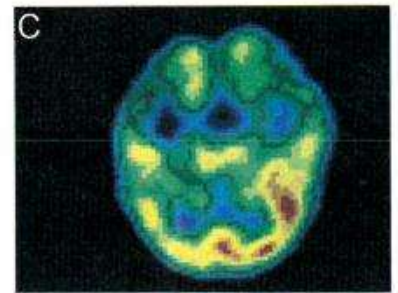
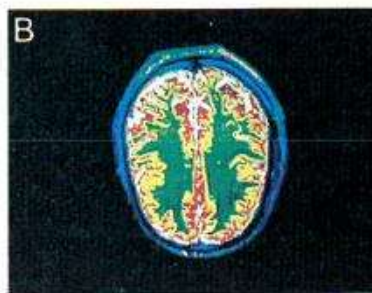
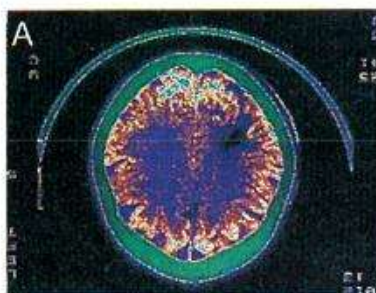
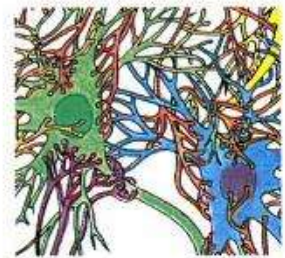


Fig. II. 21 : A- Computed Tomography Scan ; B- Magnetic Resonance Imaging (MRI) ;C- Positron Emission Tomography (PET). If a head wound has probably provoked a hemorrhage, this can be surely, rapidly and efficiently confirmed or excluded by MRI.

Exploitation of the document

1. What is the functional interest of these different techniques?
2. What do the tomography techniques reveal?
3. Are all the brain functions known?

Document 2 The processing of sensory information

The body sensations originate at the peripheral nervous terminals. These are specialized in the transduction or the transformation of the different types of stimuli: mechanical, chemical, thermal, etc., into nervous influxes that can be transmitted only by sensory neurons to the brain. What are the routes taken by these nervous influxes and where are they processed?

1- Some observations

- A wound in the occipital cortex causes visual troubles and a temporal lesion causes hearing troubles...
- An extensive lesion in the parietal cortex along the central sulcus eliminates the sensation of the opposite moiety of the body. A localized lesion eliminates the sensation of a specific region of the body.
- A lesion within the parietal zone eliminates the ability to identify objects.

There are specialized cerebral zones containing an area of projection where sensory information is received and an area of psychosensorial association that interprets the information received. A sensation corresponds to the arrival of a sensory message to the cerebral cortex, whereas a perception is the result of a more complex cerebral elaboration that identifies the information.

2- The sensory pathways

- The sensitive neurons proceed along the posterior root of the spinal nerves and terminate in the posterior horn of the spinal cord. Directly or shortly after making a synapse, they continue towards the encephalic nervous centers: however, their trajectory may differ, depending on the type of sensibility that is considered.
- Receptors transmitting information may have two functions: they can warn of modifications in the external or internal environment, and they can provide information on the positioning of the body. These two types of sensation proceed through two different routes within the white matter.
- In the spine or the bulb, all fibers come across the median axe before ending in the nervous centers. The pathways of sensibility (and motility) are in fact crossed in humans: i.e., if we injure our right hand, the information is transmitted to the left hemisphere.

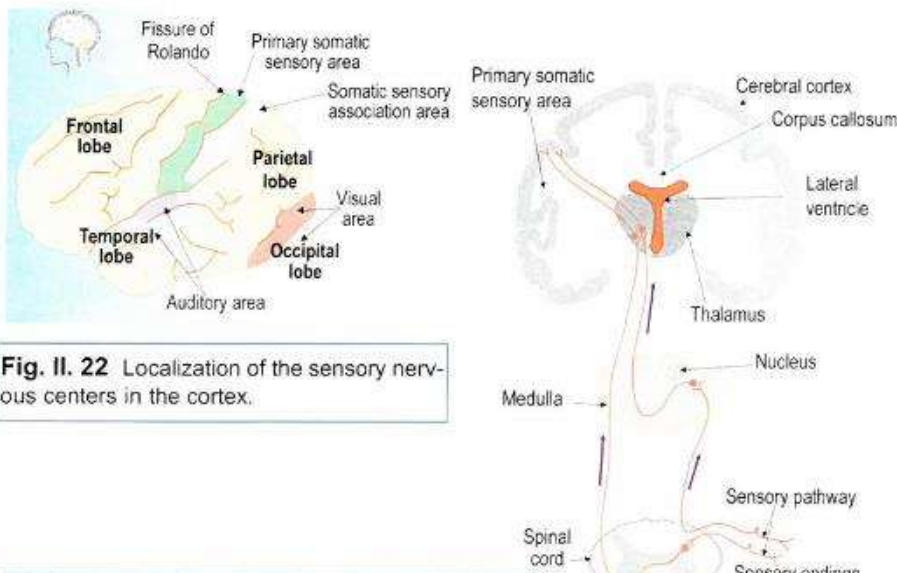


Fig. II. 22 Localization of the sensory nervous centers in the cortex.

Fig. II. 23 Ascending sensory pathways; the information is filtered in the synapses.

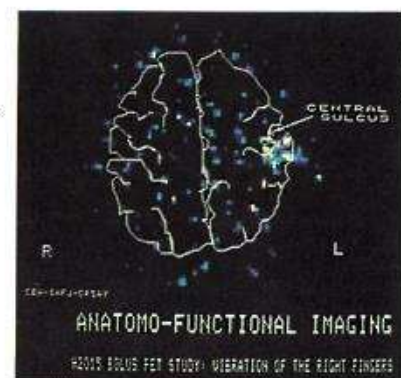


Fig. II. 24 Activation area in the left sensitive cortex following the application of a vibration on a finger in the right hand (PET). The representation of each part of the body does not reflect its actual surface, but the sensory receptors density within that part.

Exploitation of the document

1. What is the importance of sensory receptors in the processing of information?
2. Can we assert that the processing of information is done only in the general sensibility area?
3. Give some examples to illustrate the notion of sensation and perception.

Document 3 *The voluntary command of the gesture*

An individual is capable of deciding when to make a gesture. This voluntary motion is initiated without the intervention of sensory stimuli. The translation of a thought into intentional movement is very complicated. Many cortical and sub-cortical centers are implicated from the start of the movement to its completion. What are the cerebral structures that are specific to the motor act and how do they execute their functions?

1- Some observations

- Motor disorders in patients are often related to cerebral lesions discovered during a surgical procedure or after autopsy.
- An electric exploration of the cerebral cortex is possible through surgical interventions with the help of surface electrodes. The stimulation of a site in the motor area, located in the frontal lobe along the central sulcus, triggers the contraction of specific muscles situated in the opposite body moiety to the cortical zone stimulated.
- A lesion in the zone situated in front of the motor area leads to an inability to perform a movement despite the fact that the muscles are not paralyzed.
- The current exploration techniques of brain functions using PET allow the visualization of cortex regions that are implicated in motor activity.

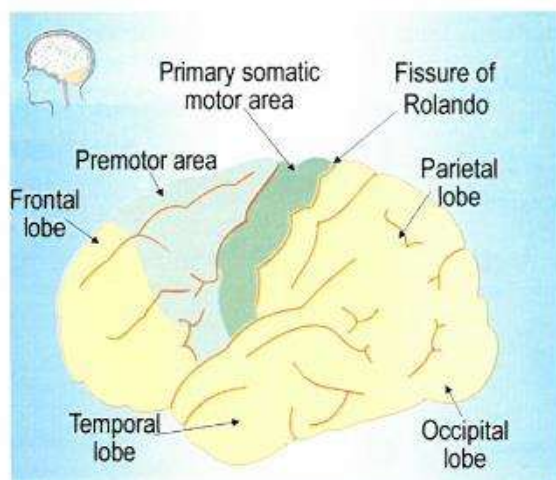


Fig. II. 25 The motor areas of the brain.



Fig. II. 26 Projection of the various parts of the body on the right hemisphere of the motor cortex. (homunculus).

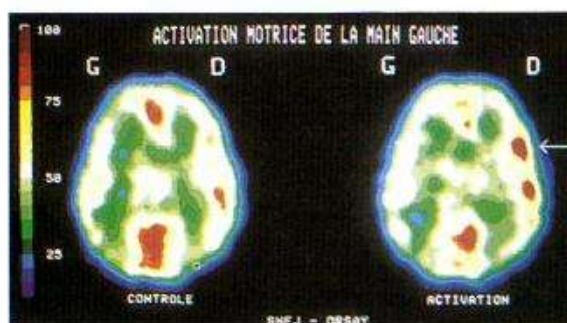


Fig. II. 27 The voluntary movement: a gesture that is well prepared



Fig. II. 28 Activated cortical zones during a movement of the hand.

Summary diagram

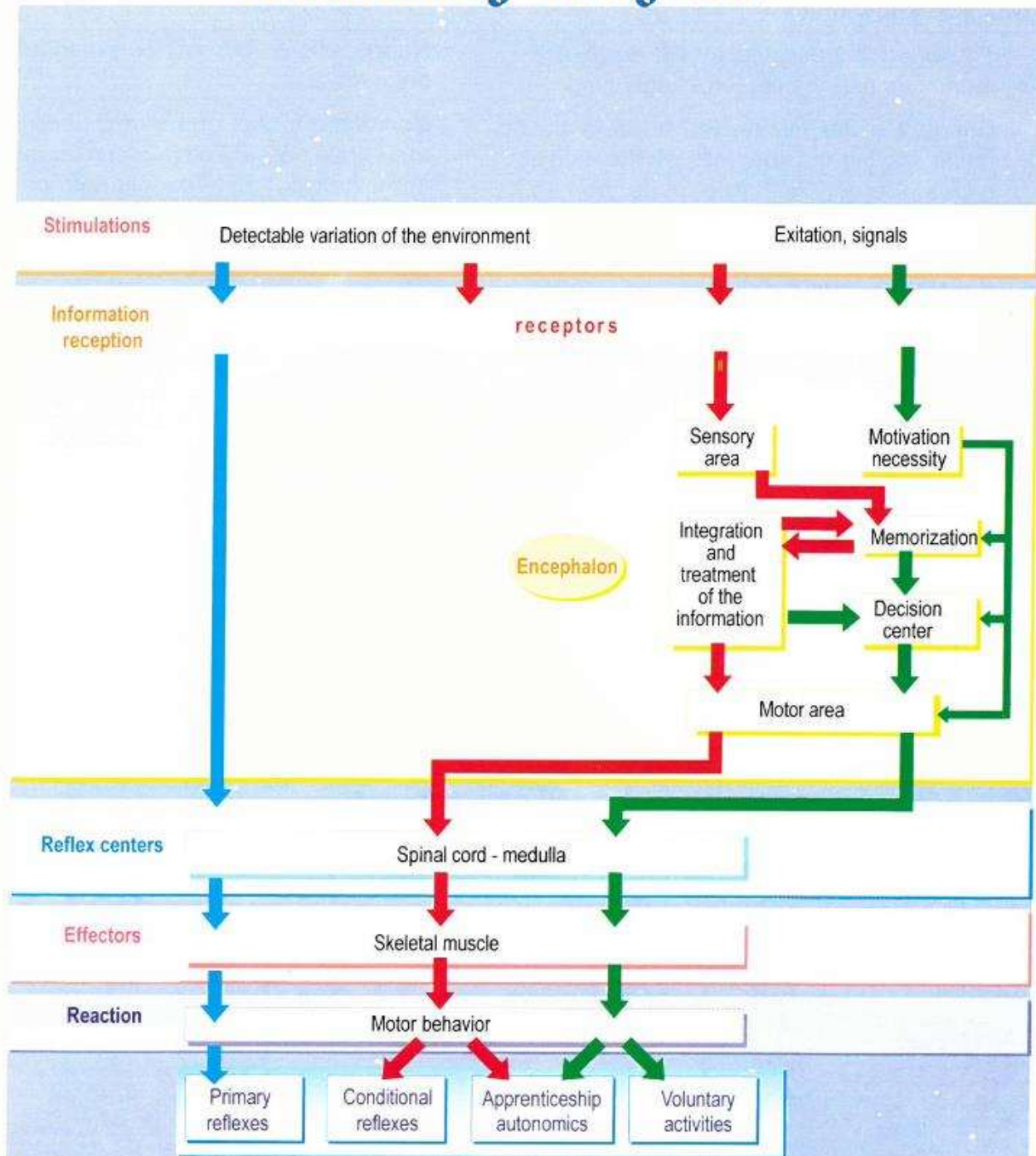


Fig. II. 37 From stimulation to reaction there is a long road, but the treatment of the information is very rapid.

Document 4 Reflex, protection and adaptation

When touching a hot object, our hand is retracted immediately. We have therefore performed an involuntary movement. Once we are accustomed, we drive a car without consciously thinking about it. What is the significance of these involuntary movements that we call reflexes?

1- Innate reflexes

The somatic nervous system is in principle voluntary, but in many instances it responds to external stimuli in an automatic and involuntary manner. The autonomic nervous system that regulates our internal functions (such as the heartbeat) is not voluntarily controlled: these reflexes are called innate. They exist from birth and are constant

throughout life and are identical among individuals of the same species. By integrating the various items of information received, the reflex centers ensure, independently of the brain, a response that protects the body and helps it adapt to the needs of the moment.

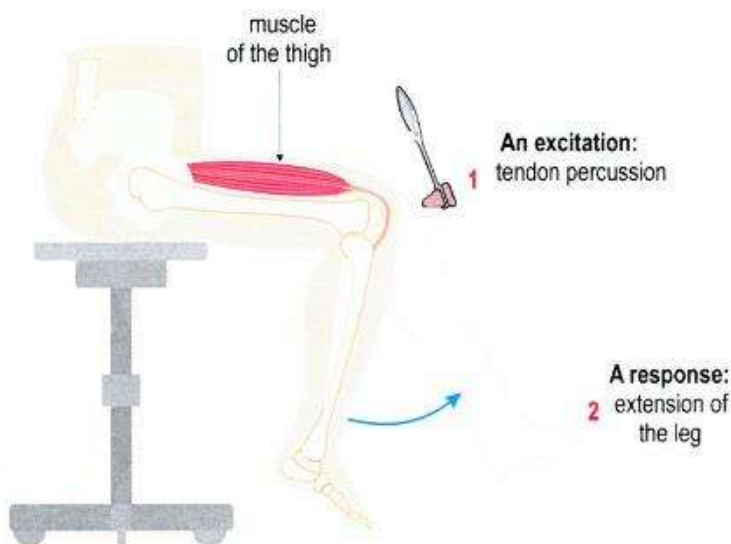


Fig. II. 31 Patella reflex



Fig. II. 32 A frog with its brain removed retracts a leg after contact with acid.



Fig. II. 33 From archaic reflexes to coordinated movement. Maintaining one's balance involves making adequate movements in response to stimuli coming from equilibrium receptors indicating the position of the body in space. The acquisition of this automation necessitates several trials.



2. From innate to conditional reflex: liberating the brain from certain tasks

In 1902, Pavlov (a Russian physiologist) did an experiment where he triggered an audible signal just before he introduced a piece of meat in a dog's mouth. The dog salivated when receiving the meat. After several trials, the emission of the audible signal without the administration of the meat was enough to trigger salivation. The audible signal (neutral stimulus) became a condition for salivation.

To maintain a conditioned reflex:

- the chosen stimulus must be neutral vis-à-vis the absolute stimulus,

- the neutral stimulus must be isolated from all other stimuli,
- the neutral stimulus must be preceded by the absolute stimulus, and it must be maintained.

Animal training is based on the principle of conditioning. In humans, learning allows the acquisition of automation, as in driving, talking and reading...

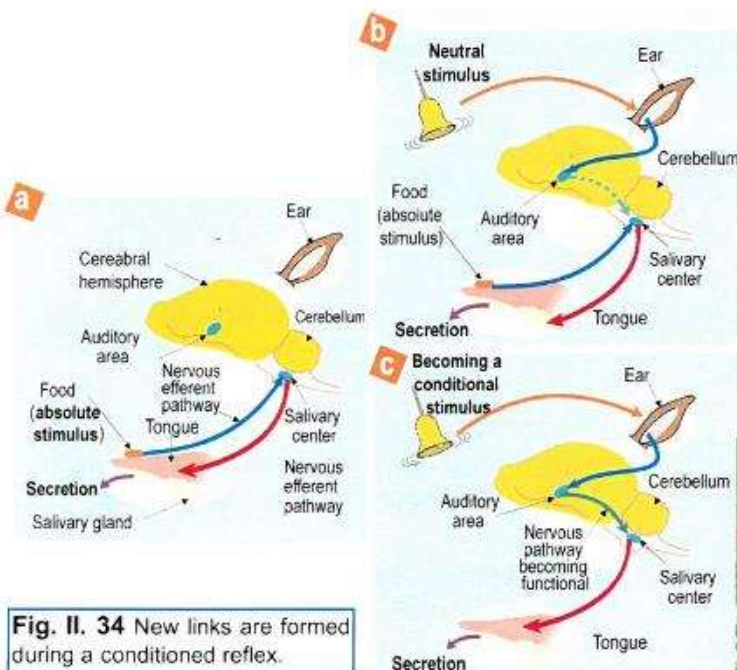


Fig. II. 34 New links are formed during a conditioned reflex.

Fig. II. 36 Frequent conditioning: training.

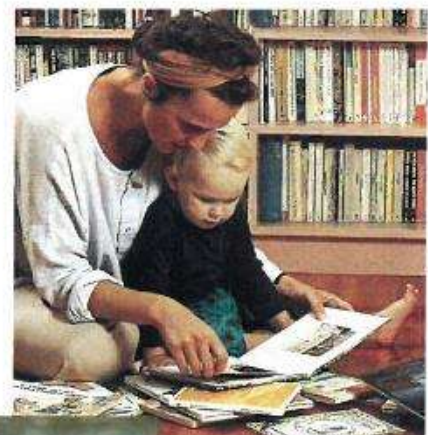


Fig. II. 35 A rich, diversified and stimulating environment that highly solicits the cerebral networks contribute to the learning of this child. However, in order to develop properly this child's brain requires a peaceful, happy and reassuring environment.

Exploitation of the document

1. Name some innate reflexes that we possess. What is the role of each one?
2. Can we have control over our reflexes?
3. What is the importance of conditional reflexes in our lives?
4. What is the absolute stimulus of the salivary secretion in Pavlov's dog?

Essential Points

Cerebral activity and conditional reflexes

The brain

- It is composed of two cerebral hemispheres that are formed of a cortex and central nuclei (basal ganglia) that are made of gray matter containing the bodies of neurons, and a white matter containing the extensions of neurons.
- The cortex contains hundred of billions of neurons. Each receives around 100,000 synapses.
- It is divided into lobes: occipital, frontal, parietal, and temporal.
- The cortex is responsible for all conscious activities and many non-conscious activities; it is formed of specialized areas controlling sensation and motility of each part of the body.

Conscious sensation

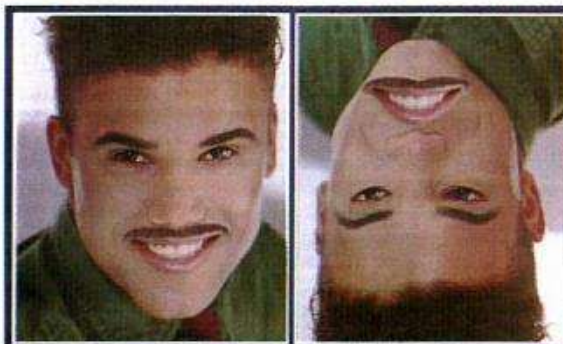
- After a stimulus, the nervous influx that originated in a specialized receptor of a sensory organ, or any other organ, follows a nerve, proceeds through the spinal cord and reaches the gray matter of the opposite moiety of the brain.
- The area of sensory projection in the brain initiates an elementary sensation.
- Association areas are responsible for the processing of the information received by the corresponding sensory area we are conscious of what we see, hear and touch...

Voluntary motricity

- After a decision, or in response to a sensory stimulus, the influx originates in the motor area of the cerebral cortex directly or indirectly connected to motor neurons that are linked to muscles.
- The area of motor projection in each hemisphere commands the contraction of the opposite moiety of the body.
- Each zone of this motor area corresponds to a specific organ whose dimension does not reflect the muscle mass but the precision of the movement accomplished by the group of muscles.
- The association motor area is implicated in the programming of the motor commands.

Reflexes

- An innate reflex is an automated and involuntary response to a stimulus in order to adapt functions of the body to its needs.
- Conditioned reflexes are individual responses acquired by learning to associate an absolute to an environmental stimulus.
- These responses are selective reactions. They can disappear if they are not maintained.
- Conditioned reflexes always favor a better adaptation of the individual to its surroundings and to its social environment.



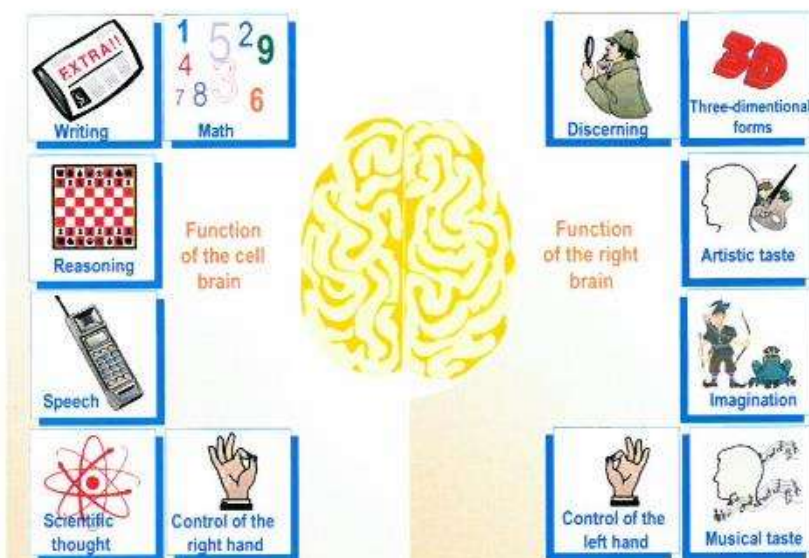
We see with our brain. Thus, we can attribute one meaning or another to the same figure. In other words, the same visual sensation may have two different interpretations.



Additional Information

Did you know that?

- The brain is protected by the skull, the meninges and the cerebro-spinal fluid.
- During the 3rd month, the brain of the fetus, as big as the rest of the body, grows at a very high rate, more than 15 million new neurons per hour.
- Among the 100 million neurons, 1 % are in communication with the environment; these are sensory and motor neurons. The other neurons are to circulate information.
- The total surface of the cerebral hemispheres fully spread out measures 2 m².
- There are between 100,000 to 1,000,000 billion synapses: if we counted 1000 synapses per second, it would take 3,000 to 30,000 years to count them all.
- More than 40 million individuals worldwide have mental disorders.
- The brain consumes 18 % of all the body oxygen.
- Noises above 115 decibels can trigger brain lesions.
- Compared to that of a normal individual, the brain of Einstein (conserved in formaldehyde) has deeper intraparietal sulci and two parietal lobes that are more convex, containing 15 % more gray matter.
- Saying that we use only 10 % of our brain is not really true. If at time (t), only a few parts of the brain are used, soon after many others are stimulated. To understand the brain, we must analyze its functions in time and not reduce its capacities to a simple percentage of utilization.
- Many powerful international firms finance studies to come up with ideas that stimulate the brain of a customer to induce a desire to buy.



The brain is symmetrical with two lobes that ensure the unification of images; however, certain intellectual functions are characteristic of one hemisphere. Language and logical operations are functions of the left while emotions and arts are functions of the right hemisphere.

The enemies of the nervous system-

Bacteria, viruses, protozoa

- Chromosomal abnormalities
- Tumors, hemorrhages, thromboses, cranial trauma
- Overwork, intense noise, psychological stress

- Nutritional deficiency in proteins, vitamins and certain minerals
- Lack of oxygen
- Excess of alcohol, smoke, drugs
- Hypertension, aging, diabetes



Cellular telephone, a danger!

While the link between the waves emitted by a cell phone and neurological troubles is not established, it is advisable to avoid prolonged conversation, especially in the car, to restrict its usage by children, to leave it in a bag and not to carry it directly on the body... cellular phone relay stations are also risk zones.



This three month old baby contributes to research.

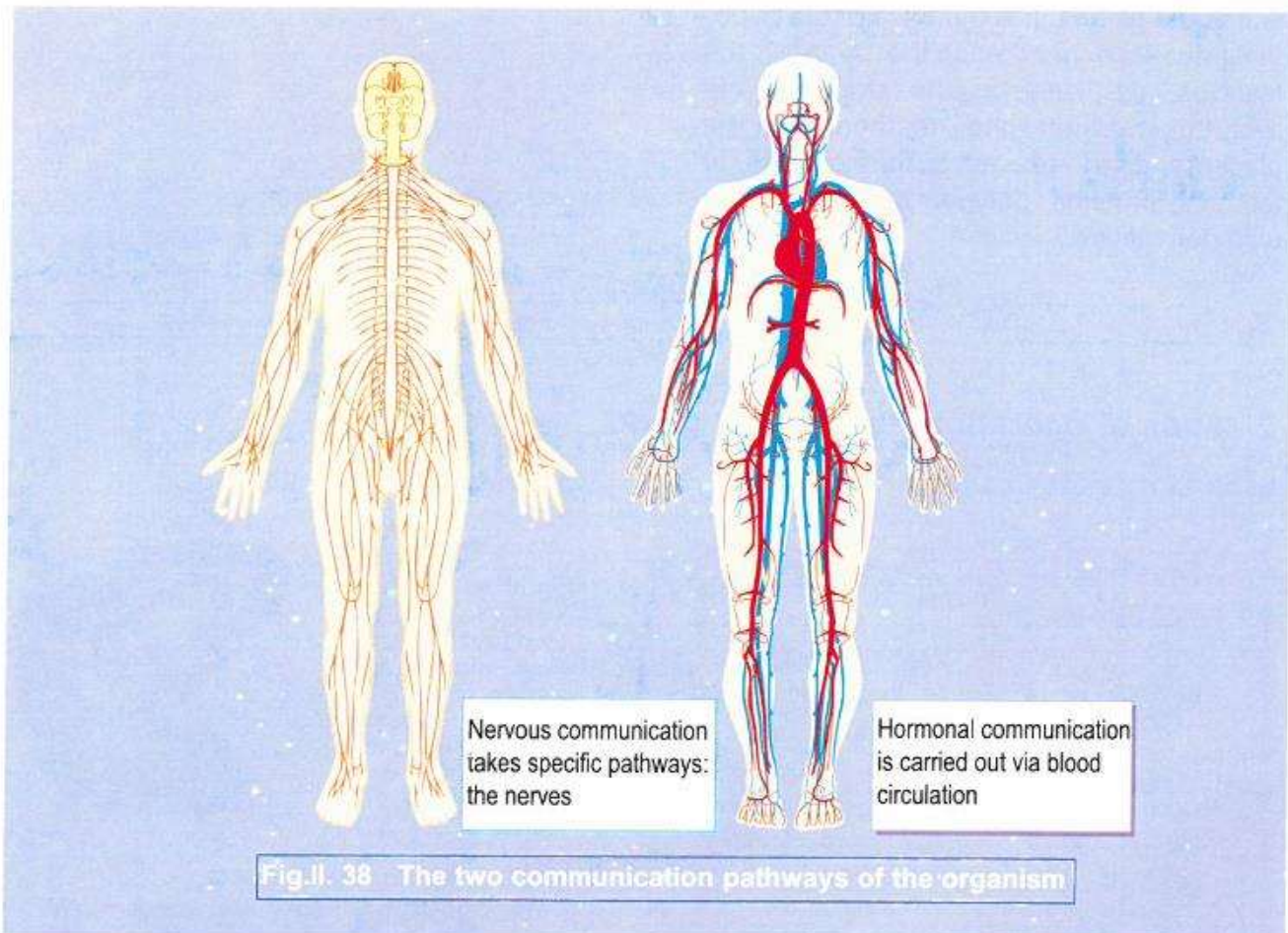


During identical language exercises, depending on the sex of the individual, different zones of the brain are utilized.

Chapter 3

The Endocrine System and Hormonal Communication

The endocrine system regulates and coordinates the functions of all the organs of the body in partnership with the nervous system. The nervous system acts rapidly through the nerve impulses conducted along the neurons and reaching the effector cells. The endocrine system is made of several glands that release molecules called hormones. These circulate in the blood, and act on target cells equipped with selective receptors. This neuro-endocrine system of communication maintains a balance in the body by ensuring nutrition, defense, reproduction, etc.



In this chapter, we shall answer the following questions:

- 👉 How is the endocrine system constituted?
- 👉 Where are the hormones made and how do they circulate in the organism?
- 👉 How do hormones act?
- 👉 How do target-cells respond to the hormonal message?
- 👉 Is there cooperation between the nervous and the hormonal system?

The formation of the thyroid hormones

The thyroid cells take their constitutive matter from the blood (iodine and amino acids) in order to form a substance called colloid. This substance makes the thyroid hormones T3 and T4 which are then released in the blood.

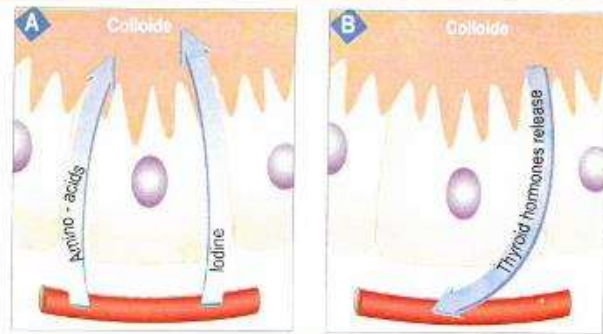


Fig. II. 43 A- Synthesis, and B- secretion of the hormone: hormones are released in small quantities.

Transport of thyroid hormones

The thyroid hormones bind to specific carrier proteins in the plasma. This association between the carrier protein and the hormone is reversible. The reversible association protects the organism against excessive concentration of the hormone and delays its degradation.

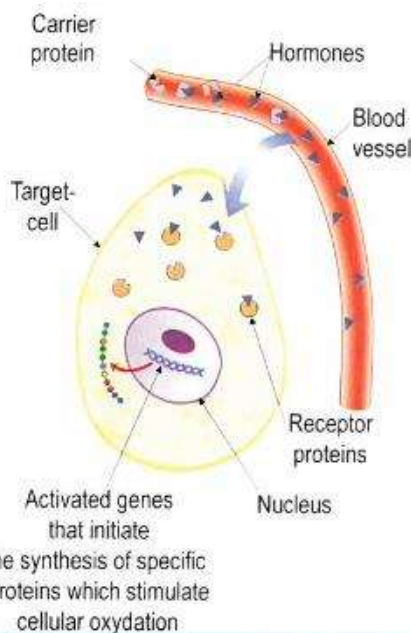


Fig. II. 44 Stages of hormonal action.

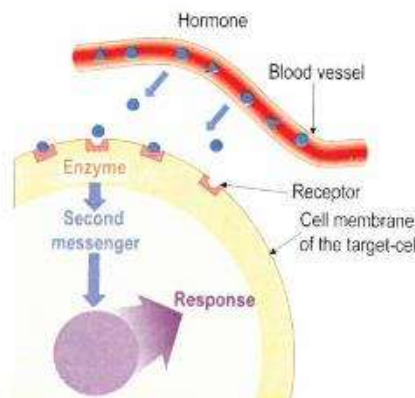


Fig. II. 45 Several hormones bind to receptors located on the cell membrane and produce an effect on the cellular functions via intracellular mediators.

3- Mechanisms of the hormonal action

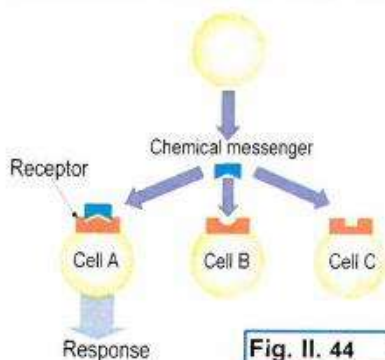


Fig. II. 44 The notion of a target cell.

Only some cells have specific receptors to the hormone. The attachment of a hormone to a receptor triggers after translation of the hormonal message, a series of cellular reactions that modify its activity.

Exploitation of the document

1. In spite of the distance that separates glands, they can still communicate. How is this possible?
2. What can you conclude from the results of figure 40?
3. According to figures 43 and 44, what are the successive stages starting from the formation of the hormone until its action on the target cell.
4. Name some foods rich in iodine. What effects do they exert on the thyroid gland?

Document 1 *La communication chimique*

Endocrine glands are not in anatomical continuity, but they form a functional system. This makes the endocrine system different from all others. How do endocrine glands communicate, and what is their role?

1- The endocrine glands

The endocrine system consists of the following glands: the pituitary, the thyroid, the parathyroid, the adrenal gland, and the epiphysis. In addition, several organs of the body contain cells that secrete hormones without being exclusively considered as endocrine glands. It is the case of the hypothalamus, the pancreas, the ovaries, the testicles, the kidneys, the stomach, the liver, the small intestine, the heart and the placenta. The released hormones include steroids, proteins, polypeptides and amino acid derivatives.

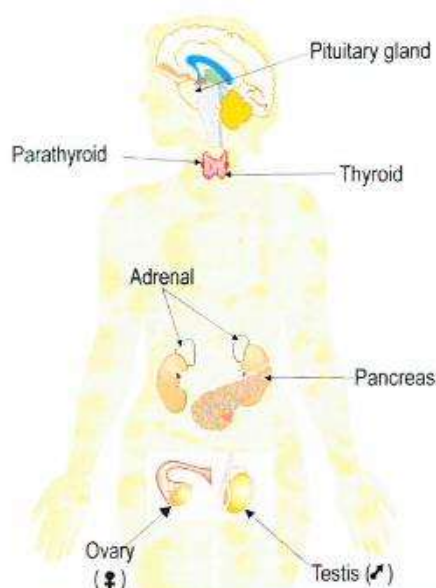


Fig. II. 39 Important endocrine glands.

2- Mode of operation: the thyroid gland

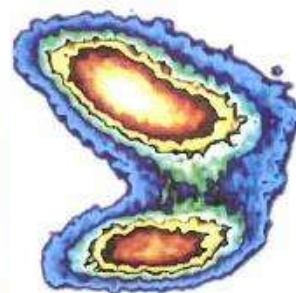
Laboratory experiments

Experiments	Resultats
1- Thyroid gland ablation 	Loss of the aptitude to fight cold
2- Ablation then graft of the thyroid gland 	Regaining the aptitude to fight against cold after graft.
3- Ablation of the thyroid gland and injection of thyroid extracts 	Regaining the aptitude to fight against cold after each injection

Fig. II. 40 The role of the thyroid gland in fighting cold.

Clinical observations

Fig. II. 41 Scintigraphy allows the exploration of endocrine glands. In order to detect dysfunction of the thyroid gland, radioactive iodine is injected and its incorporation in the body is monitored.



Microscope observations

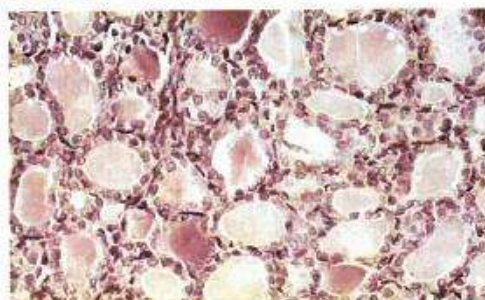


Fig. II. 42 Microscopic preparation of human thyroid. Endocrine glands lacking secreting ducts are formed of secretory cells that release their product in the extracellular spaces before their passage into the blood.

EVALUATION EXERCISES

A- Choose the correct answer (s)

- 1- The white matter of the encephalon and the spine are composed of:
 - a) meninges,
 - b) neurons extensions,
 - c) neurons cellular body.
- 2 - The sensory information received by the receptors from the right side of the body are interpreted by:
 - a) the left cerebral cortex,
 - b) the right cerebral cortex,
 - c) both cerebral cortexes.
- 3- The functional study of the brain shows :
 - a) primary projection areas and secondary association areas,
 - b) a cross of afferent and efferent information,
 - c) a cortex that is formed of one layer of neurons,
 - d) cortical surfaces proportional to the precision of movements or to the precision of gestures.

B- Choose the right statement(s) and correct the false one(s)

- 1) Maintaining the standing posture is a voluntary act.
- 2) In the human brain there is a million neurons.
- 3) The cellular bodies of all the cerebral neurons forms the cerebral cortex.

- 4) The bundle of fibers of the right side of the body passes through the left side of the cerebellum.
- 5) Pre-motor area intervenes only in the preparation of the movement.
- 6) Messages issued from the pre-motor area go directly to motor neurons.

C- Think about it

- 1- The stimulation of a precise site in the motor area triggers contractions in specific muscles. Is this a controlled motricity ?
- 2- When we see an object approaching our face quickly, our reflex is to withdraw. Indicate the structures involved in this process.
- 3- A rat in a cage presses at random on a pedal, and it receives food. Quickly this action is renewed more frequently due to the reward effect. This behavior can go on for a certain period, even in the absence of a reward. However, it should be maintained from time to time by a periodic association with food, causing reinforcement. What is the difference between this kind of reflex and the Pavlovien reflex ?
- 4- When we open a box of cream crackers, a cat comes running to the source of the sound we make. Do all cats come running when we shake the box?
- 5- Can we consider vomiting and fainting to be reflexes ?
- 6- Motor neurones receive at all times, several messages from inter-neurones; some are excitatory while others are inhibitory. What is the aim of these messages ?

Document 2 Neuro-hormonal stimulation

Endocrine glands secrete their products in response to a modification in the surrounding environment. What is the effect of these modifications?

1- The stimulation of the thyroid gland

The thyroid hormones increase in a rat's blood after its exposure to cold. However, there is no hormonal increase following the sectioning of the nerves leading to the thyroid or its perfusion with a cold liquid.

Further, the destruction of the anterior pituitary gland or the stem which connects the hypothalamus to the pituitary gland in this animal leads to thyroid hypo-activity.

2- Various stimulations

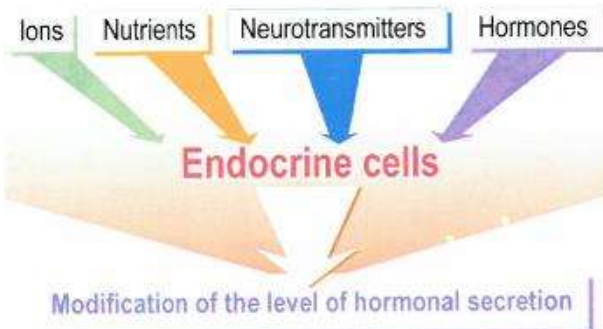


Fig. II. 48 Many commands.

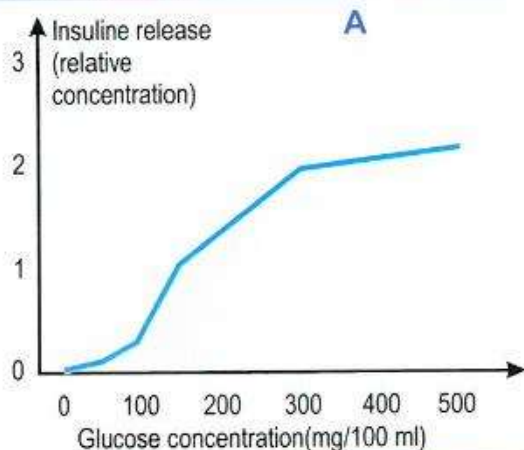


Fig. II.49 a- Experimental stimulation by a nutriment: glucose. b- Nervous command: secretion of adrenaline in the adrenal medulla.

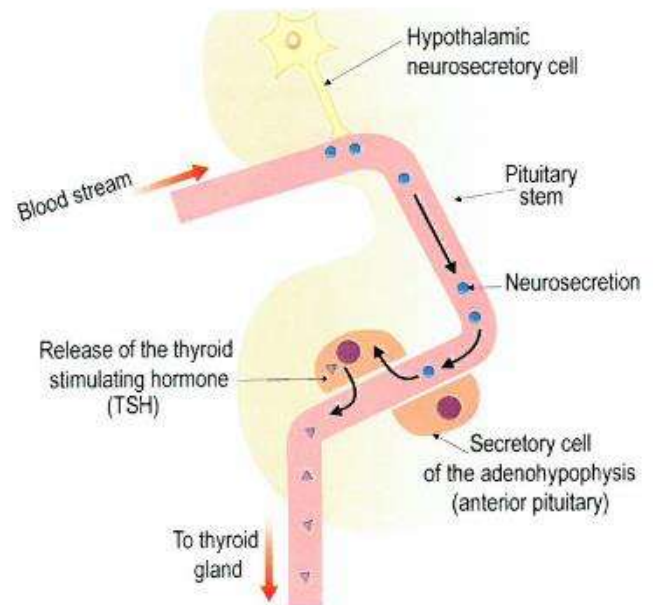
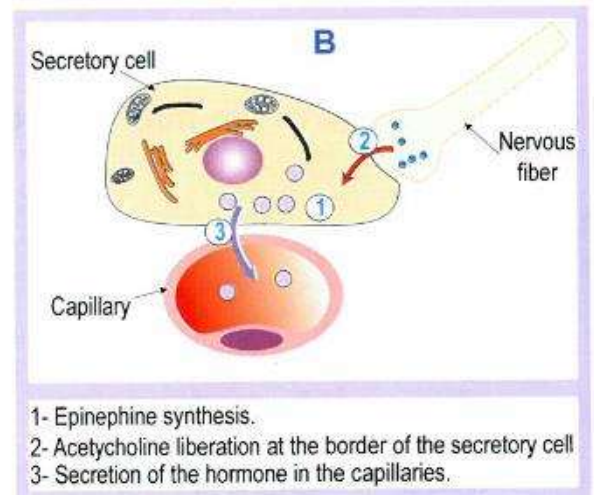


Fig. II. 47 Intervention of the hypothalamus in the regulation of the pituitary and thyroid activity.



Exploitation of the document

1. What can you deduce from the experiments on thyroid gland stimulation?
2. What do figure 48 and 49 suggest to you?
3. From figure 50 deduce a mode of action common to endocrine glands.

EVALUATION EXERCISES

A- Choose the correct answer(s)

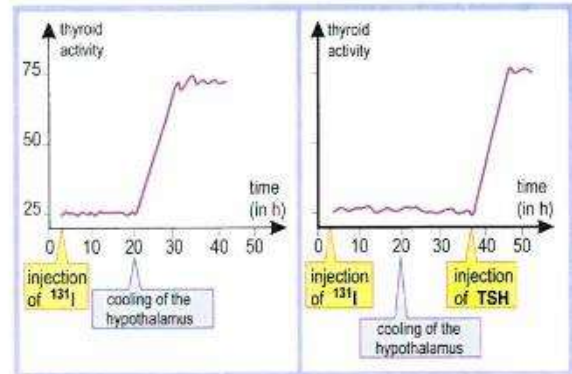
- 1- A hormone is a chemical substance secreted by :
 - a- an exocrine gland,
 - b- an endocrine gland,
 - c- an endocrine tissue,
 - d- all the above.
- 2- The hormone released in blood modulates the activity of :
 - a- only one type of cell,
 - b- all the cells,
 - c- certain target cells,
 - d- several types of target cells.
- 3- In hypothyroidism, the basal metabolism is :
 - a- decreased,
 - b- increased,
 - c- maintained at a normal level,
 - d- unable to be measured.

B- Choose the right statement(s) and correct the false one(s)

- 1- A neuro-hormone is a hormone acting on the nervous cells.
- 2- The same cell can carry several types of receptors that are specific to several hormones.
- 3- The life span of a hormone can vary from few minutes to few hours.
- 4- All the hormones have a receptor inside the cell-target.
- 5- Some hormones do not need a stimulus for their release.

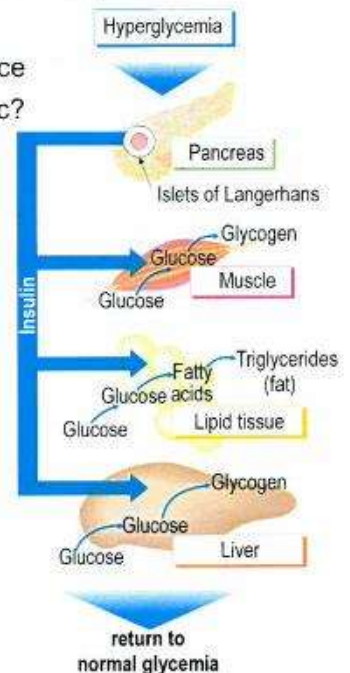
C- Think about it

- 1- After the injection of an animal with radioactive iodine (^{131}I), the iodine gets incorporated by the thyroid hormones during their synthesis. The measured dose of the incorporated radioactivity allows the monitoring of the thyroid gland's activity.
 - After the injection of ^{131}I , the hypothalamus is submitted to a cold treatment.
 - The pituitary gland is removed and the animal is injected with ^{131}I . This is followed by cold treatment of the hypothalamus and an injection of thyroid-stimulating hormone (TSH).



- What conclusions can you draw from these experiments?

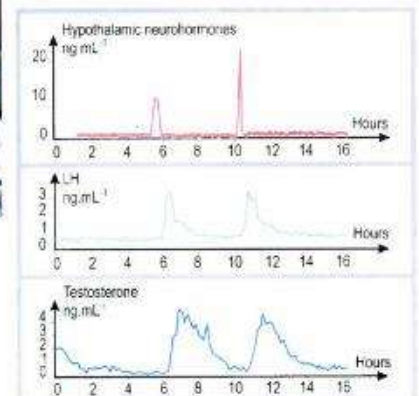
- 2- What do you deduce from this schematic?



- 3- a. What causes the morphological, physiological and behavioral modifications in this teenager?



- b. Comment on these layouts?



Essential points

- The endocrine system includes glands and many organs that contain an endocrine tissue
- The endocrine glands secrete hormones in small quantities directly into the blood.
- The hormones are chemical messengers (like proteins, polypeptides and the steroids) and many are linked to plasmatic proteins.
- They regulate the internal environment, the metabolism and the balance of energy, and play a role in growth and development.
- They affect only the target-cells, which have receptors that are able to recognize a given hormone, and bind to it.
- The attachment of hormones to the mem-

brane receptors of target cells modifies their activity. Other hormones bind to receptors inside the cell, (ex T_3 and T_4)

- The hormonal secretion is controlled by signals coming from the nervous system, the chemical changes and other hormones.
- A neural stimulation induces the release of hormones synthesized in the hypothalamus. This leads to a secretion of anterior pituitary hormones. These, monitor several other endocrine glands that act on various physiological organs.
- There is a regulation of the hormonal secretion in order to prevent an excessive or a deficient secretion.

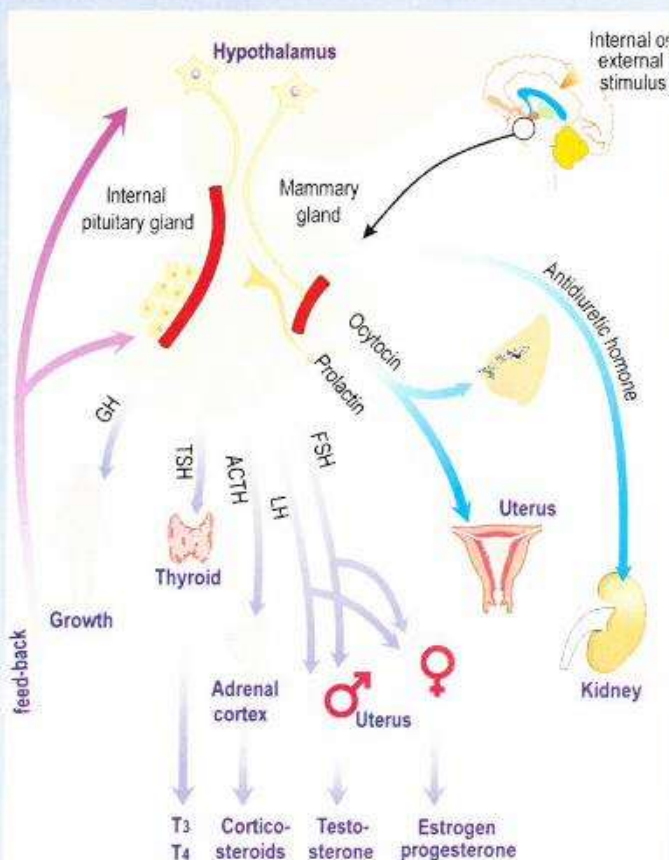


Fig. II. 50 Some activities of our organism, such as the contraction of the skeletal muscles, are controlled exclusively by nervous mechanisms. Other activities such as maintaining the level of blood glucose constant are exclusively controlled by hormonal mechanisms. These two modes of nervous and hormonal communication rarely intervene in separate ways. Thus the majority of the physiological phenomena imply a close collaboration between nervous and hormonal mechanisms. This association constitutes a neuro-hormonal integration.

In certain instances, the pituitary gland follows a hypothalamus command and acts directly on some organs (bone, kidney, and breast). In many other cases, it is the center of command of other endocrine glands (thyroid gland, cortico-adrenal, testicle, and ovary). The resulting hormonal blood level modulates the secretion of the corresponding releasing hormone (feed back).

- Communication between hypothalamus and anterior pituitary = blood stream
hypothalamus and posterior pituitary = neurones pathway
- GH, TSH, ACTH, LH, FSH prolactin = anterior pituitary hormones

Chapter 4

Stress and Emotional Reaction

Man lives in perpetual communication with his peers. His social conduct is influenced by his relationship with others but his actions have to follow certain rules dictated by the society. This adaptation to social imperatives requires a certain tension and surveillance of the individual's natural tendencies. Life in a society affects the most elementary biological behavior.

Due to innate and acquired behaviors, man can adapt to various situations and aggressions. However in certain stressful situations, the body may not dispose of a plan to adequately control its activities, and may become disruptive. These are emotions. If the stressful situation persists for a long time the body may weaken, which then leads to grave consequences.



Fig.II. 51. Psychic crisis of a man, with its anxiety, and frustrations, doubts and aggressions, results in changes that exceed the possibilities of biological and cultural adaptation.

In this chapter we shall answer the following questions:

- ❏ What are the consequences of living in a society?
- ❏ What are stressful situations?
- ❏ How does an organism respond to stress?

Document 1 *Reaction to stress*

We are constantly confronted with stressful situations, in the office, at home, or in the street. The term stress means "force" or physical constraint but it can also designate the sum of all biological and physical perturbations provoked by an aggression and the body's response. In life, physical aggression constitutes an important source of stress. However, there are also many emotional constraints. Emotions are sudden crises, more or less a transitory phase accompanied by bodily reactions that are specific and highly varied.

Can we always control our behavior? How does the organism react to stress?

1- Instinctive and social behavior

An individual can have a feeling of attraction or repulsion towards another individual. All social interactions are based on inter-individual communications (competition, conflict, cooperation). Aggression remains an innate behavior, a natural tendency of the individual, which society is trying to control. Even in children, the fundamental attitudes vis-à-vis others are attitudes of hostility and jealousy. The aggressive instinct (defense) is already associated with the aptitude for biting or chewing solid foods. There exists however an instinct of sympathy prior to every social behavior.



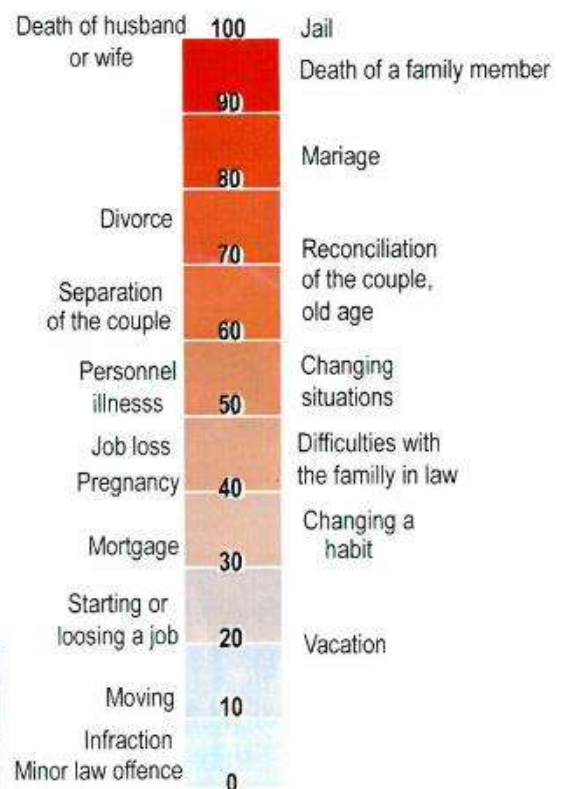
Fig. II. 52 In every society there is a hierarchy. A higher ranking person can morally impose his or her views on subordinates. This is a form of dominance that should not be confused with a desire to use excess force where it becomes slavery

2. Stress in our life

Emotional constraints	Physical aggression	Sudden changes in the environment
- fear (violent), physical menace - before taking an exam, getting mad, anxious - before a surgical operation	- trauma and pain - intense physical exercise - infectious diseases, intoxication	- sudden variation in temperature - dehydration, drop in oxygen intake

Fig. II. 53 Many situations can be stressful.

Fig. II. 54 Arbitrary order of stressful events happening in life. Highest values represent difficulties and are hard to adapt to situations. However, what can be stressful for one person may not be for another, and what can be stressful today may not be stressful tomorrow for the same individual.



Summary Diagram

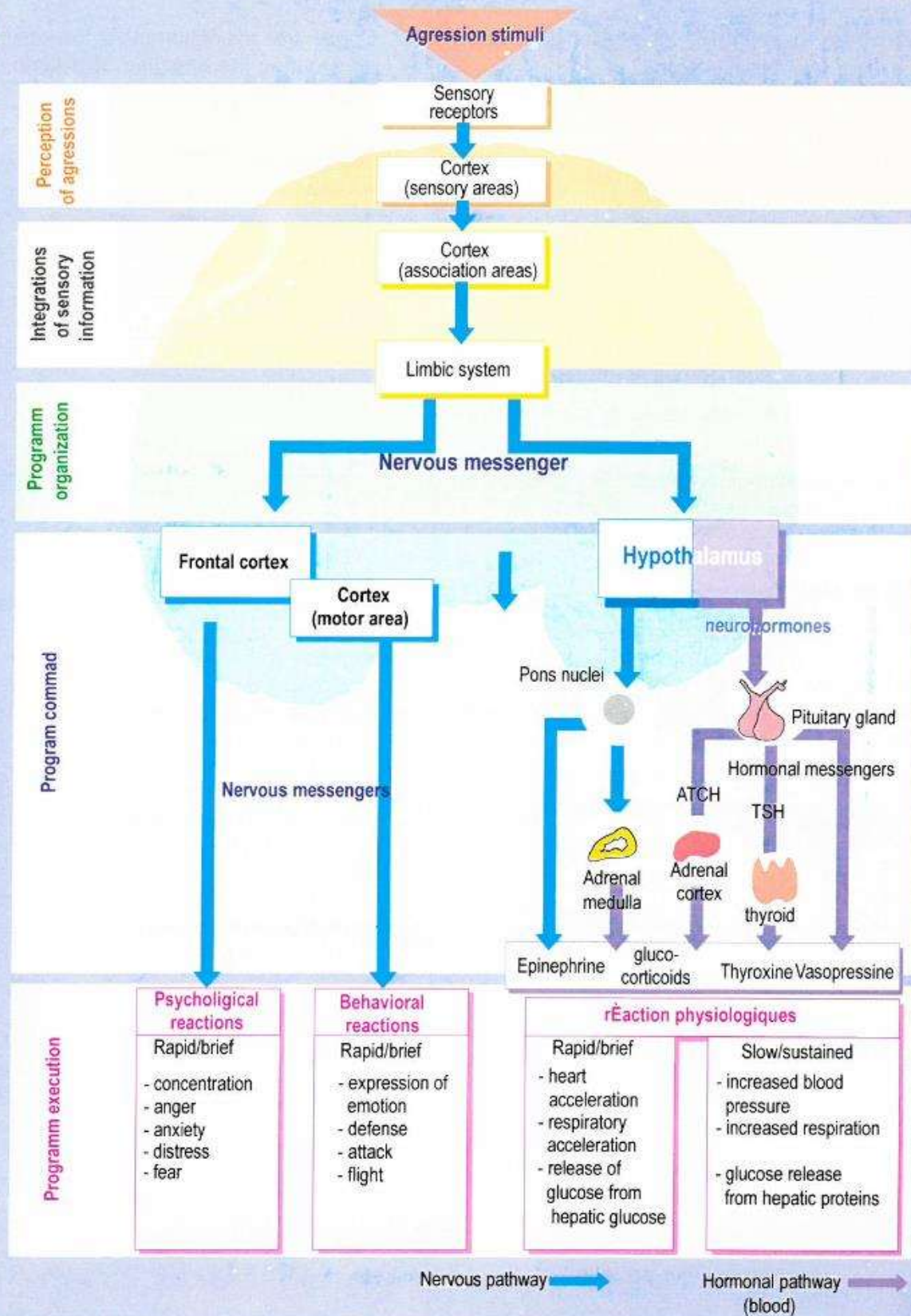


Fig. II. 65 : Against aggression, a full program, a total strategy, many possible responses.

Document 2 Mechanisms of stress response

A behavioral act, in response to aggression, involves both the nervous and the endocrine systems. How is the response of the effector organs, coordinated in order to ensure protection, integrity and equilibrium to the body?

What are the structures implicated and what is their mechanism of action?

1- Control by the nervous system

Some observations

- Stimulation of the hypothalamus in the animal provoke an exaggerated emotional behavior, and the isolation of this zone from the rest of the limbic system eliminates emotions.
- In humans, tumors of the frontal and pre-frontal cortex leads to character perturbations.

- Patients with lesions in the cortex, but with intact hypothalamus, have strong emotional crises.
- We obtain an emotional indifference in aggressive subjects after their limbic system is destroyed.

The emotional brain is situated in the limbic system and it is linked via axons to the cortex and other sub-cortical zones.

The nervous system intervenes by mobilizing

- **Sensory receptors** : eyes, ears, skin receptors as well as the receptors for blood pressure, pH, etc.
- **the nervous centers** : The limbic system is made up of several centers and neural networks situated in a strategic position between the hypothalamus and the cerebral cortex. External as well as internal information is compared with prior experiences (learning, memory). Hence there is a coordination of urgent reactions to adapt the organism to its new situation
- **Affected organs** : Facial muscles, blood vessels, glands, etc.

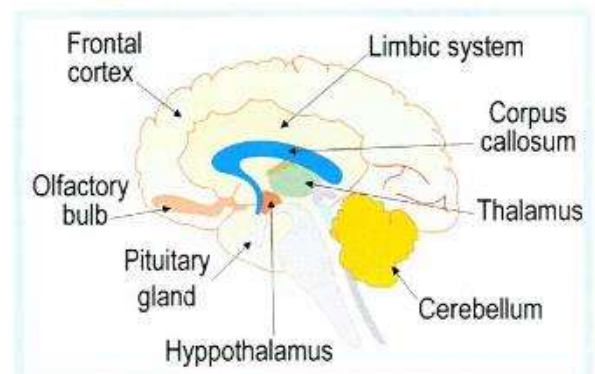


Fig. II. 60 The limbic system.

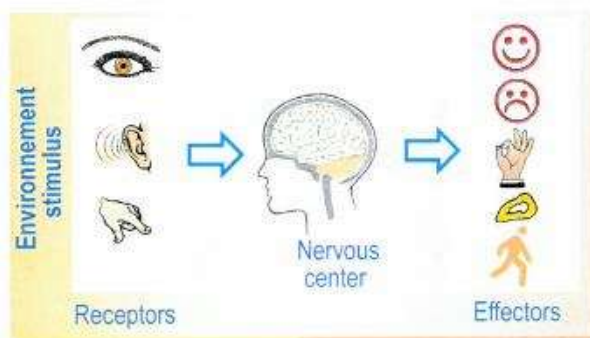
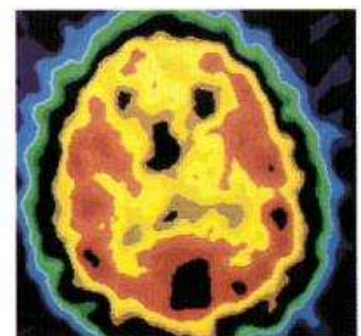


Fig. II. 62 A classic reflex arc.

Fig. II. 61 Anger illustrated above in yellow, a region where serotonin, one of the neurotransmitters of emotion, is metabolized.



Essential points

Stress and emotional reactions

- Man lives in society and he communicates with his peers.
- Relations among individuals are often characterized by a behavior of dominance or aggression, which in certain instances leads to social hierarchy.
- Man is constantly confronted with aggressive situations to which he has to react.
- There are physical and psychological agents of stress that provoke in individuals behavioral and hormonal changes.
- Our organs should be able to react to dangerous perturbations and adapt the body to various physical conditions.
- The alarming reaction is triggered by nervous influxes originating in the hypothalamus and reaching the sympathetic system as well as the medullo-adrenal and provoking a discharge of adrenaline. The latter increases circulation, reduces non-essential activities and favors energy production to help react to danger.
- The resistance reaction is triggered by the hormones of the hypothalamus and the pituitary gland. These stimulate the cortico-adrenal glands to release cortisol and other hormones in large quantities during a stressful condition, thereby accelerating catabolism to produce the necessary energy to combat stress.
- During intense stress, an exhaustion phase may occur which can sometimes lead to death.



Fig. II. Yoga, one way to manage stress.

3- The manifestation of emotions

Visible parameters

Becoming pale (paleness), increase in heart rate, and modification of respiratory rhythm are the most important reflex manifestations of a stressful situation. They are common to almost all emotional states of the individual. However, for the same stressful situation, inverse reactions may occur, blushing, slow heart beat, crying, screaming. Although apparently different, all these involuntary reactions are intended to help the individual adapt to his or her new situation.



Fig. II. 55
Visible manifestations during violent emotional stress.



Fig. II. 56 Exam results : Emotions change according to individuals.

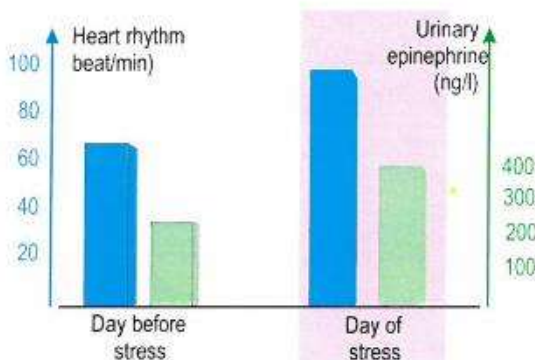


Fig. II. 57 During a public address, an accelerated heart beat and high blood pressure are observed in the person delivering the speech. Urinary adrenaline also increases (Results from personal communication of Dr R. Chahine)

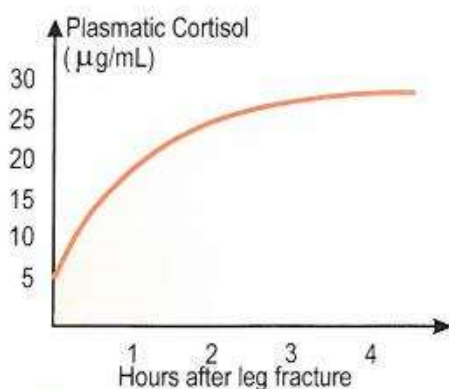


Fig. II. 58 Increase of the plasma cortisol after a surgical operation.

The hidden parameters

Specialized and sensitive blood tests performed on individuals during intense emotional stress (result of a bone fracture or having to make a public address) showed a significant increase in the level of several hormones, especially cortisol and adrenaline.

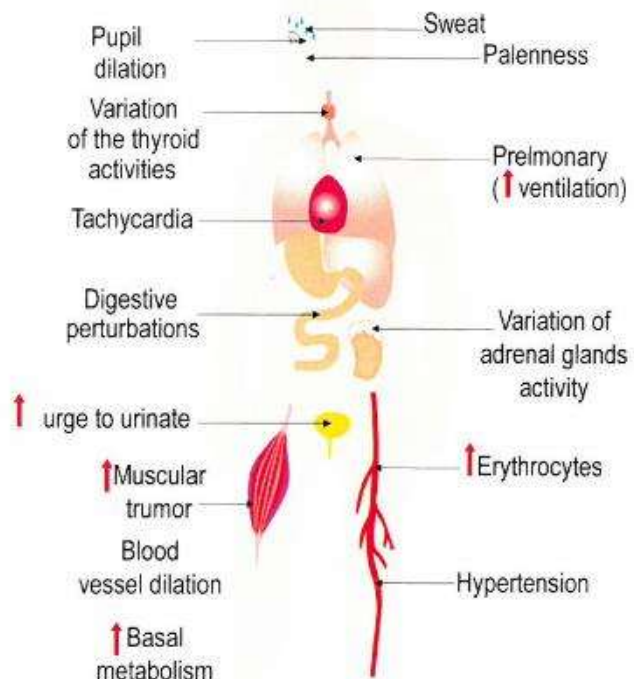


Fig. II. 59 The manifestations of violent emotions in humans.

Exploitation of the document

1. Does the social behavior of an individual mask instinctive behavior?
2. Complete this list of causes of stress.
3. What is the meaning of visible and of hidden manifestations?

2- The neuro-hormonal control

Two hormonal pathways intervene in reaction to aggression:

Adrenaline discharge

The emotional response to danger (fight or flight reaction) is regulated by the adrenaline secretion that results from the stimulation by the sympathetic system of the medullo-adrenal gland. The aim is to increase the frequency of cardiac contractility, to dilate the pulmonary bronchioles in order to increase the exchange of oxygen, to increase the concentration of glucose in the blood and to redistribute blood to the muscles.

Glucocorticoid liberation

The hypothalamus regulates via hormones the pituitary gland, and this latter stimulates the cortico-adrenal gland to release glucocorticoids. This raises the blood sugar level, and lowers the excretion of water from the kidneys. It also has an anti-inflammatory effect that helps the organism in resisting aggressions.

Reaction to stress or adaptation syndrome

An alarm reaction (fight or flight) to danger is immediately triggered by nervous influxes that originate in the hypothalamus. This mobilizes the body's resources in its reaction to stress. In order to anticipate and perceive the danger, the brain must remain alert and the muscles ready to intervene at any moment. Therefore, the presence of oxygen and glucose is constantly required.

Another long-term reaction is triggered, largely by the hormones of the hypothalamus. Its aim is to combat a stress provoking agent. It constitutes a resisting phase. If it is prolonged it may lead to an exhaustion phase with total loss of potassium ions and of glucocorticoids and can sometimes lead to death.

Human social behavior is such that the reaction of fight or flight is not always possible, even though the body is prepared to do so. If the aggression is frequently repeated, the accumulation of hormones may have a negative effect in the long run.

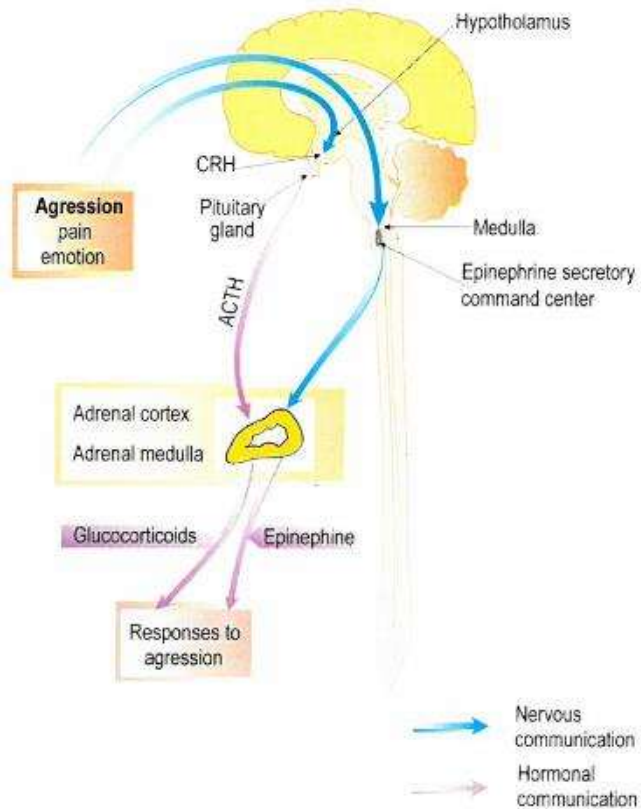


Fig. II. 63 Two hormonal pathways under nervous control

Exploitation of the document

1. Taking into account the importance of the CNS in the control of stress, what do you think of the following citation: "You should know how to manage your stress".
2. Is the term adaptation syndrome justifiable?
3. What is the difference between a reflex and an adaptation syndrome?
4. What is the link between the behavioral and hormonal changes after stress and the reaction of fight or flight. What is the role of human behavior in this situation?
5. What are the effects of adrenaline in the reaction of fight or flight?

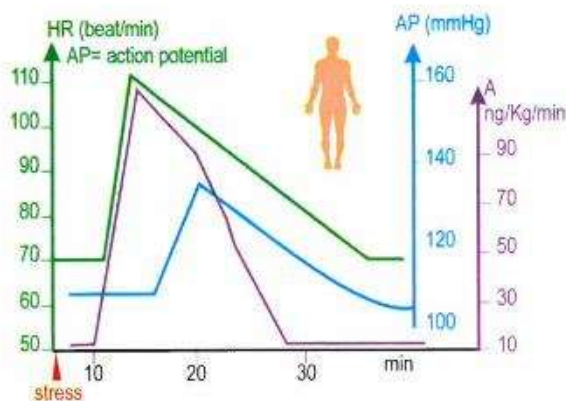
EVALUATION EXERCISES

A- Choose the right statement(s) and correct the false one(s)

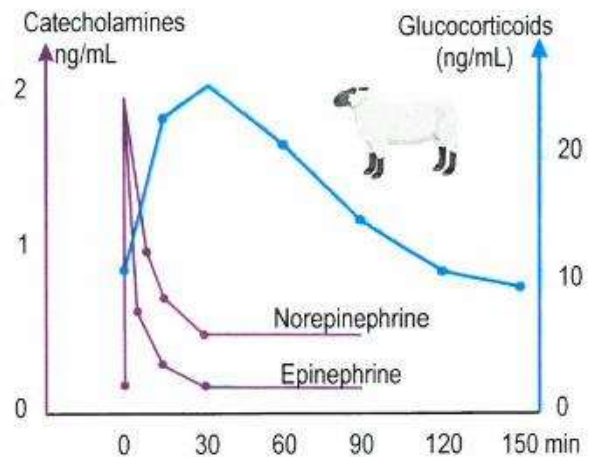
- 1- Agents of stress are always of a physical nature.
- 2- Reaction to stress does not need a sensory receptor.
- 3- The body responses to stress have an adaptable function.
- 4- There is no reaction of "fight or flight" in humans
- 5- Adrenaline is a dangerous hormone
- 6- The release of stress hormone is under nervous control
- 7- The hypothalamus organizes the program of action during a stressful situation.

B- Think about

- 1- During intense stress we measure in an individual the frequency of heart rate (HR), the arterial pressure (AP) and concentration of adrenaline (A) in the plasma. Comment on this graph.



- 2- Concentrations of hormones have been measured during the shearing of a lamb. Analyze these graphs.



- 3- Comment on this text:

"Our lives are closely shaped by many surrounding factors, people, objects, and circumstances. We are an integral part of a social-environmental structure to which we belong. With the incessant dialog between us and the world around, we are modeled by it and we transform it with each act that we perform. In this perpetual movement, non-adaptation can give rise to violent emotions that can affect the organic function of the body (this constitutes what we call psychosomatic disorders). Every society is autonomous, it functions differently and interactions between individuals and society occur according to different norms.

Document 2 *The drugs' mode of action*

Drugs disturb the biochemical mechanisms of the nervous system. The similarity that exists between their molecules and that of neurotransmitters eliminates, or prolongs, their effect depending on the case. This involves a disturbance of the mental equilibrium and the normal operation of the organism. Drugs produce irreversible lesions of various cell types: nervous, hepatic, pulmonary, renal, etc. In addition, drug addicts frequently get into a state of depression which may bring them to take an overdose of the drug, which is often fatal. What is at the origin of these perturbations?

Drugs interfere with the pre or post-synaptic receptors for the various neuro-transmitters. This leads to their beneficial usage in medicine: morphine is given to calm the pain, cocaine is a local anesthetic.

In the central nervous system within the pathways of pain, the modulating neuron releases enkephaline (endogenous morphine), which binds to receptors of the pre-synaptic neuron; this results in the inhibition of the secretion of a pain mediator substance (P). Morphine binds to enkephaline receptors on the modulating neuron. However, enkephaline is degraded by an enzyme soon after its release to prevent its accumulation, whereas morphine will have a more durable action because the body cannot degrade it.

Dopamine, a molecule secreted by the brain, is responsible for the sensation of pleasure. Drug users will have great sensations of pleasure because the drugs they use have a direct effect on the dopamine receptors in the brain. The effect of dopamine is at the center of the state of dependency. The pleasures obtained from the use of drugs push the user to use more drugs. The frequent use leads to drug tolerance which in turn results in addiction. This is the vicious circle of drug abuse.

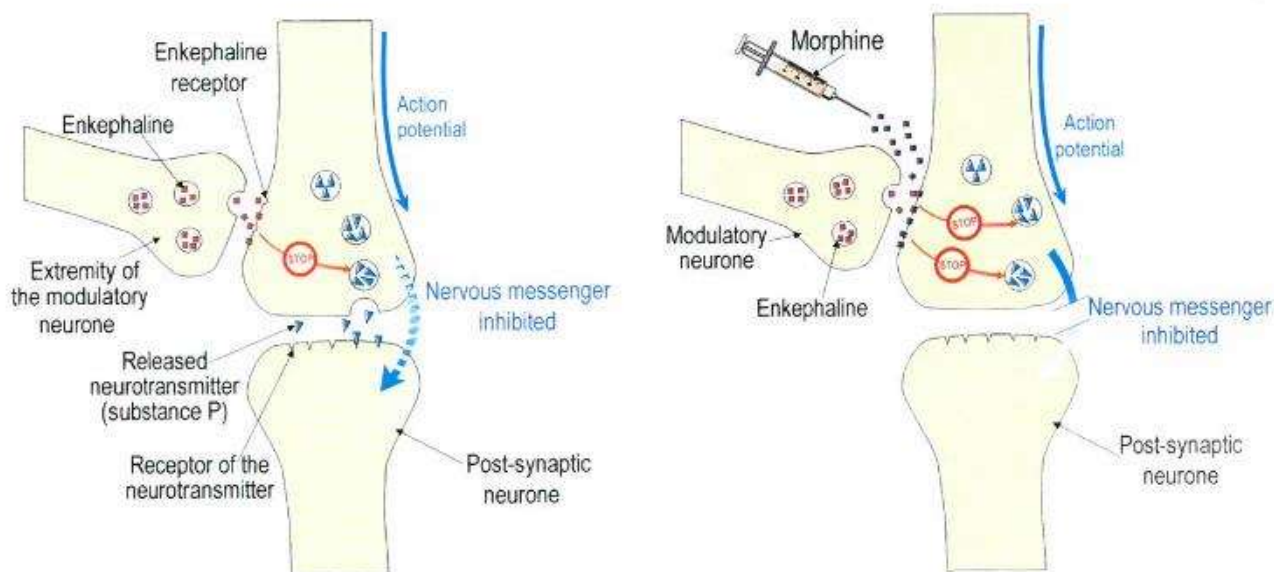


Fig. II. 70 The action of Morphine.

Document 1 Drug-addiction, an artificial paradise

Inhaling, smoking, chewing or voluntarily injecting a substance that is toxic to the body that can lead to a state of dependency is considered as drug addiction

1- Some characteristics of drug addicts:

- Irresistible desire to obtain the drug no matter what the cost is.
- Tendency to increase the dose.
- Physical and/or psychic dependence.
- Physical or psychic symptoms when there is an interruption of consumption.
- The individual's tendency to cause harm to him or herself or the society.

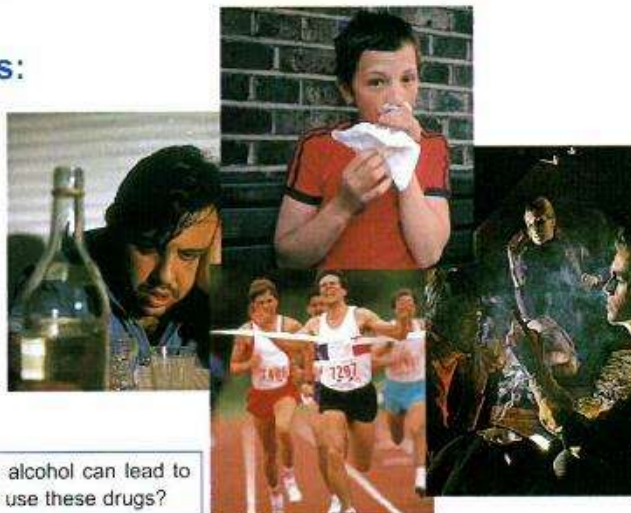


Fig. II. 66 Heroin, tobacco, sleeping pills, even alcohol can lead to addiction. What is the reason that pushes people to use these drugs?

2- Classification of drugs

Products	Effects and dangers
Opiates Opium and derivatives: morphine and heroin. Opium: secretion of a plant, the poppy Morphine: chemical substance extracted from opium. White powder.	• Impression of euphoria, dreaming state. • Double dependence, psychic and physical, somnolence and apathy. • Suppression of pain (medical use). • Double dependence, psychic and physical. Same effects as opium, but more extreme. Risk of overdose (large amount that the organism cannot tolerate; it results in a quick death). • Symptoms: respiratory and cardiac troubles, coma, risk of HIV infection (transmission by syringe).
Heroin: chemical derivative of morphine. White powder Hallucinogens: Cannabis: herb (marijuana), resin (hash). LSD 25: (acid). Liquid or crystallized	• Impression of intense, very brief pleasure (flash). • Double dependence, psychic and physical. • Risk of HIV infection (syringe), physical pains, and risk of overdose; physical and psychic forfeiture. • Impression of euphoria. • Psychic dependence, weak physical dependence. • Risk of anguish, memory disorders and lack of concentration. Sometimes vision or auditory troubles. State of panic. • Exaltation of perceptions. Excitation of the brain. • Possible psychic dependence. sensory disorders and hallucinations. • Significant risks: state of panic, psychiatric disturbance. States of abatement and anxiety (prolonged nightmare)
Psycho-stimulants: Cocaine Amphetamines	• Stimulation effect. • Physical and psychic dependence. • Anguish, depression. • Reduction in the sensation of tiredness, feeling of euphoria. Psychic and physical dependence.
Barbiturates	• A weak dose: induced sleep. • When used frequently: dependence and need for increasing the amounts. • Risk of serious accidents, even death, when diverted from their medical use.
Solvents Ether Trichloroethylene Glue	• Drunkenness, euphoria impression, dizziness. • Psychic and sometimes physical dependence. • Risk of serious accidents, even death.

Fig. II. 67 Drugs: natural or synthetic substances.

3- Dependence, habitual usage and tolerance

The pleasant feeling when the drug is first used encourages the user to ask for more. This may develop into a state of psychic dependency. Hard or soft, all drugs lead to a state of dependency. The whole life of the user becomes centered around drugs despite their harmful effects on health, and the consequences they have on loss of family, friends and work.

Frequent usage of drugs may also lead to a state of tolerance, i.e. with the adaptation of the organism to the repeated doses. The drug may lose its effect, and the user must unceasingly raise the doses to obtain the desired effect.

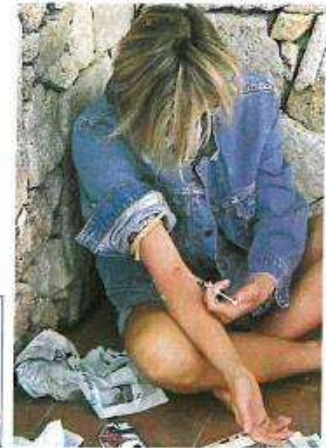


Fig. II. 68 After a while the drug user will need a dose every half-hour. A physical dependency has developed



Fig. II. 69 Hallucinations and pleasure at the beginning, but depression at the end.

Physical dependency

This is a biological adaptation to the product used. The user can no longer live without it and interruptions in taking the drug will result in severe physical troubles. This is called drug withdrawal. All drugs (morphine, heroin), even high consumption of alcohol and tobacco, induce a withdrawal syndrome when the quantity of the drug circulating in the organism decreases, because of its elimination by the kidney and the liver (or lungs for alcohol). The terrible pain and discomfort of withdrawal is due to the fact that the body has adapted to the massive presence of the drug in the body. This state of discomfort is what attaches the user to the drug, which leads to the state of dependency.

Exploitation of the document

1. Draw two significant conclusions about drugs from the table of figure 3.
2. Why is drug use prohibited by the law?
3. Is the pleasure obtained from drugs real?
4. What are the harmful effects that a drug may have on health?

Chapter 5 Drugs and drug-addiction

Experts of the World Health Organization (WHO) define drug-addiction as "a state of physical and/or psychic dependence, vis-a-vis a product, due to periodic or continuous use of this product". Drugs have been used throughout history and are common to all societies. People use drugs in order to fulfill a need or to escape some of the pressures of daily life. Usually an occasional usage leads to dependency and soon the health of the user is compromised.

All drugs have a negative effect on the normal operations of the nervous system. Drugs are harmful by themselves, but they also cause a state of dependency which is disastrous to the whole person.

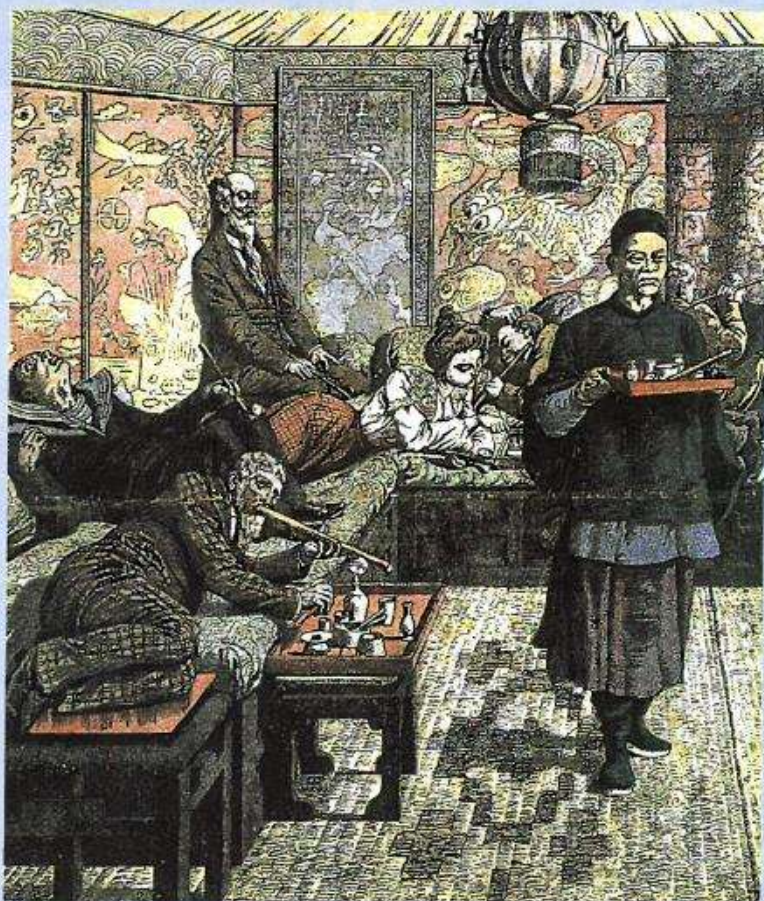


Fig.II. 65 The hallucinogenic effect of certain plants has been known for a long time. Some people use these plants to access an artificial paradise.

In this chapter we shall answer the following questions:

- ☞ What are the various drug categories ?
- ☞ What is their mechanism of action?
- ☞ How is physical and psychic dependence established?
- ☞ What are the consequences of such a dependence?

Chapter 6

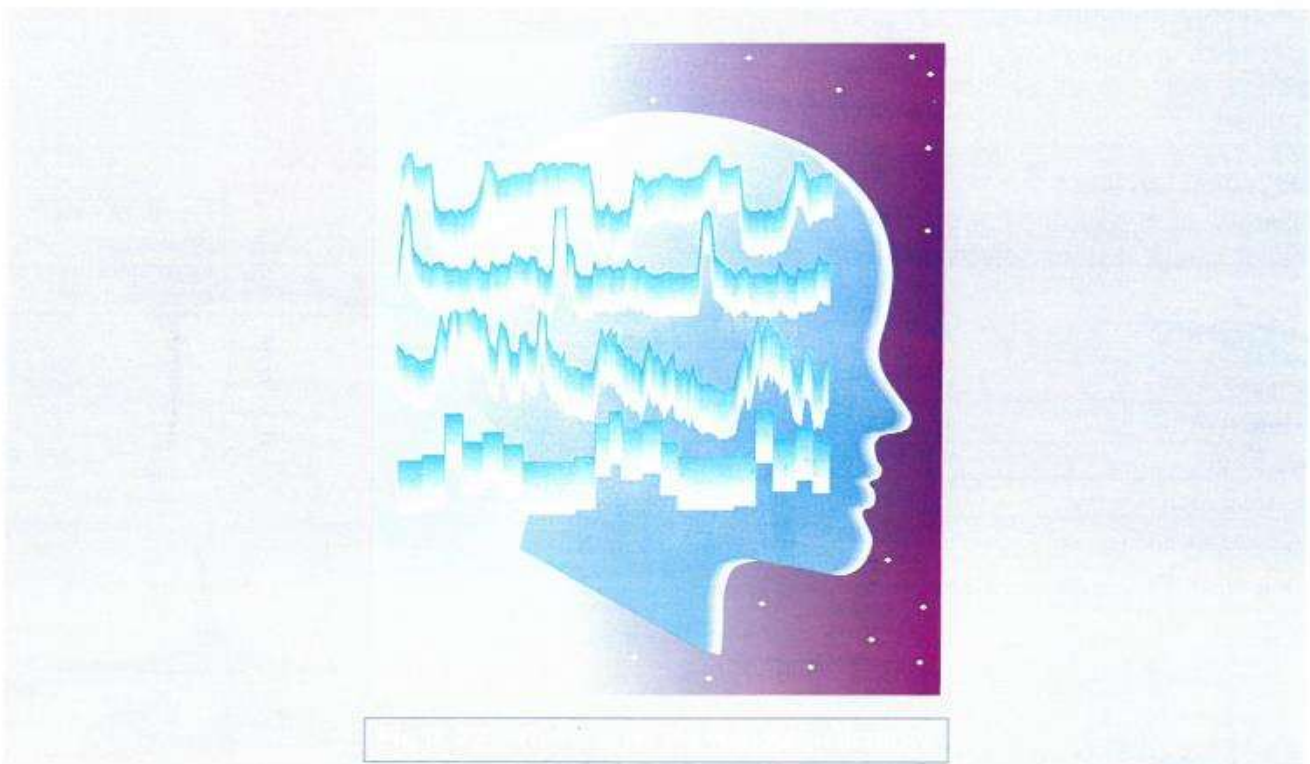
Biological Rhythms

Periodicity is a fundamental property of living matter

We wake up every day at the same hour, we feel tired and we go to bed also at the same time, we get hungry at certain hours of the day and we rarely urinate in the middle of the night. When we observe several functions of the organism, we can see that the arterial pressure, the core body temperature, and the secretion of many hormones vary with daily rhythms. These variations are described as the circadian rhythm.

At first sight, these phenomena could be explained as a simple cause and effect relation. For instance we wake up because "we have slept enough" or we go to bed because "it's dark outside", we urinate less at night because we don't drink. But the most simple experiences are enough to show that this reasoning is wrong. Many have experienced a night of work or a jet lag; in both cases the body tries to follow its usual rhythm even though it is inappropriate to the real conditions.

The study of Chronobiology or the biological rhythms and their perturbations is useful to the comprehension of several pathological processes.



In this chapter we shall answer the following questions :

- ☞ What are the principal cycles of our life ?
- ☞ What is the importance of the sleep-wake cycle ?
- ☞ What is the origin of biological rhythms ?
- ☞ What are the applications of Chronobiology ?

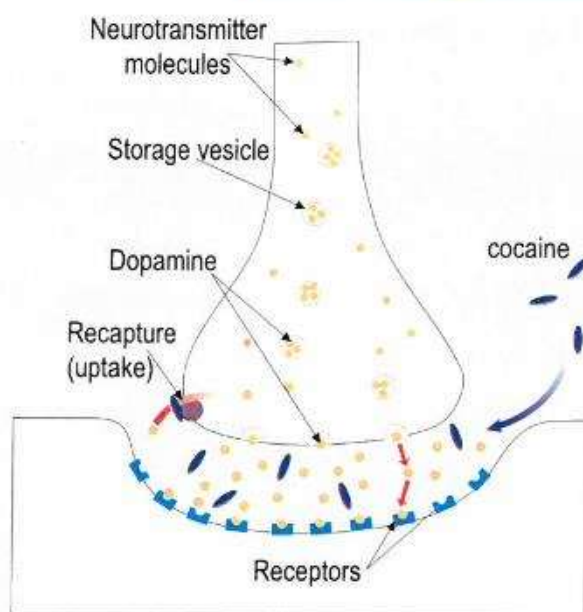


Fig. II. 71 The cocaine trap . Cocaine blocks the normal process of recapture of dopamine by the transmitting neuron. Dopamine remains trapped inside the synapse and thus its activity is more intense. Like almost all drugs, cocaine acts not only on dopamine but on several other neuro-modulators.

Drug addiction and youth

According to a 1999 survey some 14 million teens in the USA, age 12 to 17 are at moderate to high risk of substance abuse, this means they have friends who use marijuana, friends who drink regularly, have a classmate or a friend who uses LSD, cocaine or heroin. More than 4 million kids age 12-17 are current smokers. Three thousand kids start to smoke every day and more than 5 million children under the age of 18 today, will eventually die from smoking-related disease, unless current rates are reversed. According to the U.S. Centers for Disease Control and Prevention, starting to smoke when you are a teen means trouble in the future. Early smok-

ing can trigger genetic changes in your body that put young smokers at higher risk for cancer even if they quit later on.

After marijuana, high school teens cite LSD (23%) as the most popular drug and cocaine (10%) the second most popular. Middle school students rank cocaine higher (11%) compared with 7% for LSD. Most teens agree that marijuana leads to use of harder drugs like cocaine, heroin and LSD.

"Make your own decisions -- that's an important part of becoming an adult. It's hard enough to make choices when you're healthy -- why make it harder by drinking or using drugs?"

Exploitation of the document

1. Morphine is used in medicine, to reduce pain. Justify the use of this substance in the case of intense pain. What are the dangers of using morphine without medical supervision?
2. Comment on the text above.

a change of behavior and dependence involving a state of tolerance. Chronic alcoholism can involve degeneration of the dopamine secreting neurons. To preserve a normal behavior, an alcoholic needs a certain amount of alcohol in the blood: there can be physiological dependence on alcohol. This dependence can also become psychological.

Need for treatment:

The number of drug users in the world has been decreasing since 1979. Despite this the number of drug-related deaths remains high, therefore there is a need to apply an efficient drug treatment. The United Nations and the World Health Organization view treatment as rehabilitation. For them, rehabilitation programs must include several governmental and non-governmental sectors. Health and education sectors as well as law enforcement agencies and others must all participate in the development of such programs.

Physical rehabilitation

Is it enough to free a person from his drug dependency?

This is not enough to eliminate the problem. A treatment may abolish the physical dependency, but it does not cure the behavioral dependency that is behind this condition. One of the most effective drug therapeutic strategies is to avoid the situations that led to the consumption of drugs. It is not easy to be freed immedi-

ately from drug dependency; however one can exercise some form of rigorous abstinence. There are now substances called " substitution products ", able to limit the subjective effects of drugs. These substances are prescribed in addition to a psychological treatment to help drug addicts regain self-control.

Curing drug dependency does not imply only treating the physical syndrome of dependency, but especially eliminating the psychic dependency definitively, which no drug to date has been able to achieve. In Lebanon, there are several public and private centers like the Anti-drug Youth " in Jbeil, that offer help young people with drug related problems

Risk of relapse:

The psychic dependency results in a high number of relapses, whatever the product consumed; 80% of the rehabilitated heroin users relapse at the end of a few years; 85% of the smokers who stopped smoking go back to their habit after a while and 70% of the rehabilitated alcoholics go back to drinking



EVALUATION EXERCISES

A- Choose the right statement(s) and correct the false one(s)

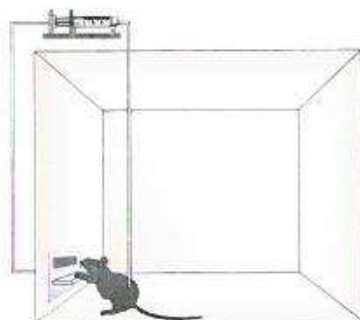
- 1- The use of drugs is based on a true market with very powerful producers, consumers and intermediaries.
- 2- Drugs are derived by the modification of natural or industrial products that have a stimulating effect.
- 3- All drugs act on the transmission of the nervous impulse.
- 4- Tranquilizing drugs or barbiturates are not dangerous.
- 5- Tobacco and alcohol are not drugs.
- 6- The use of drugs from time to time does not lead to dependency.
- 7- Drugs may provoke death by overdose.

B- Think about it

- 1- A drugged rat !

In an experiment, a tube connects the heart of a rat to a syringe. A lever enables the rat to absorb the content of the syringe.

Experiment No.1: When the syringe contains cocaine, the rat quickly learns how to inject itself with the drug by pressing on the lever. This is followed by an increase in its motor activity that is a form of intoxication. The animal gradually increases the amounts injected by pressing more often on the lever. When the apparatus is disconnected, the rat shows several signs of disorder and presses the lever more frequently. The disorders disappear if the auto-injections of drug are allowed again.



Experiment No. 2: When the syringe contains alcohol, the auto-administration does not take place. An

animal previously trained to press on the lever refuses to inject itself with more than 1 to 2 mg each time it presses on the lever. But if one

directly injects amounts of alcohol that exceed the threshold of toxic manifestations, the pressures exerted on the lever increase by 20 to 30 fold.

Experiment No. 3: One group of rats is provided with syringes containing heroin, and a second group is provided with syringes containing cocaine. The rats can auto-inject the drug at will. After 15 days, 10% of the first group and 70% of the second died. At the end of one month, mortality reached 36% in the first group and 90% in the second, and individuals in the latter group became completely anorexic.

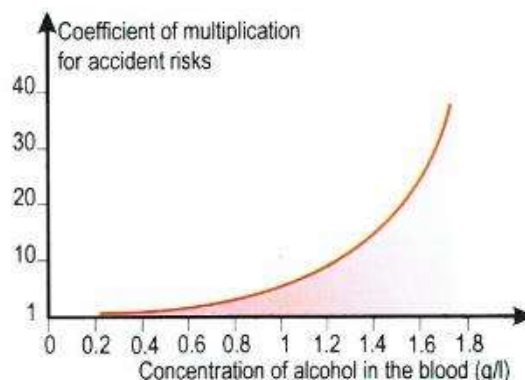
Analyze the experiments above and draw the adequate conclusions.

- 2- In an attempt to show the effect of alcohol consumption on the speed of nervous transmission, the quickness of a driver's reflex to stop his car when faced with an obstacle is tested.

The results are shown in the table below:

Distance covered before stoppage

Speed	Alcohol content in the driver's blood	
	0g/l	0.8g/l
60Km/h	35m	43m
80Km/h	57m	68m
100Km/h	85m	99m
120Km/h	116m	132m
140Km/h	150m	170m



Did alcohol consumption influence the speed of the reflexes in the motorists?

What are some of the non-desirable effects on driving that originate from alcohol consumption?

Does alcohol consumption have an effect on the frequency of car accidents?

Essential points

Drugs and Drug Addiction

- Drugs generally act by disturbing nervous communication in the organism.
- Drugs induce a state of dependency, an unstoppable desire to continuously take drugs in a desperate attempt to eliminate physical and psychic discomfort.
- Drugs act specifically on the dopamine receptors in the brain, generating an effect of pleasure, which is the origin of the syndrome of dependency.
- The phenomenon of tolerance occurs following the adaptation of the organism to drug consumption.
- The regular use of drugs leads the user to gradually raise the amounts consumed in order to obtain the desired effects. The consequences are often harmful and can lead to death.
- The state of drug withdrawal underlines a physical dependency or a need for the massive presence of the drug in the body.
- The beneficial effect of certain drugs lies in their ability to block nervous transmission, which desensitizes the patient during a surgical procedure. However, they should never be used without close supervision by qualified people.

Additional Information

Caffeine

Caffeine is a chemical substance in coffee, chocolate and tea. Its absorption is very fast. It crosses the blood-brain barrier quickly and its effect on the central nervous system is at its maximum one to two hours after absorption. It stimulates the brain, the cardiovascular system and the respiratory centers. It increases heart beat and stimulates the secretion of acid into the stomach. Caffeine is a stimulating factor, which leads to dependency that can start by a daily consumption of 4 cups (the equivalent from 400 to 500 mg of caffeine).

Tobacco and nicotine

The nicotine in tobacco stimulates directly the production of dopamine. It results in a surge of dopamine in the brain to

which the smoker becomes a slave. Regular use of nicotine requires only a little time. Quitting smoking becomes harder with the increase of the duration of the smoking period. In fact, "smoking" is not simply a practice, but a true dependency on nicotine. On average, a cigarette contains 10 mg of nicotine, of which 1 to 2 mg go to the lungs. Smoke crosses the blood-brain barrier easily. Nicotine is metabolized in the liver, but after regular consumption it accumulates in the body. Michel Moal said: "to smoke 2 packets of cigarettes per day without managing to stop is a disease"

Alcohol

Alcohol increases the secretion of dopamine. Excessive alcohol consumption produces a temporary intoxication, an emotional state characterized by

The Essential Points

Biological rhythms

- We find periodicity in all kinds of living activity.
- There are very short biological rhythms such as the heart beat, and long ones such as seasonal or annual rhythms, but the most important ones are the circadian rhythms (24h.)
- The waking-sleeping rhythm is the most studied rhythm.
- Sleep is divided into two principal phases which repeat themselves during the night, slow sleep and paradoxical sleep.
- Slow sleep has a repairing function; paradoxical sleep corresponds to an integration of new information.
- The biological rhythms are endogenous in nature. There are many synchronizers or internal "clocks", the principal one located in the hypothalamus and influenced by one or many environmental synchronizers.
- The most important external synchronizer is rhythm of day and night.
- The biological rhythms are an additional component of the regulating systems of the organism and are classified under the frame of the predictive adaptations that allow our organism to face the environmental variations, such as the alternation of light and darkness or the seasonal variations...
- The application of chronobiology is useful in fields of human activity: the work rhythms, sleep rhythms, the time at which medicine should be taken, etc. One can imagine the benefit to be obtained from more self-knowledge.

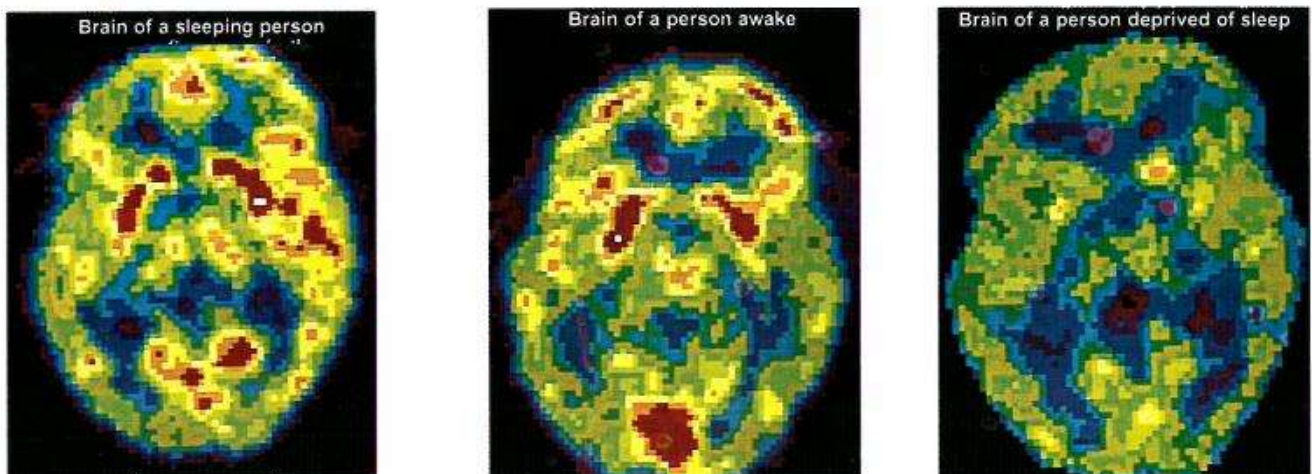


Fig. II. 82 Cerebral activity is at its lowest levels during sleep deprivation. Active zone in red. Resting zone in blue

Document 1 *Life, a series of rhythmical variations*

The examination of every living system shows that its activity is not constant. There are phases of high and low activity that occur regularly and in an alternating manner. Where, when and how do these phenomena occur?

1- A varied periodicity

Elementary rhythmical activities

The rhythmical phenomena constitute a mode of normal activity of each excitable cell. So, the heart beats, the breathing movements are controlled by cells which depolarize and repolarize in a cyclic manner. This activity may occur spontaneously and persist when the system is isolated, but placed in favorable conditions.

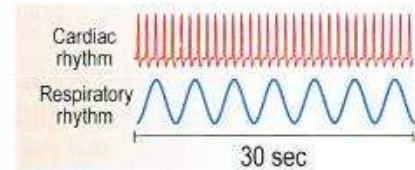


Fig. II. 73 At the very basis of our life, ultradian or high frequency rhythms.

Circadian Rhythms

Circadian or medium frequency rhythms, whose period lasts about 24 hours, are the most studied.

Seasonal Cycles

Certain physiological parameters follow seasonal or annual rhythms, of low frequency

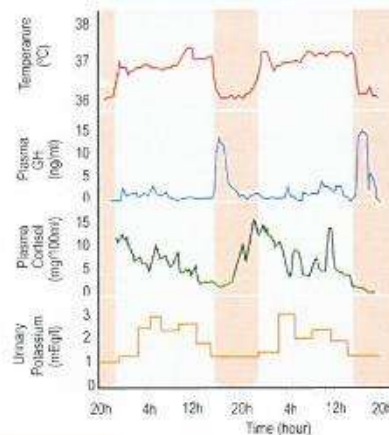


Fig. II. 74 Many physiological parameters follow a circadian rhythm.

	Summer	Fall	Spring	Winter
Heart Frequency	MAX			MIN
Growth of hair	MAX			MIN
Blood pH	MAX		MIN	
Steroid excretion		MIN	MAX	
Hb, Hc, blood sedimentation	MAX			MIN

Fig. II. 75 Some examples of low frequency rhythms.

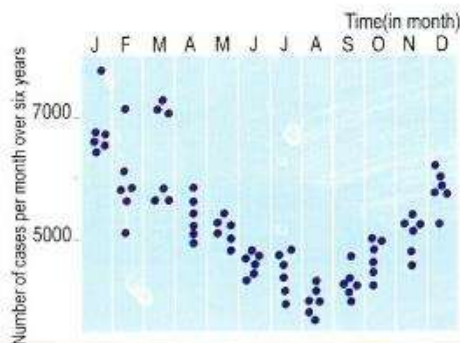


Fig. II. 76 A cyclic modification of the vulnerability of the organism toward the environmental aggressions (death by stroke).

Particular Cycles

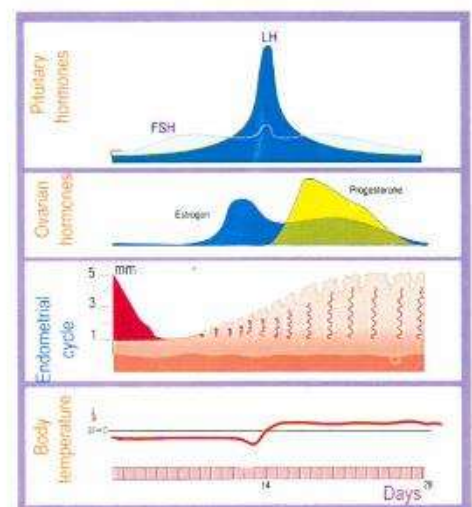


Fig. II. 77 The menstrual cycle.

Document 2 Endogenous or Exogenous rhythms

For a long time, we thought that biological rhythms were activated by an external signal. Now, we talk about endogenous rhythm, more commonly known as the "biological clock." What evidence can we show for the presence of an endogenous rhythm and what is the importance of these biological clocks in our life?

1- "Free running" rhythms

In subjects placed for several weeks in a dark room, where they can control the light themselves, the sleep-wake cycle persists but lasts for 25 hours. Such a rhythm is called "free running" rhythm. If several persons are subjected to the same experience each one alone, then each individual will have his own "loose end" rhythm. A time lag of more or less two hours is observed. But in a group, the dominant person will impose his rhythm to the others.

The light – darkness signal is the most important environment signal, but it is not the only one. The

ambient temperature, the time of the meals and other social habits constitute environmental signals.

For each function there are several endogenous rhythms. If they are coordinated and synchronized, they determine a global rhythm for a certain function or activity. The regular variations of the ambient environment – the day/night alternation for instance – intervene to coordinate and synchronize between them the phases of the different endogenous rhythms.

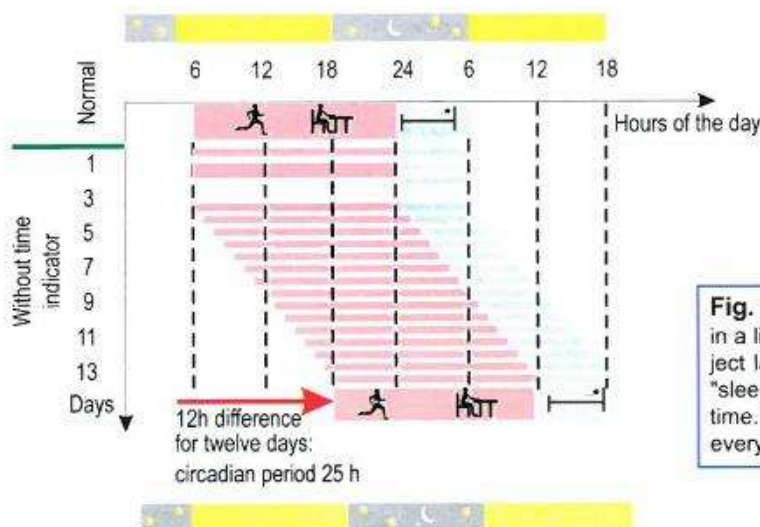


Fig. II. 81 Sleeping periods of a man kept permanently in a lighted environment. The sleeping periods of the subject lag progressively everyday behind the real time: the "sleep oscillator" turns a little bit more slowly than external time. In normal life, the internal clock synchronizes itself everyday with the external clock.



Fig. II. 82 A possible model of the internal clock organization explaining the circadian rhythms. The multicellular organisms have a set of circadian oscillators, coordinated with each other and connected to the external world by a principal oscillator located in the hypothalamus.

2- Application of the chronobiology

The rhythms of work

In all countries, the work and rest schedule is the most important synchronizer imposed by a worldwide system. It coincides with the night/day natural synchronizer. However, during nocturnal work and diurnal sleep, the working sleeping rhythm is reversed; after several days there is an inversion of the other circadian cycles of the organism, a fairly satisfactory adaptation of the subjects to the modification of their hours of activity. Certain subjects do not tolerate this inversion. What the organism tolerates badly are the changes imposed by the rotation of the work schedule, such as the 8-hour shifts. A change every 2 or 3 days is best tolerated.



Fig. II. 83 In countries, where the nights are very short indeed for several months and then very long for several months, the people adjust their internal rhythm to their working schedule.



Fig. II. 84 Nurses and air pilots are the most affected by time lag.

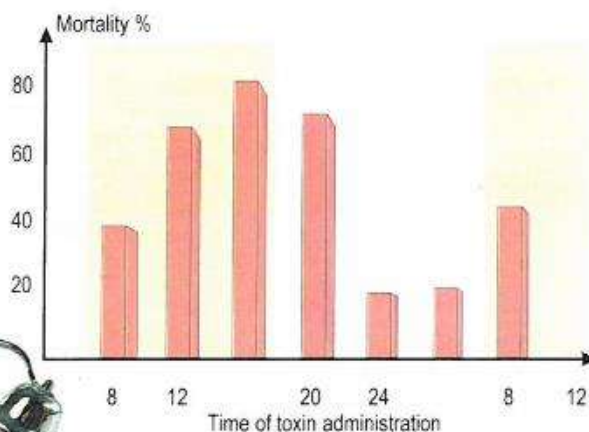


Fig. II. 85 A substance injected into two identical groups of mice, at the same dose, but at different hours.

The chronopharmacology

The knowledge of circadian rhythms is very useful and must be taken into consideration

During the interpretation of a clinical exam (temperature, pulse, arterial pressure) or of the biological exam (glucose, urea)

In choosing a time at which a medicine should be taken. Asthma crises mostly occur at night. Asthma pills should be taken at night in order to give the maximum benefit.

Exploitation of the document

1. What is the relation between the internal and the external synchronizer?
2. Why is a rotation of 2 to 3 days tolerated better than a weekly rotation?
3. Why has a medicine a different toxicity depending on the time of administration?

2. The waking – sleeping rhythm

There are deep physiological modifications between the waking state and the sleeping state affecting the organism's metabolism and all functions of the organs, but it is the activities of the nervous centers which have particularly retained the attention of the researchers.

The sleeping cycle is divided into 2 principal phases during which the electrical activity of the brain is modified.

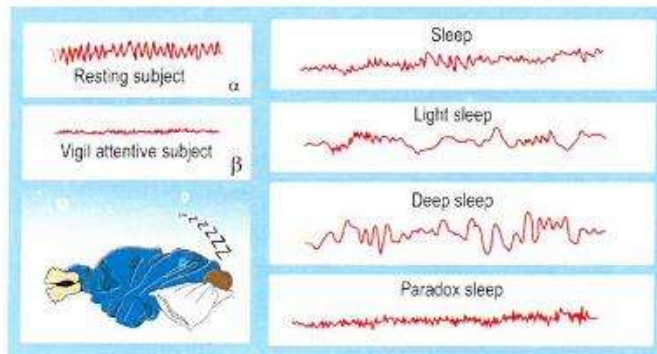


Fig. II. 78 Recordings of electroencephalograms during sleep. The waves (α) are characteristic of inattentive waking, and the waves (β) of the attentive waking or vigilance with a smaller amplitude but with a higher frequency.

SS: A brain which functions in slow motion in a mobile body.

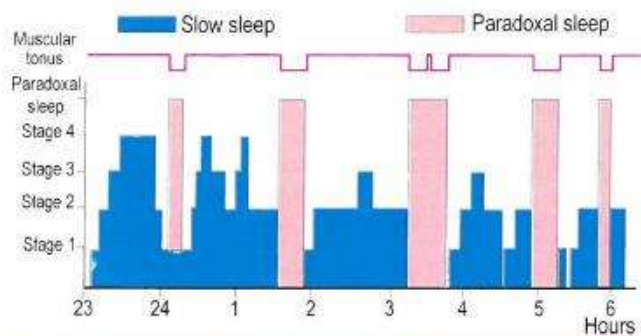


Fig. II. 79 The night's sleep of an adult (Hypnogram). Every night is made up of 4 or 5 cycles of 1 to 2 hours, each ended by 15 to 20 min. of paradoxical sleep. At the age of 60, a man would have passed 20 years of his life sleeping.

The slow sleep (SS)

It counts 4 stages:

I- Falling asleep (15 to 20 min. the sleeper still hears and feels when being touched, and he can be easily woken up).

II- Light sleep.

III & IV- Deep sleep

The paradoxical sleep (PS)

The activity of the brain is rapid and very similar to that of the brain when one is awake; the eyes have rapid movements, but the muscle tone completely disappears. So there is a paradox between the high brain activity and the low stage of vigilance. It is the sleep of the dream. It is thought that the paradoxical sleep is essential to the maturation of the brain. It is thought to facilitate the definitive integration of new acquisitions i.e. the phenomena of memorization.

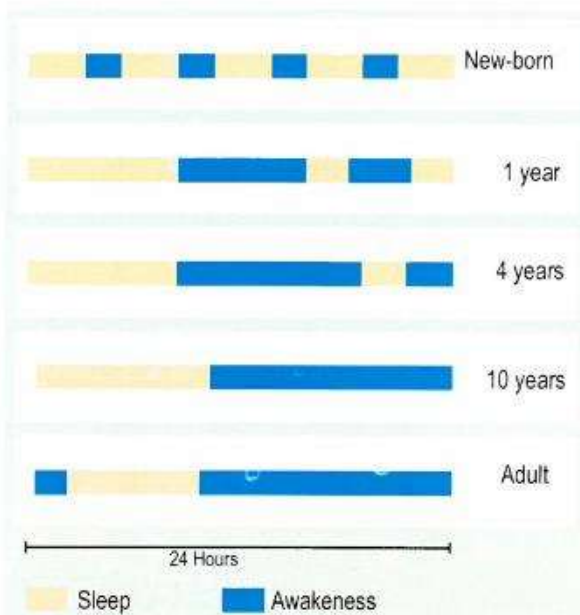


Fig. II. 80 The waking-sleeping rhythm is progressively established and evolves during lifetime

Exploitation of the document

1. Can you explain the existence of rhythms in the excitable cells? Try to imagine a physical model to represent this phenomenon.
2. Search for other examples of biological rhythms among human beings in general.
3. Can you find in this document an explanation for this quotation: We can survive longer without food than without sleep.

EVALUATION EXERCISES

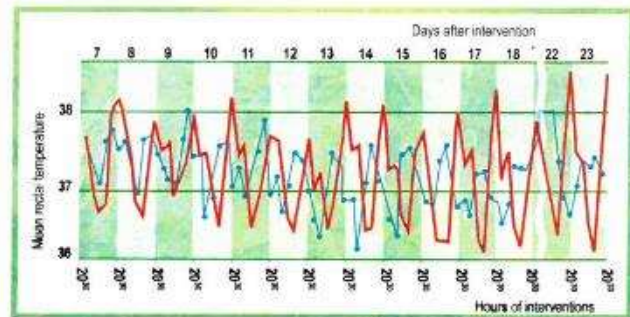
A- Choose the right statement(s) and correct the false one(s).

- 1- The circadian rhythm is a rhythm of 12 hours.
- 2- A biological rhythm has to be characterized by a periodicity always fixed for a same subject and among all subjects of the same species.
- 3- Biological rhythms have an endogenous origin.
- 4- Deep sleep is called paradoxical sleep.
- 5- The sleep-wake cycle is not hereditary.
- 6- The jet lag is a desynchronization of the endogenous rhythm.
- 7- There are several internal synchronized clocks.

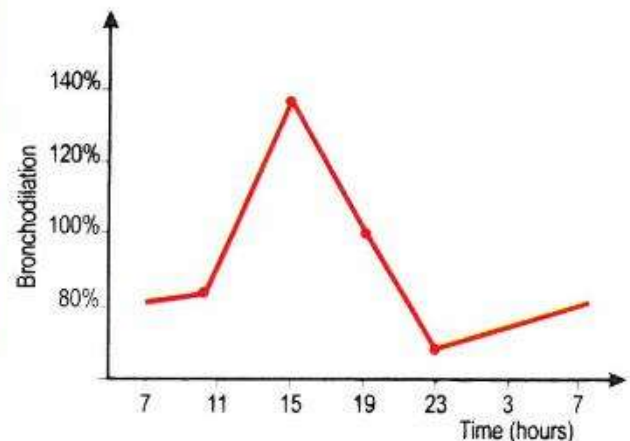
B- Think about

- 1- A national football team has to compete in a football game in the United States. What do you advise them to do so that the 7-hour jet lag with Lebanon will not be a handicap?

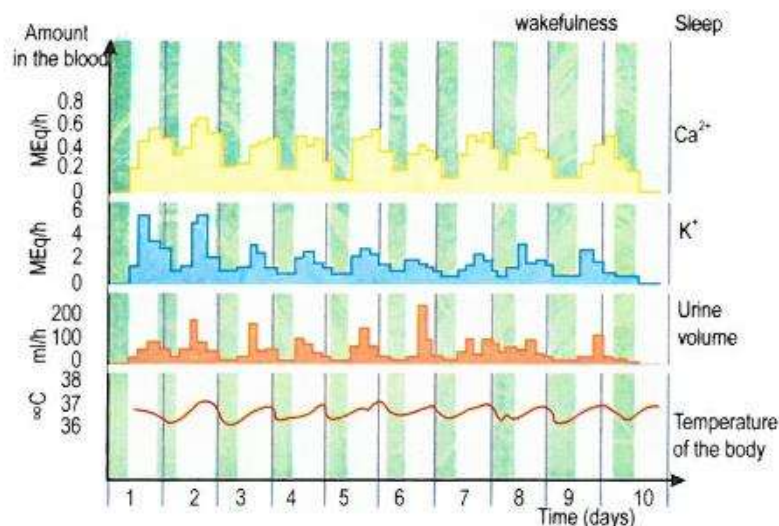
- 2- Comment on the two graphs of the rectal temperature of control mice (red) and those without ocular globes (blue).

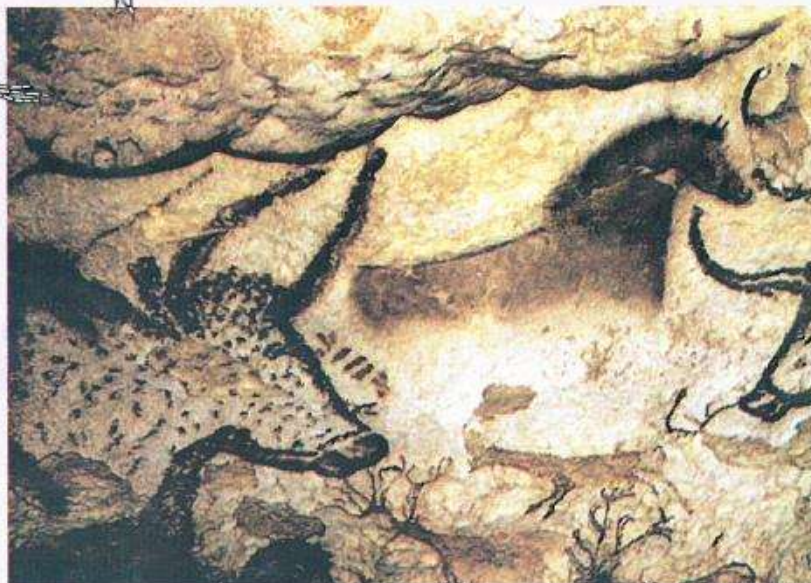
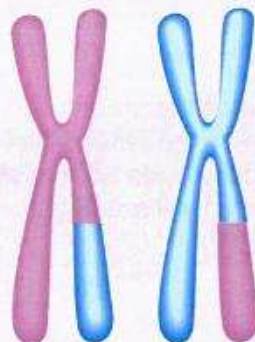
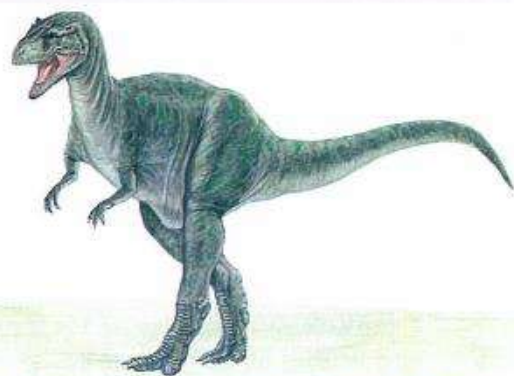


- 3- What do you think about the effect of acetylcholine on the diameter of the bronchi?



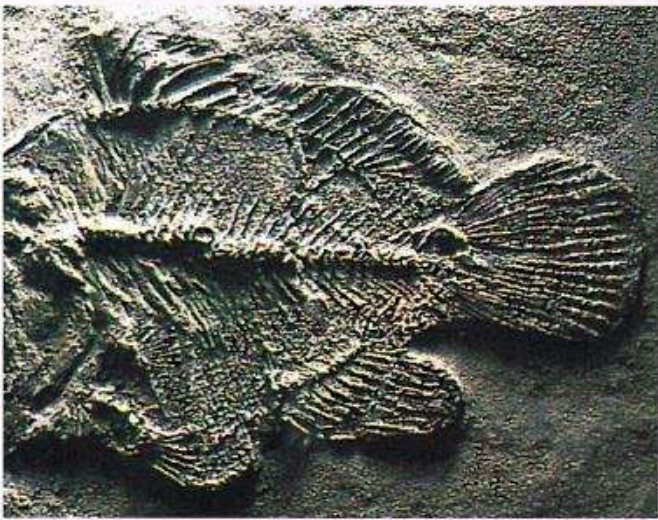
- 4- What information do we obtain from these graphs?





Unit III

Theories of Evolution



When we look at the world today, it is easy to observe a diversity of individuals going from the simple (unicellular) to the more complex (mammals).

Throughout time certain species have appeared while others have disappeared, provoking changes in populations. Despite these changes, all of the surviving species have retained some common features.

With better understanding of the morphological, and especially the genetical data that we possess today about different species and with the help of the fossil records, it is possible to establish parental links between them.

How do biologists explain a theory of evolution that opposes fixism which states that all the present species coexisted from the beginning of time?

In this unit we shall answer the following questions:

- 1 What are the proposed theories of evolution?
- 2 What is the evidence for evolution?
- 3 What is the current theory of evolution?
- 4 Where do humans fit in the evolutionary process?

Darwin concluded that those individuals that possess superior physical, behavioral, or other attributes are more likely to survive than those individuals who do not possess such attributes. By surviving, they have the opportunity to pass on their favorable characteristics to their offspring.

Because those individuals with these favorable characteristics will soon be more numerous than those without them, the population as a

whole will gradually change. Darwin called this process **natural selection**.

According to Darwin's theory of natural selection, changes in individuals occur continually within populations. A small percentage of these changes may cause the individual who has the change to produce more offspring than others. These individuals will increase in number at the expense of their less productive competitors. This process of selection

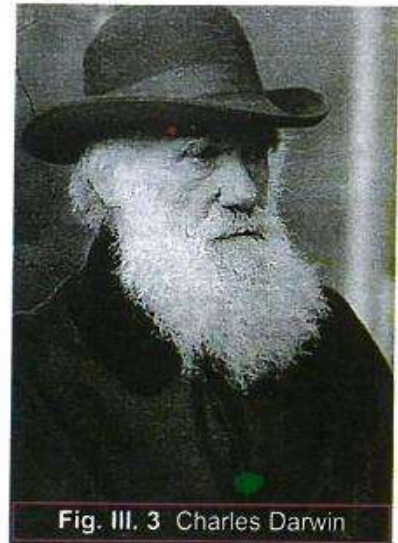


Fig. III. 3 Charles Darwin

will lead to the modification of species over time.

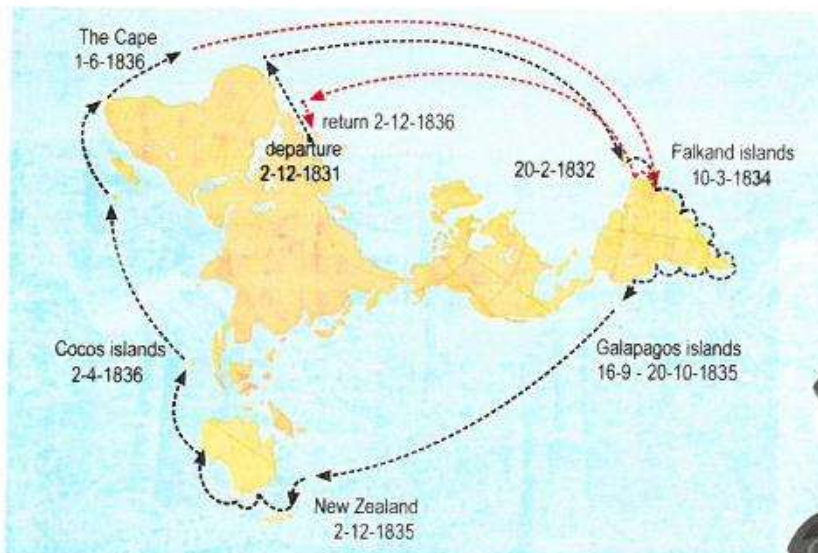


Fig. III. 4 Darwin's Finches: The resemblance between the various species of finches encountered on the Galapagos Islands suggests that they may have all evolved from a common ancestor. This ancestor may have come to the islands from the adjacent mainland of South America. Because of the lack of competition, it spread and occupied most habitats on the islands. Spreading from one habitat to another, the individual birds were subjected to diverse sets of selective pressures, which brought about their diversification.



Exploitation of the document

1. How does adaptation explain evolution according to Darwin?
2. What is the fundamental difference between Lamarck's theory of evolution and that of Darwin? How did Darwin explain the example of the giraffe?
3. How can we explain the variation of individuals by natural selection?

Document 1

The Development of Evolutionary Theory

A species is defined essentially by the resemblance between its members and their interbreeding criteria. Populations that are originally identical eventually become more and more dissimilar because of adaptive changes. Conditions under which populations exist usually vary significantly from one population to another. When selective forces are strong, for example when climate conditions are changing rapidly, either the population rapidly adapts or it may face extinction. When conditions are stable, populations may remain constant for long periods.

What are the principles behind the development of many theories of evolution?

1- The theory of Lamarck

Lamarck published his theory of evolution in 1809. He thought that when a species arises, it is initially very primitive, but it increases in complexity over time due to some internal tendency which is affected by the surrounding conditions.

The environment leads to a change

According to Lamarck, organisms are not passively changed by their environment. Instead, a change in the environment causes changes in the needs and behavior of organisms living in that environment.

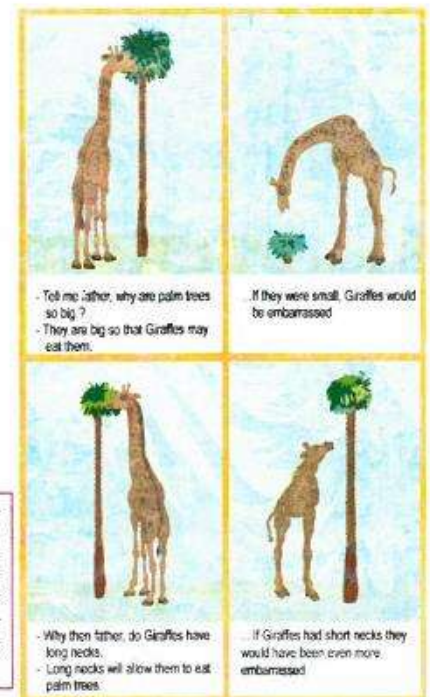
Changed behavior leads to greater or lesser use of a given structure or organ; usage would cause the structure to increase in size over several generations, whereas disuse would cause it to shrink or even disappear. For example, an animal which lives in the dark (mole) has reduced eyes, and a mammal which swallows without chewing has very small teeth (whale).

The inheritance of changes:

Lamarck also proposed that

an organism's adaptation to the environment could be passed on to its offspring. For example, he thought early giraffes stretched their necks to reach higher branches. This caused their offspring to be born with longer necks. Lamarck concluded that evolution is the continuous gradual change of a species of organism caused by its adaptation to its environment.

Fig. III. 2 "Major changes in circumstances lead to major changes in animal needs, the changes in needs necessitated changes in actions. Hence, if new needs became constant or very long lasting, animals take on new habits that are as lasting as the needs that provoked them".



2- The Theory of Darwin

During his long journey on the ship Beagle Navy (1831-1836) around the coasts of South America, Darwin observed fossils of extinct animals that were related directly to the animals that still lived in the same area.

The pattern of distribution and the relationship of organisms that Darwin observed on his voyage made him certain that a process of evolution had been responsible for these patterns.

In 1859 he published his famous book: *On The Origin of Species by Means of Natural Selection*. The ideas expressed in his book profoundly influence the way we understand the living world today.

Darwin saw that although every organism has the potential to produce more offspring than are able to survive, only a limited number actually survive to reproduce.

Document 3 Macroevolution

Paleontology, the study of fossils, shows that progressive changes underlay the evolution of each species. Molecular biology has demonstrated that those changes were due to spontaneous mutations provoking the replacement of an existing gene by a newer one. A favorable natural selection allows the predominance of these newer genes.

1- The fossil record

The most direct evidence of macroevolution is to be found in the fossil record; we now have a more complete understanding of this record than what was available in Darwin's time. Fossils are created when organisms become buried in sediments, the calcium in the bone and other tissue is mineralized, and eventually converted to rock. With the accurate dating of rocks in which fossils occur now available, we can get a very precise idea on the age of fossils.

The fossils contained in sedimentary layers of rock reveal a history of life on earth.

Fossils presented according to their age, from ancient to recent, often provide evidence of progressive evolutionary change going from simple to complex. In the evolution of horses for example, the number of toes on the front foot is gradually reduced from 4 to 1.

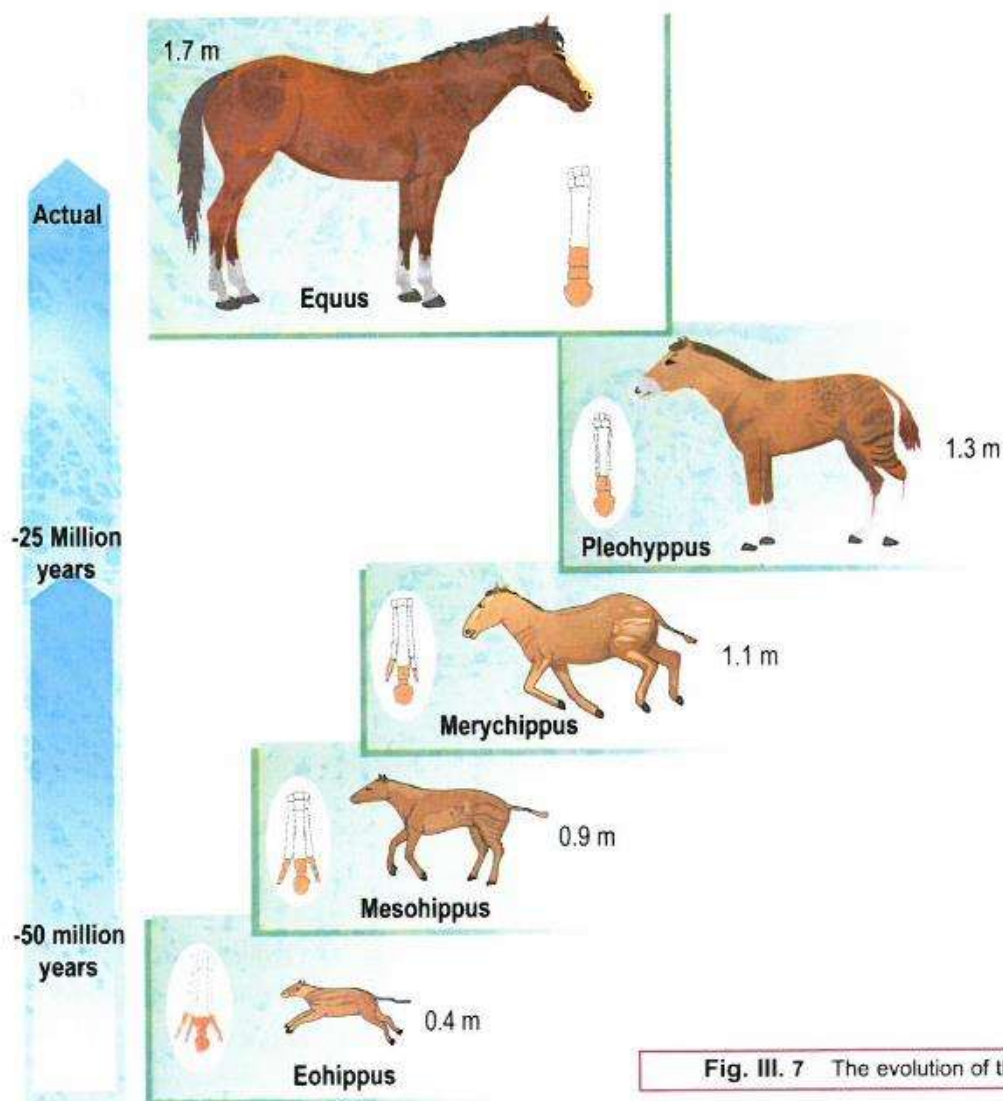


Fig. III. 7 The evolution of the horse

2. Sickle-cell Anemia

Sickle-cell anemia is a hereditary disease affecting the shape of hemoglobin in the blood, the main constituent of red blood cells.

The disease, usually severe, occurs because of a single change in the hemoglobin gene.

Individuals who are heterozygous, i.e. possess one copy of the normal gene (N), and one copy of the sickle gene (S) are said to possess the sickle-cell trait and have mainly normal red blood cells. The frequency of the sickle gene is very high in Central Africa. Individuals who are homozygous for the sickle cell gene (SS) almost never reproduce.

Why is the S gene not eliminated from the Central African population by selection?

Individuals who are heterozygous for the sickle cell gene are also much less susceptible to malaria, one of the leading causes of illness and death, especially among young children. Consequently, the sickle-cell gene is maintained at high levels in these populations because of its role in resistance to malaria.

Is adaptation the major source of genetic variation that occurs in nature?

Changes that affect the genetic make-up (hereditary instructions) of an organism may have an important impact on the ability of the organism to survive and to bear offsprings and in turn on the success of these offsprings. Natural selection favors those heritable features that better suit an organism to survive.

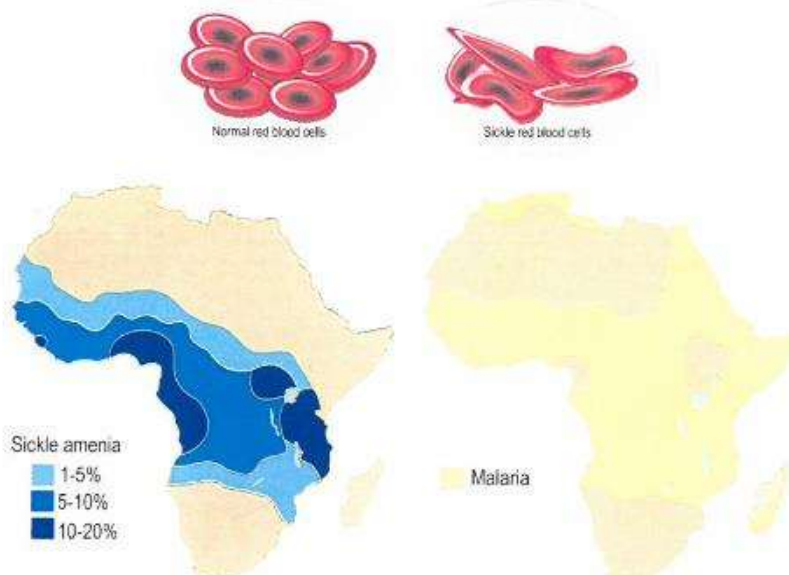


Fig. III. 6 Geographic distribution of a) Sickle cell and b) Malaria

Exploitation of the document

1. How can we explain the increase in frequency of dark moths in industrial areas?
2. Why is the sickle gene not removed from the population, through natural selection?
3. Name the factors that resulted in the selection of certain traits in the examples given above? What were the favorable traits in each example?

Document 4 *Human Origin*

The earth was formed approximately 4.5 billion years ago and *Homo sapiens* appeared some 100,000 years ago. Life and all what we see around us on this planet is the result of 3.5 billion years of evolution. What are the evolutionary changes that brought about the appearance of our species?

The evolution of our species did not occur in a vacuum, but under particular environmental and geographical conditions.

Australopithecus: appeared over 4 million years ago, most probably in Southern Africa. His jaw with massive teeth was rounded, and he was bipedal (two footed). Its brain case had a volume of 500 cm³. His feet and pelvis structure were made to support walking upright, which liberated the hands.

Homo habilis : also appeared in Africa, approximately 2 million years ago.

His larynx could have permitted the utilisation of a rudimentary language. His brain volume was 800 cm³.

Homo erectus : appeared almost 1.5 million years ago. He had long, straight legs and was an excellent walker. He was encountered in Africa, Europe and China. His brain volume was about 1000 cm³, and his skull had prominent brow ridges with a rounded jaw. The shape of his larynx suggested a capability of speech. He developed a social structure, hunted, utilized elaborate tools and cooked with fire.

Homo sapiens : appeared approximately 100,000 years ago. His tools were more sophisticated. His brain volume was about 1500 cm³. He had more complex social organization and he may have been the first human with fully modern language capabilities. He was responsible for cave paintings and other arts and accumulated the knowledge of previous generations. The species spread all over the world, cultivated and farmed, and acquired several technologies. It evolved progressively into the present human race, *Homo sapiens sapiens*.

Document 2 Microevolution

Evolution can be divided into microevolution and macroevolution. Microevolution results from changes that occur within populations of a species. Macroevolution is evolutionary change on a grand scale, a change that takes place over long periods of time, marked by major episodes of extinction, with new forms replacing old ones.

What factors are thought to bring about micro-evolutionary change?

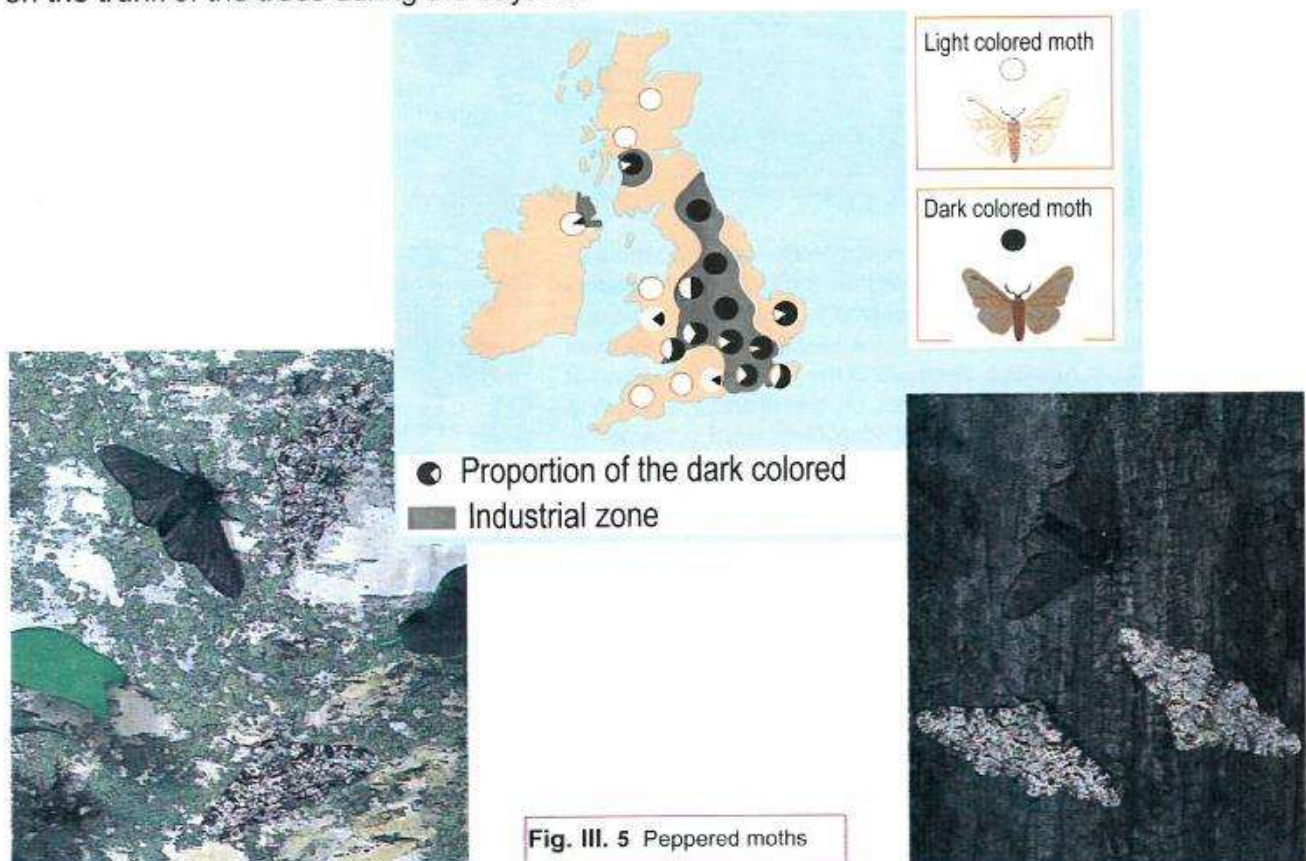
1- Peppered Moths

The English moth, *Biston betularia*, is a frequently cited example of observed evolution. In this species some individuals have light-colored wings and others dark-colored ones. Until the mid-nineteenth century, almost every individual of this species had light-colored wings. By 1898, 95% of the moths in highly industrialized areas were of the dark type. Their frequency was less in rural areas.

Can Darwin's theory explain the increase in frequency of the dark individuals?

Birds ate the moths while they were resting on the trunk of the trees during the day. The

increase in relative abundance of the dark type was due to natural selection. The late eighteenth century was the time of England's industrial revolution. Fumes from factories darkened the trees the moths landed on. Against a dark background, birds could see the lighter colored moths better and ate more of them but could not see the dark moth so easily. As a result, more dark moths survived until reproductive age and left offspring. The greater number of offspring left by dark moths is what caused their increase in frequency. This is an example of natural selection.



2- The Molecular Record: evolutionary history involves a pattern of progressive change

Today we can determine the DNA sequence of genes. Therefore, by comparing the DNA of different organisms, it is possible to determine the degree of similarity and dissimilarity between organisms and thus determine phylogenetic (i.e., evolutionary) relationships between them.

For example consider the DNA sequence of cytochrome C, a gene that is essential in the metabolism of the body. This DNA sequence varies from one species to the other. The DNA sequence is very similar in species that diverged recently. The same pattern of change is seen in several other genes such as hemoglobin or insulin etc..

1	2	3	4	5	6	7	8	9	10	11	
Gly.	Asp.	Val.	Glu.	Lys.	Gly.	Lys.	Lys.	Ileu.	Phe.	Ileu.	man
Gly.	Asp.	Val.	Glu.	Lys.	Gly.	Lys.	Lys.	Ileu.	Phe.	Val.	sheep
Gly.	Asp.	Val.	Glu.	Lys.	Gly.	Lys.	Lys.	Val.	Phe.	Val.	carp
Gly.	Asp.	Ala.	Glu.	Arg.	Gly.	Lys.	Lys.	Leu.	Phe.	Glu.	euglena

Fig. III. 8 Homologous molecule: a marker of evolution. A representation of the first 11 amino acid sequences that are coded by the DNA of cytochrome C.

3- The Modern Synthesis of Genetics and Evolution

While the Darwinian conception of natural selection was decisive in the comprehension of the global phenomenon of evolution, since 1930 a synthetic theory of evolution has been considered which encompasses

Darwinism, mutationism and population genetics.

This synthetic theory recognizes several mechanisms of evolution in addition to natural selection:

a- It states that populations contain genetic variation that arises by random (i.e., not adaptively directed) mutations. Populations evolve by changes in gene frequency brought about by natural selection.

b- It recognizes that characteristics are inherited as discrete entities called genes. Variation within a population is due to the presence of multiple alleles of one single gene.

c- It postulates that new species arise due to the gradual accumulation of small genetic changes. The Modern Synthesis is a theory about how evolution works at the level of genes, and populations.

Exploitation of the document

1. What are fossils and how do they explain evolution?
2. How can we determine the relationship between two species from their molecular record?
3. Explain the example of the peppered moth through the synthetic theory of evolution?

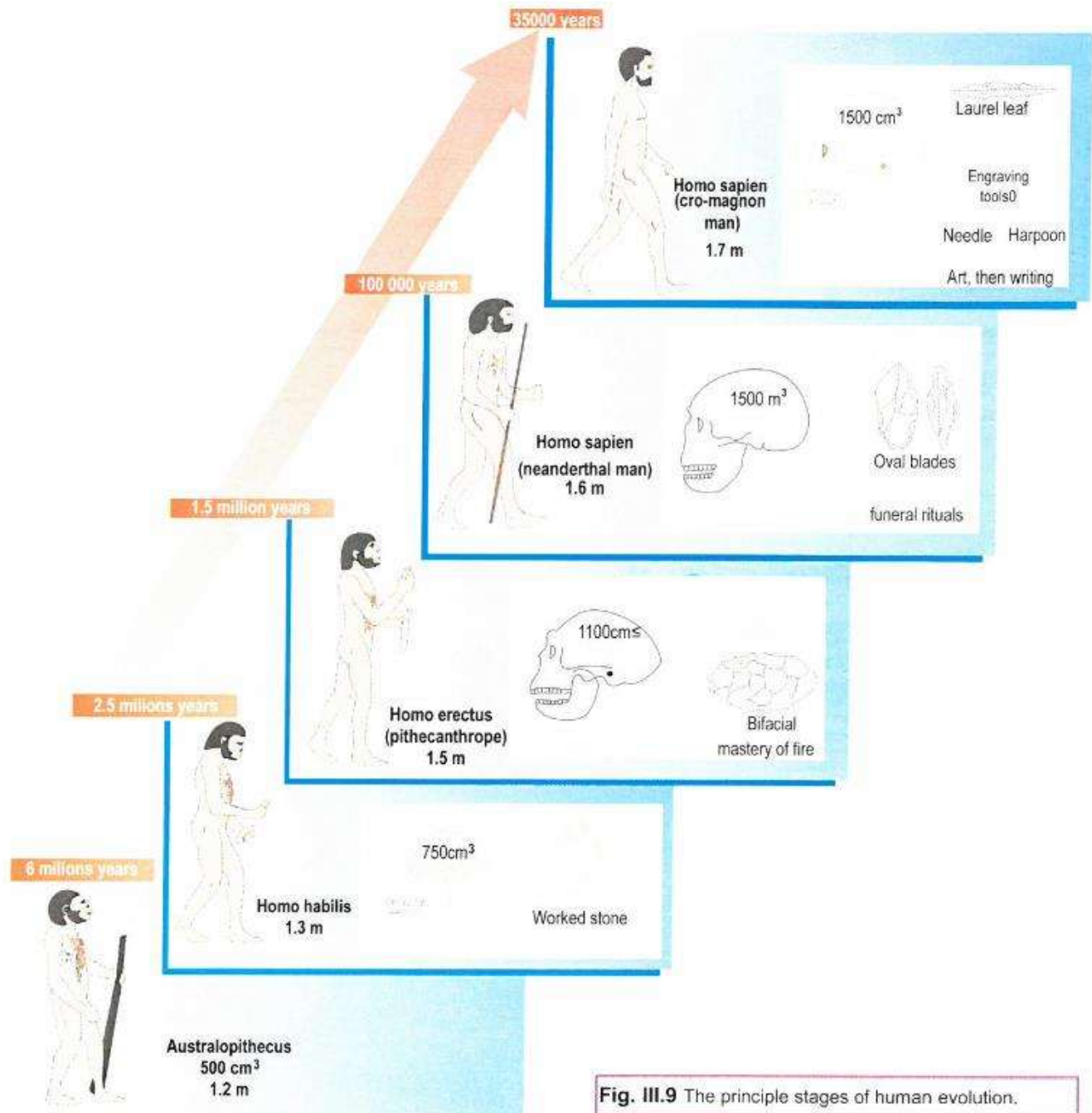


Fig. III.9 The principle stages of human evolution.

Exploitation of the document

1. What are the principle characteristics acquired throughout the human evolution?
2. How do you explain the development of tools throughout evolution?

Unit IV

SCIENCE AND ECONOMY

Biototechnology can be broadly defined as "using living organisms or their products for commercial purposes." As such, biotechnology has been practiced by human society for thousands of years in such activities as baking bread, making cheese, brewing alcoholic beverages with the help of cultured micro-organisms, selecting food crops or breeding domestic animals.

However, In the last twenty years a set of laboratory techniques have been developed and the term "biotechnology" has come to mean the use of genetic engineering, and associated techniques, in a variety of applications from medicine to agriculture.

There has been a tremendous scientific and commercial interest in biotechnology and a redirection of research efforts among established companies and universities.

Chapters

This part comprises the following chapters:

- 1 Biotechnology and immunology
- 2 Amelioration of the species, industrial farming, and biotechnology and environment study.

The Essential Points

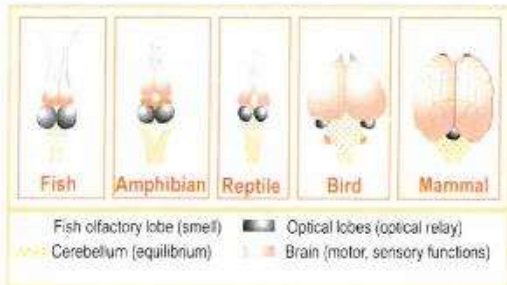
Theories of Evolution

- Lamarck proposed that environmental changes provoke a change in the behavior and characteristics of an individual. These changes are then passed on from one generation to another.
- Darwin proposed that evolution occurs as a result of natural selection. Some individuals have traits that allow them to survive better and produce more offspring in a given environment. These traits will then increase in frequency through time and generations. Different environments have different requirements, therefore different organisms exist in different places.
- Among the strong evidence that supports big changes in the course of evolution are: the fossil record, which explains a progressive change, and the molecular record, which exhibits a record of accumulated changes.
- The synthetic theory of evolution proposes that changes within populations are due to changes in gene frequencies. These evolutionary changes are brought about by natural selection.
- About 2 million years ago, the genus *Australopithecus* gave rise to humans belonging to the genus *Homo*. Our species, *Homo sapiens*, seems to have originated about 100,000 years ago.
- **Overview !**
 - Variation exists in living things via genes.
 - The role of the environment is not to produce the variations but to select the variations, i.e., the genes.
 - Genes which are of value in the environment, persist.
 - Differential reproduction occurs because of the success of favored forms, which carry the selected genes.
 - The selected genes are passed on from one generation to the other.

EVALUATION EXERCISES

A. Choose the correct answer (s)

1- The following document :



- allows us to affirm that humans descended from monkeys.
- illustrates the phenomenon of encephalization,
- allows us to find that there is a link between different vertebrates,
- explains the phenomena that intervened throughout geological periods.

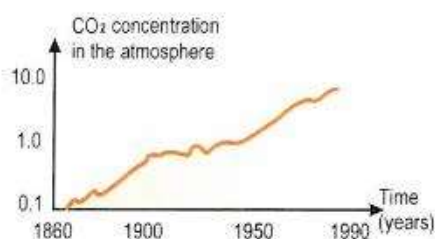
2- The appearance of new genes

- favors the diversity of a population.
- results from a modification of pre-existing genes.
- is submitted to the selective pressure of the environment.
- ensures a better adaptation to the conditions of the surrounding environment.

B. Think about it

1- Insecticides are agents that act on an enzyme intervening in mosquitoes' metabolism in order to provoke their death. Certain mutations within the mosquito may possibly modify this enzyme and render the insecticide inactive. Are these mutations provoked or selected by the insecticide? Does this phenomenon apply to antibiotics and bacterial interactions?

2- What do the following documents suggest?



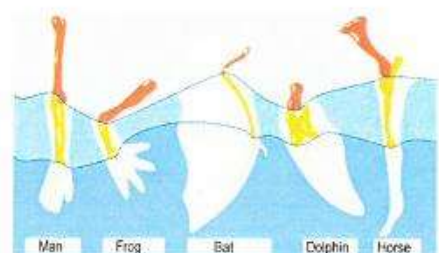
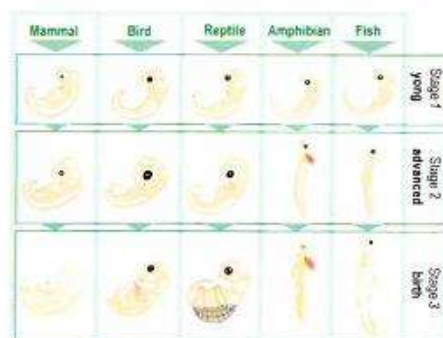
3- We observe the distribution of amino acids at the start of globin chains. The difference between any two species is indicated by the number of their differing amino acids

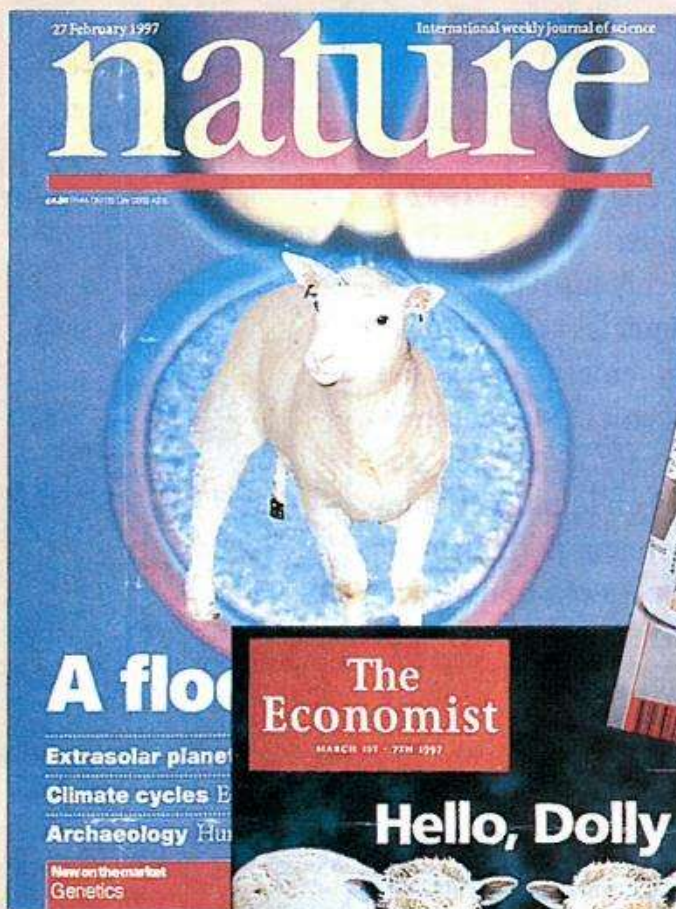
man					
gibbon	0				
macaque	0	0			
horse	2	2	0		
dog	4	4	5	0	
penguin	6	6	4	8	0
	gibbon	macaque	horse	dog	penguin

After a comparison of 25 amino acids in six vertebrates, the results indicate that

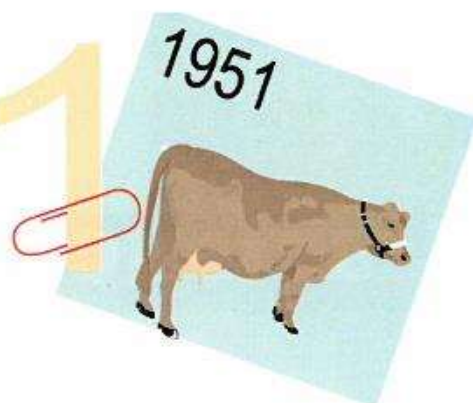
- the dog is more closely related to the gibbon than the horse.
- the horse and the dog are more closely related to the macaque than to the penguin.
- the molecular comparison permits the establishment of a genetic filiation among species.
- the penguin underwent many mutations.

3- What does the following document suggest:

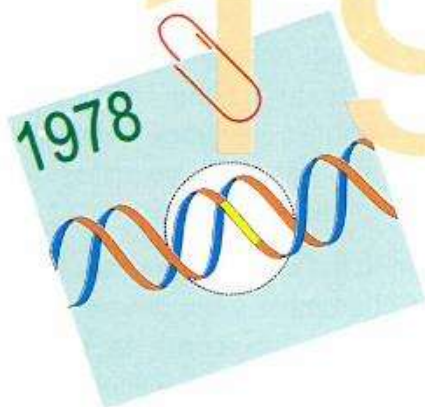




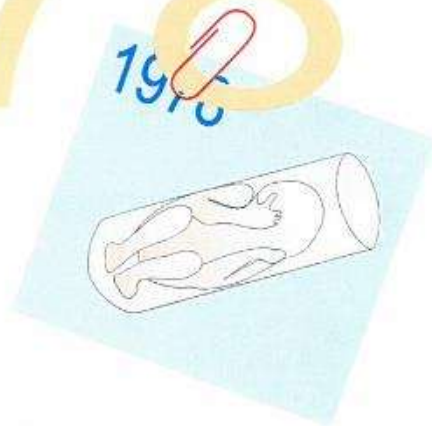
1951



1978



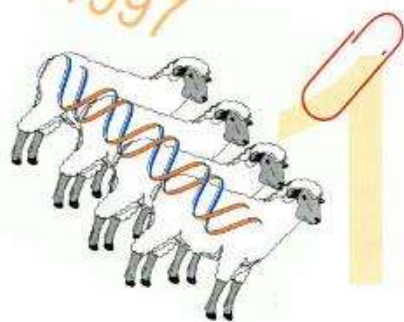
1996



1988



1997



1997

Document 1 Principles of Biotechnology

Since 1972 scientists have been able to identify the specific genes for many desirable traits and transfer only those genes into another organism. This process is called genetic engineering or recombinant DNA technology."

1- How is a gene isolated ?

The DNA fragment that contains a gene of interest is inserted into a vector or plasmid. Vectors are circular entities of double stranded DNA that can be grown inside a suitable bacterial host such as *Escherichia coli* (E.coli), where they can replicate and grow.

One approach is to cut the appropriate gene from human DNA and paste or splice it into plas-

mid DNA which will be used as a vehicle. The "scissors" used to cut the DNA are a class of enzymes called restriction enzymes. There are well over a hundred restriction enzymes, each cutting in a very precise way a specific base sequence of the DNA molecule. With these scissors, the segment of the human DNA molecule that specifies the gene can be cut and isolated.

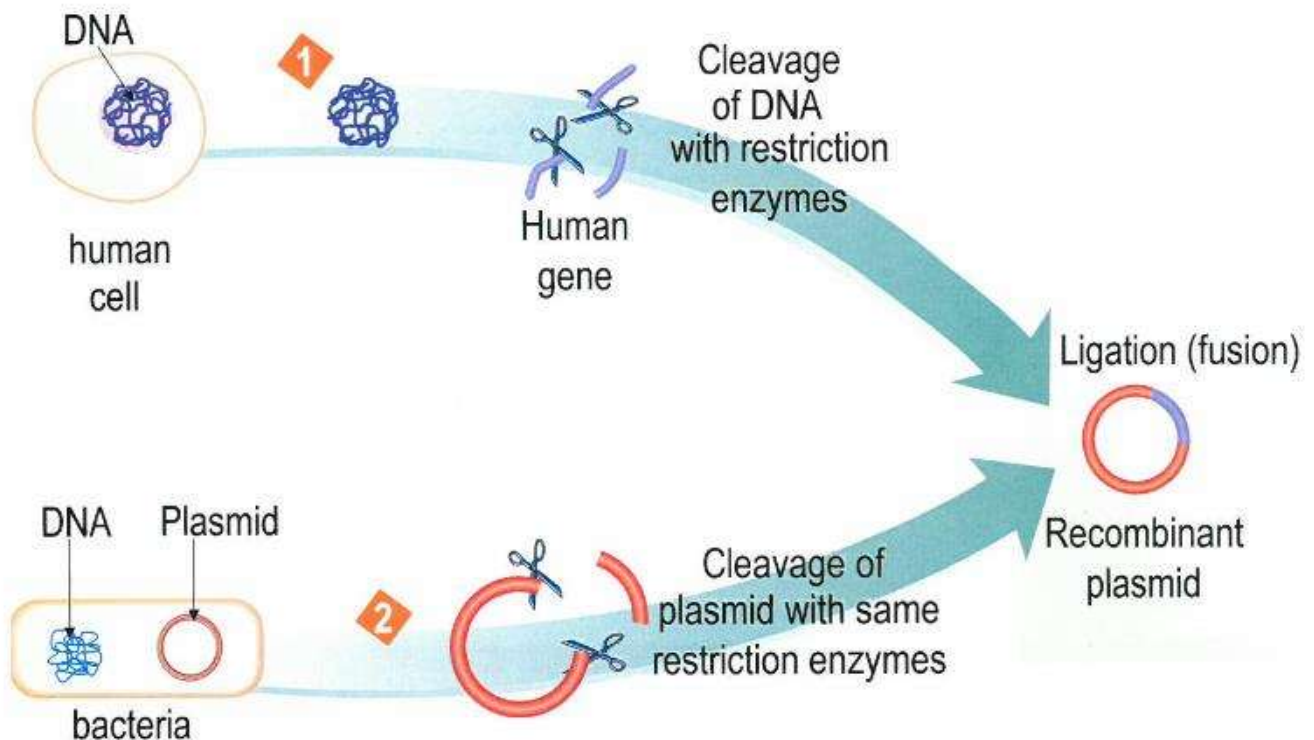


Fig. IV. 3 The insertion of a gene into a plasmid

Chapter 1

Biotechnology and immunology

All living organisms are composed of cells that contain DNA (Deoxyribo Nucleic Acid) and the characteristics of any living thing are essentially determined by the information in their DNA. Except for the sequence and number of letters in each genetic make-up, DNA from any organism is chemically and physically the same.

One of the great scientific discoveries of biotechnology is that DNA from any organism will function if it is transferred into any other organism!

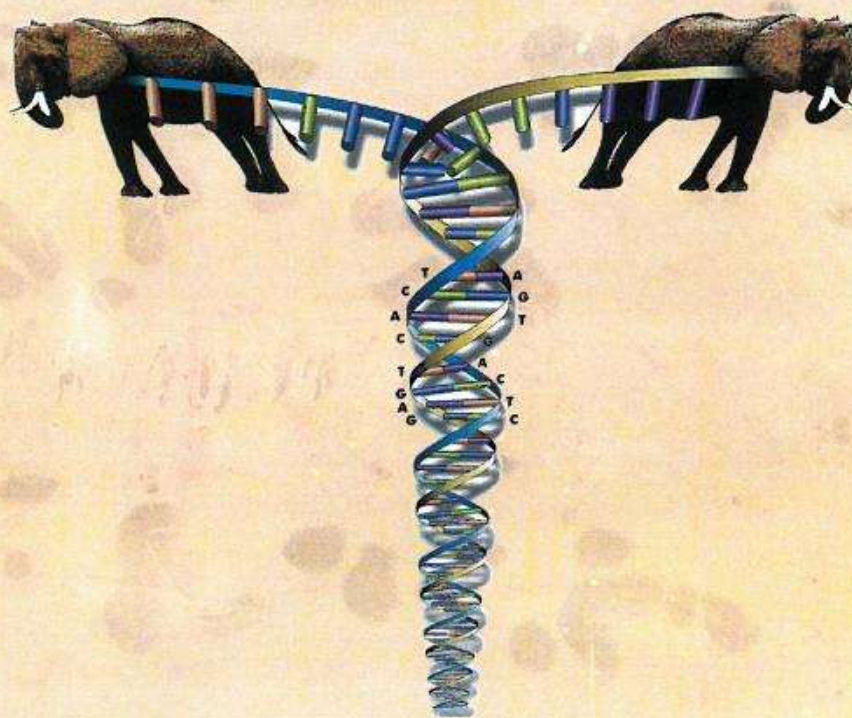


Fig.IV.2 The great impact of biotechnology on all aspects of medicine

In this chapter we shall answer the following questions :

- 🔍 What are the principles of biotechnology ?
- 🔍 What impact does biotechnology have on medicine ?
- 🔍 What are recombinant vaccines ?
- 🔍 How are monoclonal antibodies produced ?

2- How to clone a gene ?

The DNA segment is "glued" into place using an enzyme called DNA ligase. The result is a recombinant DNA (rDNA) molecule. The recombinant plasmid DNA is inserted into *E. coli* and the bacterial cell is then able to process the instructions of the recombinant plasmid and assemble the amino acids for protein production. This phenomenon is called transformation. More importantly, the new instructions are passed along to the next generation of *E. coli* cells in the process known as gene cloning.

Escherichia coli (*E.coli*) : this is a very versatile bacterium that is used in many biotechnological processes. Its genetics are the best known of any organism, with the majority of its genes known. It is by far the most common host cell for recombinant DNA work. It is also used in fermentation to make many amino acids and other products. It grows fast and can be manipulated genetically to accumulate many different proteins.

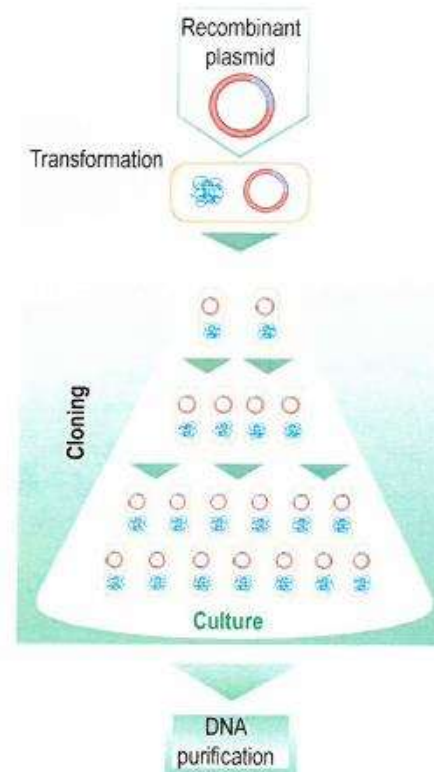


Fig. IV. 4 Cloning a gene.

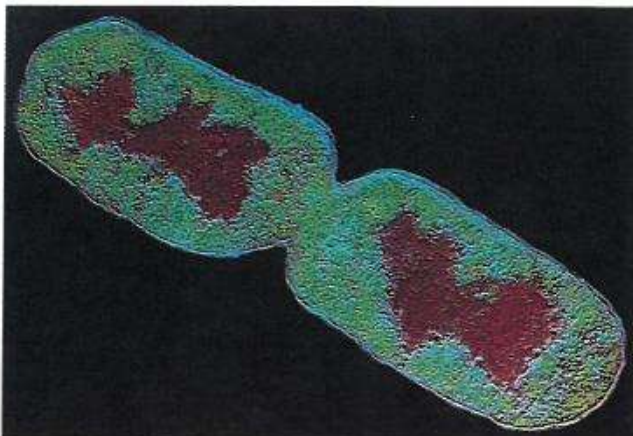


Fig. II. 5 A dividing *E.coli* is shown with its single chromosome. The powerful tool for biotechnology

This highly simplified description of recombinant DNA technology does not fully convey the enormous complexity and awesome economy and efficiency of genetic processes. But we can begin to understand how, by using recombinant DNA, it is possible to produce substances of medical and economic value.

Exploitation of the document

1. Why are restriction enzymes so important to gene cloning ?
2. What features of *E.coli* make it suitable for cloning ?
3. List and briefly describe the steps that are involved in gene cloning.

Document 2 Recombinant DNA: therapeutic drugs

Specific applications of genetic engineering are abundant and increasing rapidly in number. Genetic engineering is being currently used in the production of pharmaceuticals, in gene therapy and in many other fields.

Perhaps the most widely recognized application of all the recent advances in molecular medicine has been the use of biotechnology to make proteins.

1- Protein production

Once a gene that specifies a particular protein has been cloned it can be inserted into a suitable host which then produces the protein, which is the drug in many of these cases.

In the pharmaceutical industry, drugs normally produced by the human organism, such as insulin for diabetics or growth hormone for individuals with pituitary dwarfism, are being produced in large quantities by the fermentation of transgenic bacteria that have received

the appropriate human gene. Many of these are now in use in clinical practice.

These recombinant proteins minimize the risk of transmitting infections which was encountered with proteins purified from human or animal tissue, and unlike animal proteins they are not recognized as foreign molecules by the immune system.

The first of these proteins to be used clinically was the hormone insulin.

2- Insulin production

The gene for insulin is cut and purified from the human genome and cloned into a plasmid. The plasmid containing the insulin gene is inserted into *Escherichia coli*.

The bacterial cells divide very rapidly, making billions of copies of themselves, and each bacterium carries in its DNA a faithful replica of the gene for insulin production. Each new *E. coli* cell has inherited the human insulin gene.

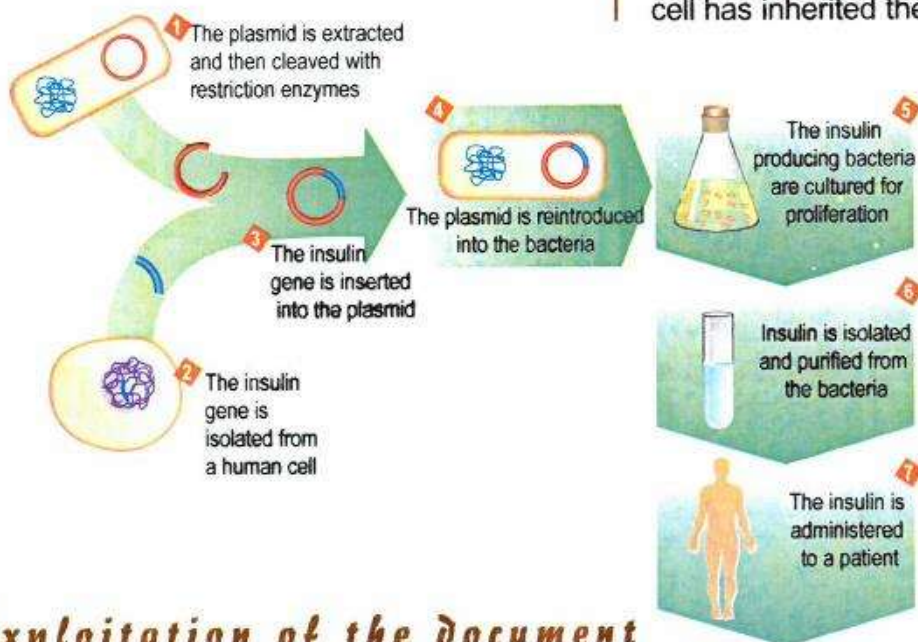


Fig. IV. 6 The production of human insulin through genetic engineering.

Exploitation of the document

1. Explain the economic importance of gene cloning in making therapeutic drugs.
2. What are the benefits that human recombinant insulin brings to medicine?
3. Explain the fact that the recombinant insulin is human even though it comes from a bacteria.

Document 4 *Monoclonal antibodies and their applications*

Antibodies are proteins made by the immune system to combat infection. Infectious agents have many antigens and each antigen has several specific antigenic sites called epitopes. An antibody is made to recognize one specific epitope of the antigen. Mammals make antibodies to almost anything that is a "non-self" molecule, i.e., which is not normally part of the body. On this basis a technique was developed that uses synthetic antibodies to detect the presence of specific antigens.

How do we make antibodies that are specific to a single epitope? How are they used commercially?

1- Monoclonal antibodies

Antibodies produced in the blood are made by a large number of different lymphocytes (B cells). Each B cell makes a unique antibody, therefore the antibodies that recognize any particular antigen are a mixture of molecules. These are referred to as polyclonal antibodies.

Monoclonal antibodies are made from a single clone of B cells which has been isolated and immortalized for growth in vitro. These are epitope specific and can be utilized to detect specific molecules

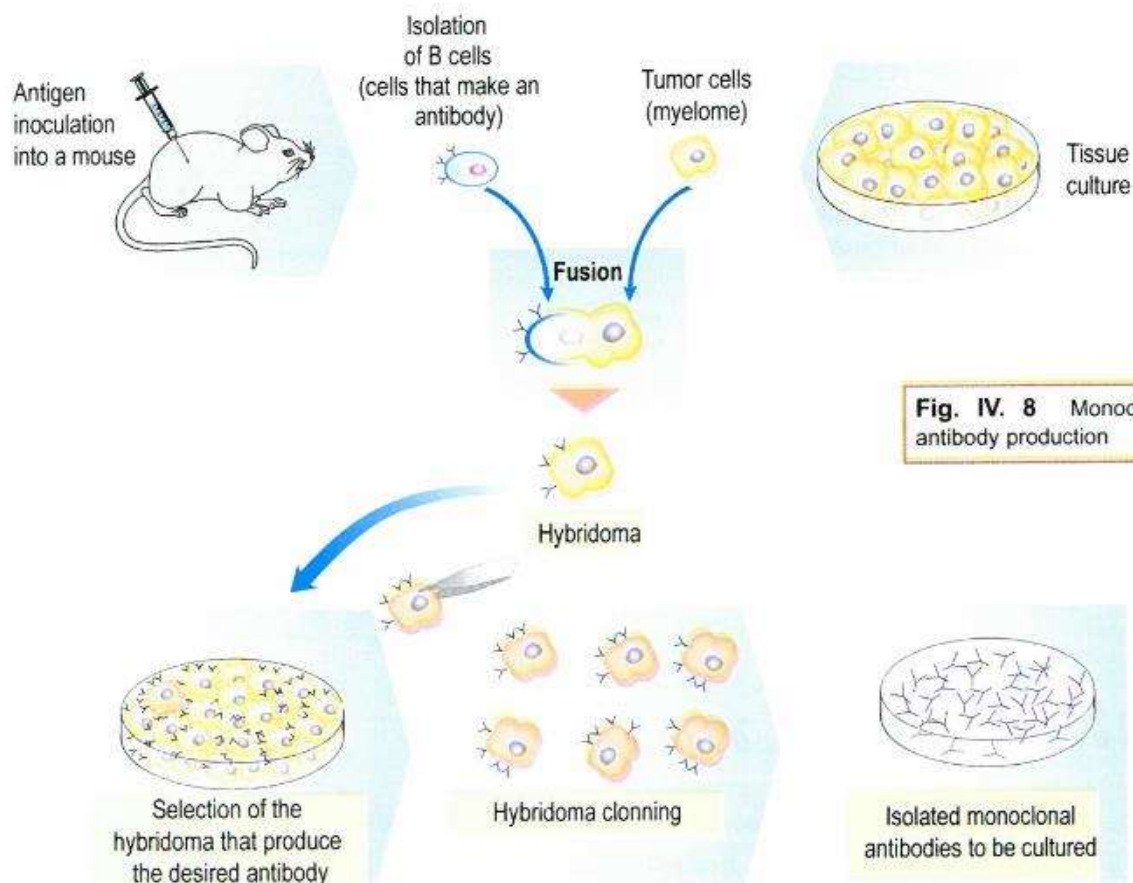


Fig. IV. 8 Monoclonal antibody production

Additional Information

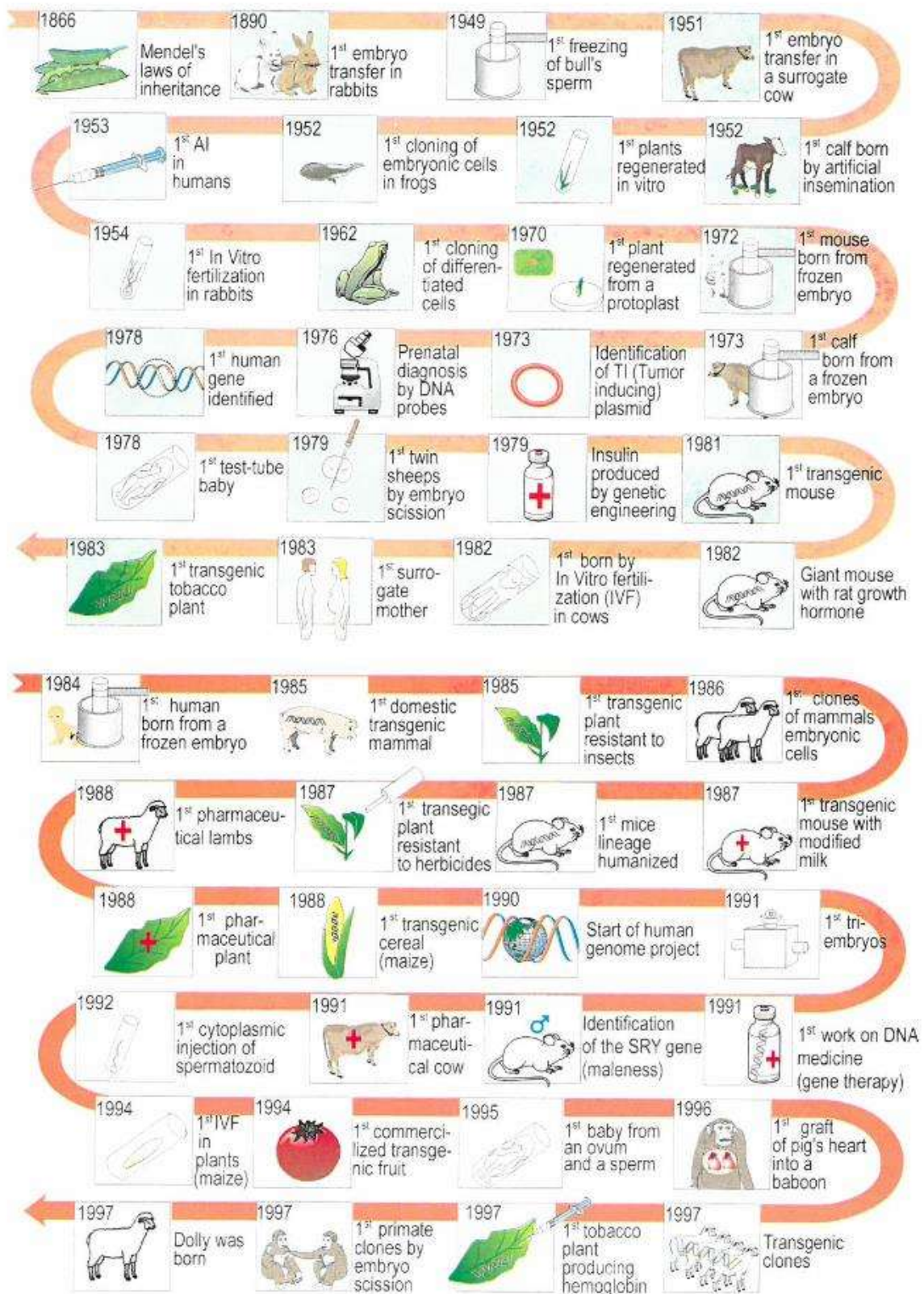


Fig. IV 11 From Mendel's laws to Dolly.

Essential points

Biotechnology and Immunology

- Biotechnology covers all techniques that utilize living organisms in making medical and industrial products that are beneficial to humans. The use of genetic engineering (recombinant DNA) revolutionized the field of biotechnology.
- The basic principle of genetic engineering consists of modifying the genetic make-up of an organism by introducing one or several foreign genes into the chromosomes of that organism.
- When a foreign gene is introduced into the genome of an organism by a process called transformation, this organism will then reproduce the foreign genes and the proteins that they code for.
- It is necessary to identify the gene of interest and isolate it from the rest of the genome, before proceeding with any genetic manipulation.
- The DNA of *Escherichia coli* constitutes the best material for several genetic manipulations :
 - The genome of the *E. coli* is known in its entirety.
 - It divides rapidly, generating a large number of cells, each producing the recombinant proteins.
- The following steps are used in gene cloning :
 - The DNA sequence which determines the gene to be cloned is selected and cut with restriction enzymes.
 - The selected fragment is ligated into a plasmid using a specific enzyme called ligase.
 - The plasmid containing the ligated DNA is called recombinant, and is inserted into the *E. coli* by transformation.
- The recombinant plasmid replicates inside the *E. coli* and is passed on to daughter cells with each round of cellular division.
- One application of gene cloning is the production of therapeutic drugs. Recombinant human insulin is very widely used.
- Another application of genetic engineering is the synthesis of recombinant vaccines. These are safe and cost effective.
- Antibodies are produced by the B lymphocytes, which have a short life span.
- A B-Lymphocyte producing a specific antibody is selected and fused with a permanently dividing myeloma cell. This fusion results in an immortalized hybridoma cell that is capable of producing the selected antibody.
- Antibodies produced by hybridoma cells are called monoclonal antibodies, because they originate from one B cell clone. They are pure, very specific and can be produced in commercial quantities.
- Monoclonal antibodies used for screening and detecting molecules are labeled with a colored or a radioactive marker that makes them detectable.
- Monoclonal antibody detection is used in pregnancy screening and in the early detection of tumors.

2- The use of hybridoma to generate monoclonal antibodies

The following steps are used in making monoclonal antibodies :

- **Immunisation** : a mouse is immunized with the target antigen. This is done by injecting the mouse with the antigen to stimulate the immune response.
- **Splenectomy** : the spleen, which is a site for concentrated B cells, is removed from the mouse.
- **Fusion** : The lymphocytes are fused with immortalized myeloma cells (cells which will grow forever in culture). The resulting cells are called hybridoma.

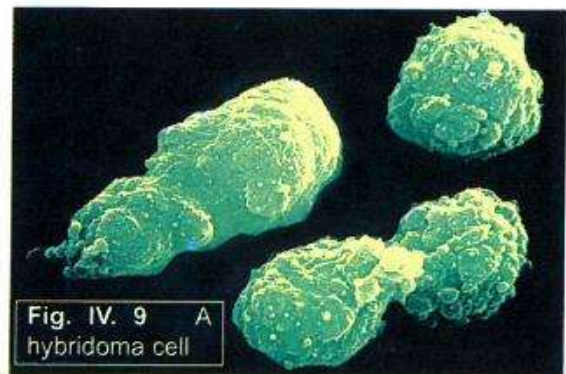
3- Applications

- **Pregnancy tests** : in this test monoclonal antibodies are used to specifically detect the presence of a specific antigen, human chorionic gonadotrophin (hCG). This hormone is secreted soon after the occurrence of pregnancy and can be detected in the urine of the pregnant woman. A urine sample from a woman who is suspecting pregnancy is incubated with hCG specific monoclonal antibodies. These antibodies produce a specific colored reaction in the presence of the hCG. The appearance of the specific color indicates a positive test.



Fig. IV. 10 Pregnancy test.

- **Selection of clones** : the fused cells are grown in a special medium where neither myeloma nor B cells can grow in it, but fused cells can. The fused cells are recovered and selected for their specific and pure antibody production.



- **Detection of tumor cells** : Tumor cells secrete specific antigens in the blood of the patient. Labeled monoclonal antibodies can be specifically used to detect the presence of these antigens in blood samples of cancer patients. This test is very sensitive and can diagnose the presence of a tumor at very early stage. Early detection increases the chance of successful treatment and full recovery.

Exploitation of the document

1. What is an antigen ? Do all antigens elicit an immune response ?
2. Why are the B lymphocyte clones fused to immortalized myeloma cells ?
3. How do we label the monoclonal antibodies used in detection?
4. What are the advantages of monoclonal antibodies in detecting cancer?

Document 3 Recombinant DNA : vaccine production

Vaccines are preparations that, when given to a patient, elicit an immune response that subsequently protects the patient from a disease. Usually the vaccine consists of the attenuated or killed organism that causes the disease, or of some part of it. Attenuation of a virus means manipulating it so that it does not lose its ability to grow but loses its ability to cause disease. There is a range of biotechnological approaches to producing vaccines.

1- Peptide Vaccines

Proteins, or sections of proteins that are the same as the proteins in a virus can be made by recombinant DNA methods as vaccines. This is the standard biotechnological method of making vaccines; it has the advantage that there is no possibility that the resulting vaccine will contain any live particles.

Other methods are also used to produce peptide vaccines :

- Producing single protein vaccines in recombinant microorganisms.
- Producing vaccines in transgenic plants (bananas). Plants will not have any immune response to a human virus, and will not get ill from it. They can produce quantities of the protein and could in principle be used as food to provide oral vaccines.

2- Live Virus Vaccines

Live vaccines consist of live viruses. Clearly the virus itself cannot be used, as that would simply give the disease; instead, one of two biotechnological methods is used to produce a virus that elicits an immune response but will not cause the disease :

- The virus is engineered so that it is harmless but can still grow in cultured cells. The resulting virus cannot mutate back to its original harmful form because entire genes are deleted from the virus and replaced by genes with completely different genetic material.
- The gene of protein from the pathogenic virus is cloned into another harmless virus, so that the result looks like the pathogenic virus but causes no disease. The protein chosen must not itself be dangerous.

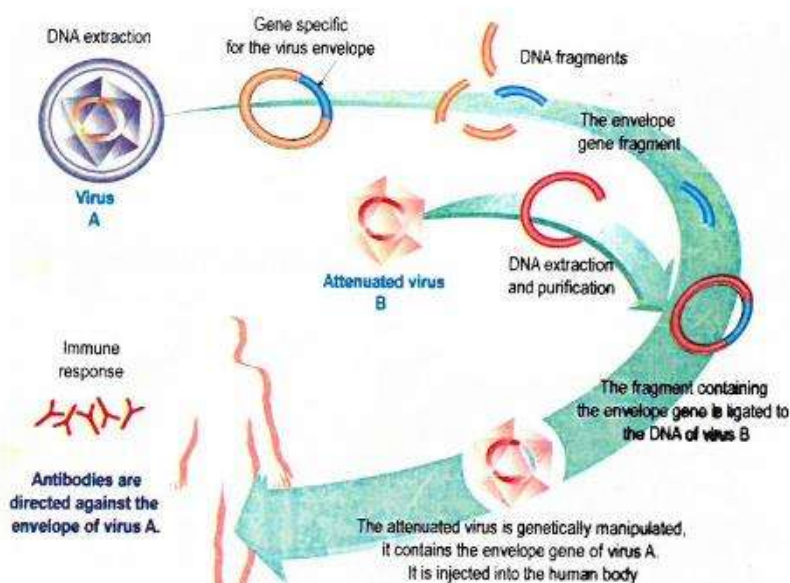


Fig. IV. 7 Construction of live vaccine.

Exploitation of the document

1. Differentiate between the two types of vaccines produced by recombinant DNA technology ?
2. How is it possible to administer to oneself a vaccine in eating a banana ?
3. What could be a potential problem when giving live vaccine ?

EVALUATION EXERCISES

A- Choose the correct answer(s)

- 1- The process of gene cloning includes :
 - a- the transfer of a chromosome from one organism to another.
 - b- the transfer of a gene from one organism to another.
 - c- the multiplication of a gene before it is transferred.
 - d- the multiplication of a gene inside the host cell.
- 2- Some of the necessary tools for gene cloning are :
 - a- transcription enzymes,
 - b- restriction enzymes,
 - c- ligation enzymes,
 - d- hydroxylases,
- 3- E.coli is a bacteria used in different biotechnological processes because :
 - a- it has a well-known genome,
 - b- it contains only one chromosome,
 - c- it's DNA genome is circular,
 - d- it is a site for protein synthesis,
 - e- plasmids can multiply in E.coli.
- 4- A hybridoma cell is produced by the fusion of :
 - a- any two cells,
 - b- two lymphocytes,
 - c- two genes,
 - d- one lymphocyte and one cancerous cell,

B- Select the correct statement and correct the false ones

- 1- Monoclonal antibodies allow the identification of different antigens.
- 2- Vaccines are produced from foreign antibodies.
- 3- Plasmids are circular DNA molecules used to transform human cells.

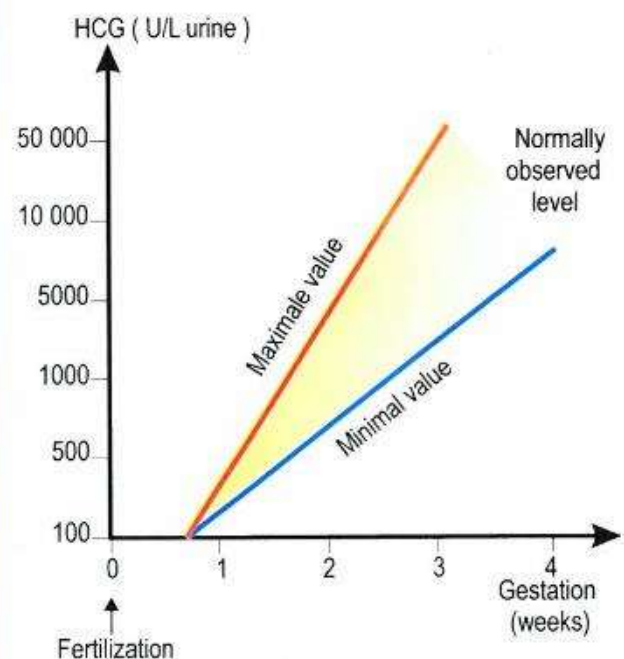
C- Put in order the following steps that are used in insulin production

- 1- Ligation of the insulin gene into the plasmid
- 2- Selection and isolation of the insulin gene from the rest of the genome

- 3- Selection of the plasmid
- 4- Replication of the host cell
- 5- Gene transcription

D - Think about it

- 1- What is the advantage of using immortalized myeloma cells in making monoclonal antibodies ?
 - 2- Human chorionic gonadotrophin (hCG) is produced at a very early stage of gestation (from day one). Its production doubles every 48 hours and reaches its peak after two months of pregnancy. Pregnancy tests used in the past could only detect the hormone when it reached very high levels in the body (1000-2500 U/L). The use of monoclonal antibodies allow the detection of very small amounts of hCG (50-250 U/L).
- a- What is the advantage of using monoclonal antibodies in the detection of hCG ?
- b- The graph below shows the evolution of hCG during pregnancy. At what time during pregnancy can each test be used ?



Document 2 The transfer of genes

Cloning in animals and plants has allowed the conservation of characters. In the last few years, the progress of biotechnology in plants and animals has made it possible to transfer genes from pre-existent varieties to new ones, in order to obtain varieties of significant economic interest. Various techniques are applied in transgenesis or gene transfer.

1- Transgenesis in plants

Biotechnologists can now apply a whole range of new interventions to modify the genetic make-up of plants. The most revolutionary is transgenesis, which consists of transferring a foreign gene into an embryonic vegetable cell, grown in a nutritive medium. The ability of these genetically modified cells to differentiate allows them to generate a plant with a new genetic make-up. We can thus correct deficiencies and manufacture improved transgenic vegetables.

Direct transfer of genes

A protoplast is a vegetable cell without a cell wall, isolated from the external environment

only by its cytoplasmic membrane. This cell can then be made easily permeable to DNA by subjecting it to a series of electric shocks. The DNA migrates to the nucleus and becomes integrated into the chromosomes. The protoplast is then regenerated into a plant. These techniques allowed the transformation of rice and maize. However, it is now possible to insert DNA directly into the cell in spite of the pecto-cellulose wall, particularly in soja.

Transfer of genes by agrobacteria

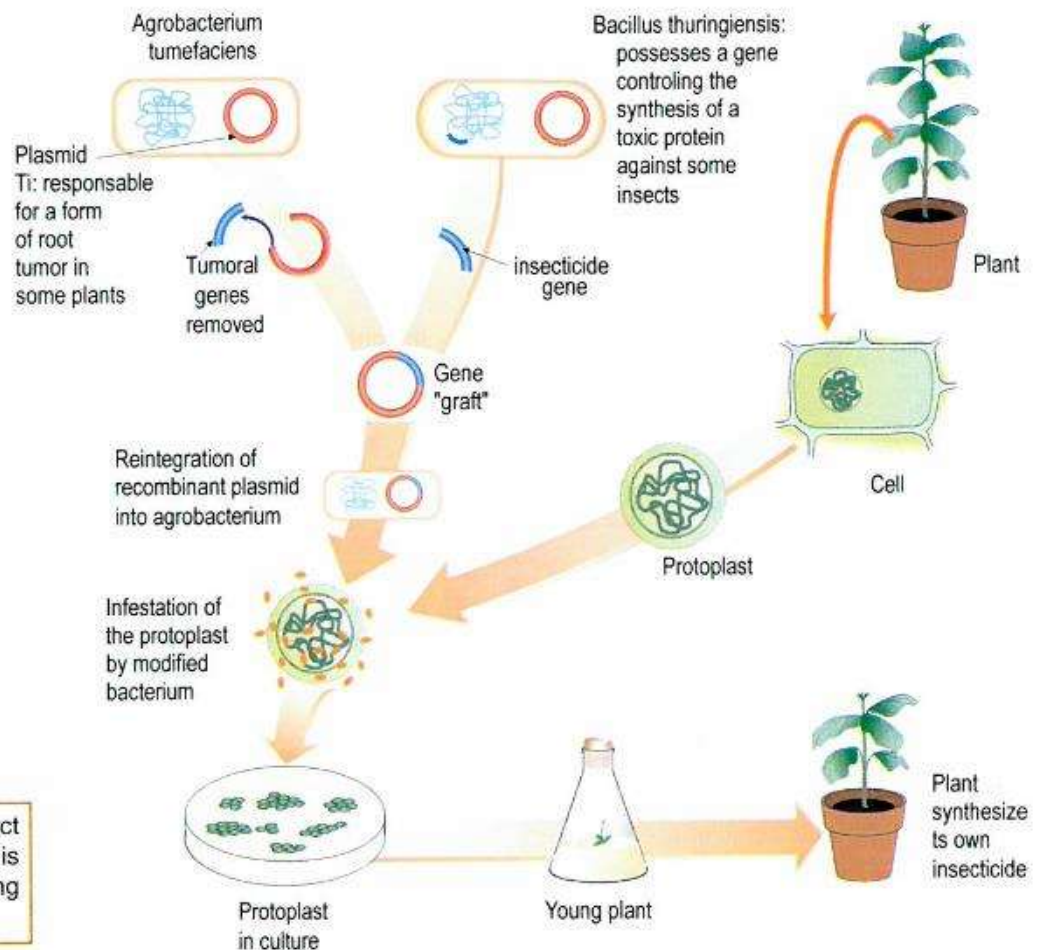


Fig. II. 20 Indirect transfer of genes is now possible by using pathogenic bacteria.

Document 1 Searching for performing species

Sexual reproduction results in a diversity among individuals of the same species. Man exploits this diversity to obtain, select and improve certain varieties of plants or animals. He can also obtain and select new varieties from different races for cross-breeding. However for any experimental intervention, it is necessary to subject the races in question to a thorough genetic study to ensure that they can be interbred and adapted to the conditions of the environment. Thus, controlled cross-breeding of two plant or animal varieties aims at obtaining a variety with new characteristics having a beneficial economic value. How is it possible to improve a species ?

1- Species variability

- A species possesses qualitative and quantitative characters, which vary in a continuous way from one extreme to another. Quantitative variations are measured and exploited for the production of a better variety of species.
- New phenotypic characters can appear in a culture or spontaneously during breeding experiments. When these are favorable, they are considered for their beneficial economical value.
- The performances of the individual in a species are related to its genetic make-up and the conditions in which the species is bred and cultured.
- The phenotypic variability depends on the genetic make-up as well as on the surrounding environment of each species; however, the genetic factor is the only component considered when improving a species.

	Qualitative variations	Quantitative variations
Animals	- Skin, feather (color) - Presence of horns - Maternal behavior - Susceptibility to disease	- Milk (quantity, composition) - Meat (% of carcass, composition) - Weight of the newborn - Interval between gestation - Rate of growth, - Daily gain - Weight of the eggs
Vegetables	- Aspect of the flowers and fruits - Aptitude to graft - resistance to dryness, sensitivity to diseases	- Number and weight of fruits and seeds (% of carbohydrates & proteins) - Yield - Flowering date and maturity - Number of branches

Fig. IV. 14 Some variable morphological and physiological criteria.

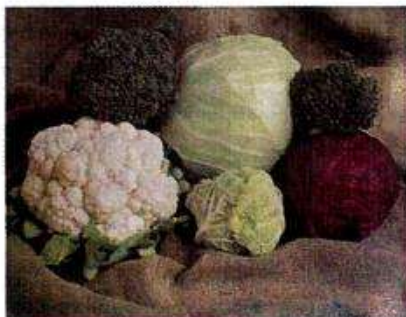


Fig. IV. 15 Some cultivated varieties of cabbage.

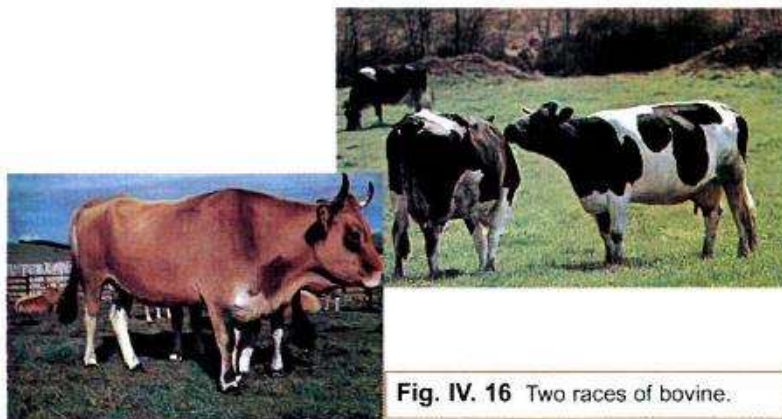


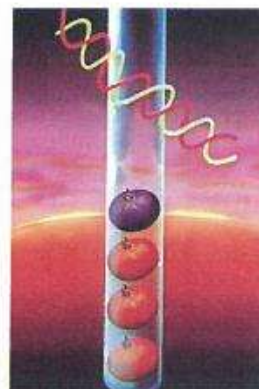
Fig. IV. 16 Two races of bovine.

Chapter 2

Biotechnology, agriculture and environment

Improvement of the species

Fig. IV. 11 Since the beginnings of agriculture, man has sought to improve the plants that he cultivates and the animals that he raises according to output or quality standards corresponding to his needs. At first the techniques used were empirical but evolved rapidly, owing in particular to the contribution of genetics. The development of biotechnology permitted scientists to cross the barrier separating two different species and the inter-species crossing became possible in order to obtain more advantageous individuals.



Industrial breeding

Fig. IV. 12 In view of the growth of the world's population, some hens from the farmyard or a flock pecking ducks in a field could not satisfy the demand for their eggs and meat. An explosive transformation took place during the second half of the twentieth century, when rearing was industrialized and races improved greater productivity and profitability.



Biotechnology and environment

Fig. IV. 13 "Health is a natural state of well-being, threatened by the condition of the artificial life imposed by civilization". The Ozone Layer hole, the Chernobyl disaster, the "mad cow disease" crises, are sad examples that make this affirmation credible. We cannot deny, unfortunately, that our environment is degrading fast. That is why it is legitimate to be on alert, and to be worried from the risks presented by scientific and technical innovation, when the well being of the environment is not considered. Can we condemn, without appeal the power of man's intervention over his environment ?

In this chapter, we shall answer the following questions :

- 🔍 How can we improve the plant and animal species ?
- 🔍 What are the conditions for industrial breeding ?
- 🔍 What is the interest of agronomic research ?
- 🔍 How does industrial and agriculture progress affect pollution ?
- 🔍 What should we do to protect our environment ?
- 🔍 What is the positive side of biotechnology ?

2- Improvement of the species

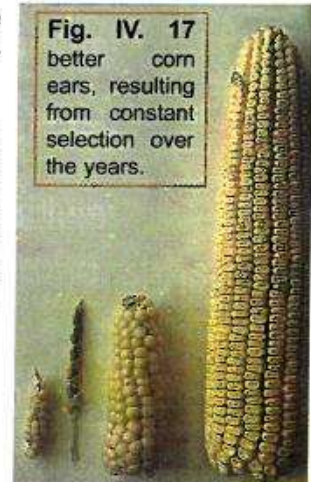
Selection : Selection is a technique often applied to agriculture and breeding. It consists of choosing the best individuals in a population, cultivating them or raising them and making them reproduce in order to improve the species. The resistance of certain insects to chemical substances and of some bacteria to antibiotics are examples of natural selection.

Research in plant breeding (maize, wheat), during the last several decades, has resulted in the production new varieties. The seeds obtained from the best plants are kept and used in the following year. The improvement by selection of

crop plants is responsible for the increase in agricultural production.

As for the stock breeding, animals with the best yield or performance are chosen for mating, so that each generation shows improvement over the previous one.

Fig. IV. 17
better corn ears, resulting from constant selection over the years.



Hybridization breeding : For a long time, man has used hybridization breeding in order to obtain individuals having the desired characters and has applied it to the improvement of both animals and plants. Cross-breeding two varieties of wheat makes it possible to obtain a new variety. Figure 18 represents the specific characteristics of each variety that is inbred and the new properties that are introduced by cross breeding the two varieties.

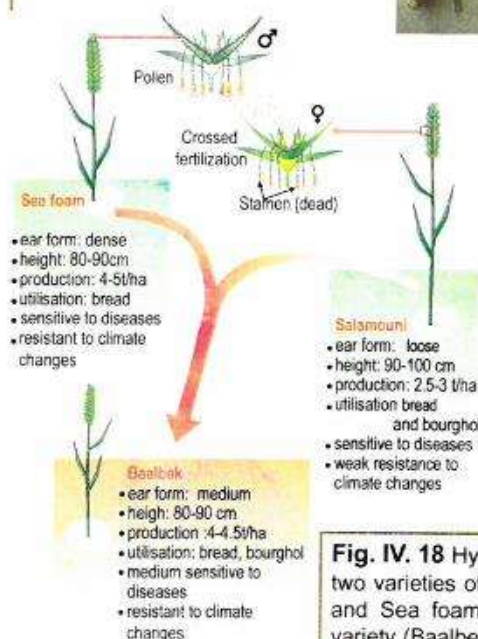


Fig. IV. 18 Hybridization between two varieties of wheat (Salamouni and Sea foam) to obtain a new variety (Baalbek).

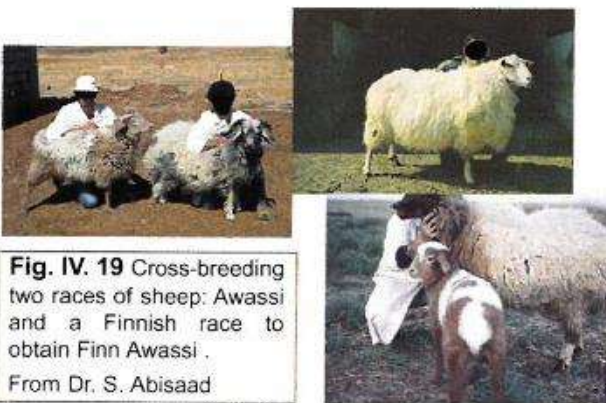


Fig. IV. 19 Cross-breeding two races of sheep: Awassi and a Finnish race to obtain Finn Awassi .
From Dr. S. Abisaad

In animals, some cross-breeding experiments were made in Lebanon on two races of sheep, sheep of the Middle East and the European sheep. Each race had its desirable qualities. They were crossed to obtain a new variety more or less adapted to the needs of consumers. The new generations differ from their parents by the acquisition of new characters that are superior to the average of both parents. These relate to the quality of meat, growth rate, the percentage of meat in the carcass, and adaptation to the environment.

Exploitation of the document

1. What is the economic importance of the characters observed in fig.14 ?
2. When you compare the members of each race of wheat or sheep among themselves and with their parents, what can you deduce ?

Document 3 *Industrial breeding, a controlled production*

A genetic selection combined to a high-energy value diet has allowed the breeder to produce at lower prices. However, progress leading to commercial and intensive breeding remains confined to only a few countries. Such progress requires industrial, financial and scientific resources, which are far from being universally available. In many places animals are still mated in a haphazard manner which prevents people from being better fed. The contrasts in breeding methods reveal clearly the different rates of progress in different countries. What are the conditions for industrial breeding?

The conditions for industrial breeding

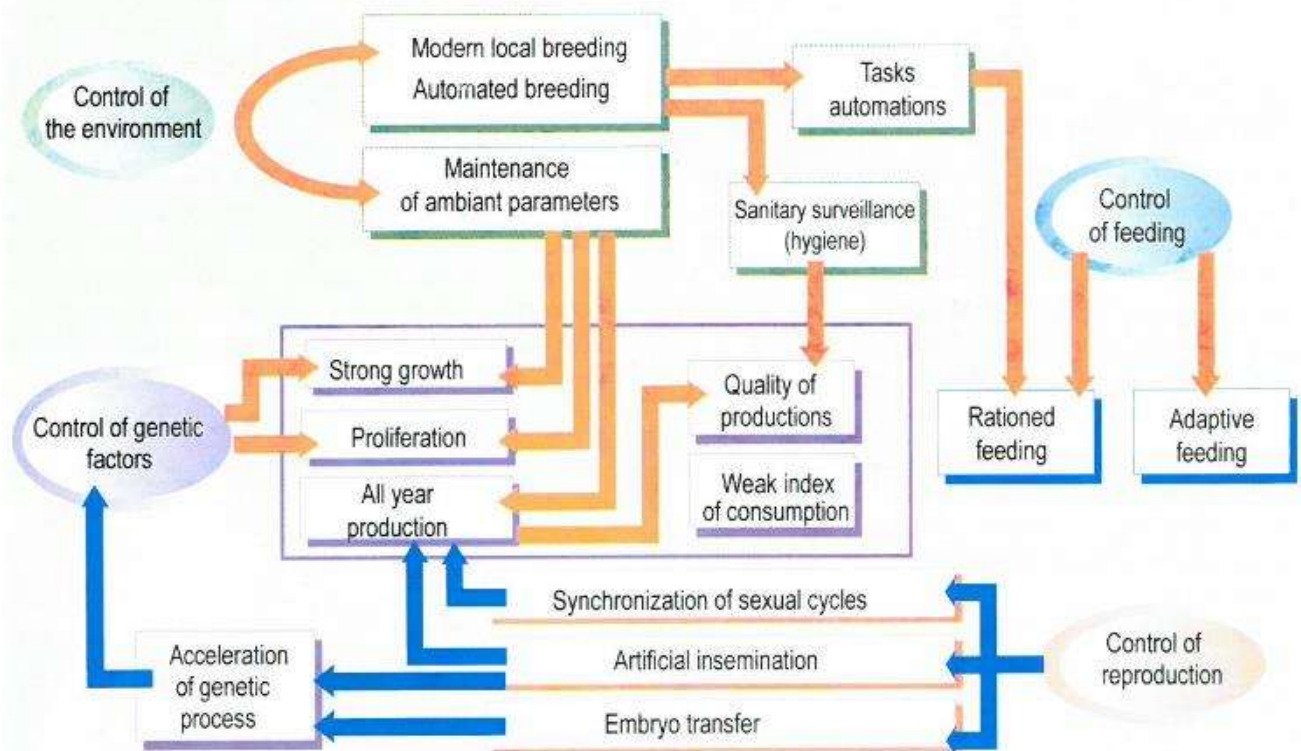


Fig. IV .24 Diagram showing the relation between the different degrees of progress in industrial breeding

Exploitation of the document

1. Analyze the diagram and try to relate these different factors. What conclusion can you draw ?
2. Try to find the meaning of the following terms: rationed food, low rate of consumption and prolific.

Document 4 Animal food

To satisfy the increasing needs of consumers especially in meat, breeders try to raise the nitrogen content in the food of the livestock in order to improve quantity and quality ?

1- A search for poultry food

The grains of certain plants such as pea, bean, or lupin are rich in proteins. Their roots have nodules containing nitrogen-fixing bacteria. Consequently, these plants do not need large amounts of nitrate fertilizers for development, thus reducing costs. Residues of oil extraction from seeds of soya, sunflower or colza give by-products rich in proteins. However, these are expensive and require pre-treatment before being fed to the animals.

In a study, 3 groups of egg-laying hens were fed for 4 months, each group with a ration slightly different from the others. The effect of the different rations on egg production are represented in the fig 25.

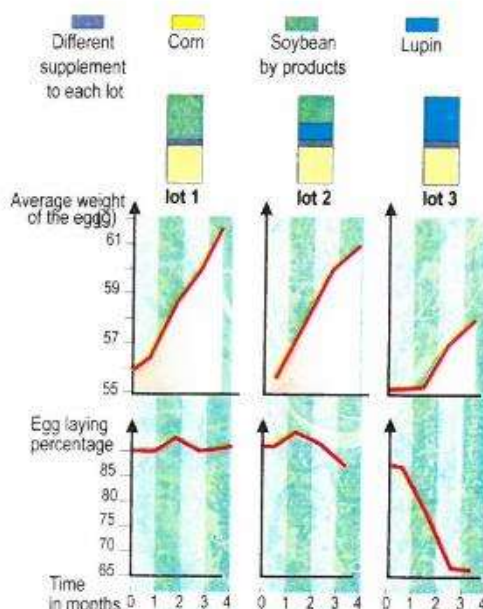


Fig. IV. 25
Effect of
different
rations on
egg pro-
duction.

2- A search for caprine food

Eighteen 3 month old "Baladi" goats are divided into 3 groups each receiving a different ration during 4 months: traditional, untraditional or a

mixture of both (fig 26). The results of the different rations on some growth and reproductive data are represented in figure 27.

Ingredients	traditional	mixture	non traditional
corn	57	48,2	39,4
barley	23	11,5	-
soy	10	5	-
wheat bran	9	4,5	-
fava	-	5	10
peas	-	4,8	9,6
lentils	-	5	10
lentil bran	-	15	30
NaCl	1	1	1
Total	100	100	100

Fig. IV. 26 Different rations for goats.

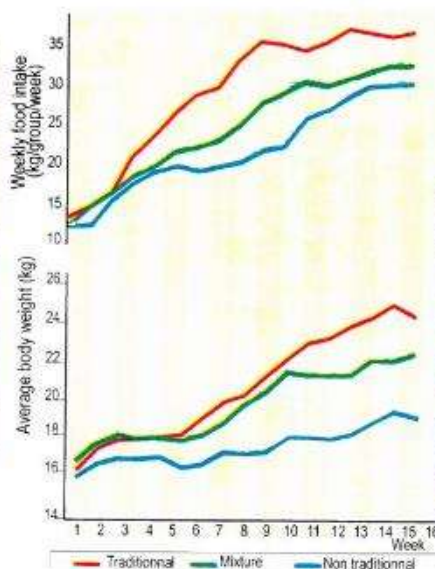


Fig. IV. 27
Effect of dif-
ferent rations
on some
growth and
reproductive
data in
goats.
(results com-
municated
by Dr S. Abi
Saab.

Exploitation of the document

Analyze the results of fig. 25 and 27 and give your comments concerning the growth of the goats in relation to the different food rations.

4- Modern agriculture

Industrialized agriculture necessitate the utilization of massive quantities of nitrate fertilizers. In fact, many plants, notably the cereals, are not capable of fixing free nitrogen. They can only assimilate it as mineral compounds, essentially as nitrates. Farmers have a tendency to use much more nitrogen fertilizers than is required, and when there is heavy rain, the excess drains off into the rivers and underground water bed.

The following facts should therefore be noted :

- A plant sometimes needs nitrates. Fertilizers applied at inappropriate times are sources of pollution.
- Each cultivation leaves behind a certain amount of nitrogen. The order of succession of cultivation is important.
- Grounds that are left uncultivated in winter loose their nitrogen in underground water.

Water containing more than 50 mg of nitrates per liter is no longer drinkable. In fact, it is better not to exceed 25 mg/l because newborn babies have a bacteria present in the stomach which can easily transform nitrates (NO_3) into nitrites (NO_2). The latter can be released in the blood, where it combines with hemoglobin and prevents oxygen fixation, leading to suffocation and death. Nitrites can also produce nitrous derivatives that are carcinogenic, by combining with derivatives of organic nitrogen derived from foods.



Fig. IV. 33 Chemical fertilizers; it is not the quantity that counts !

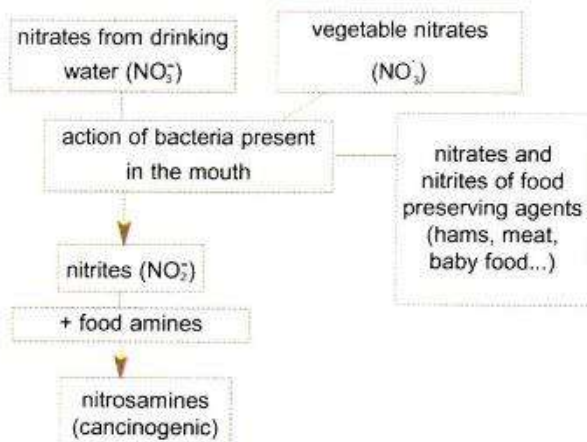


Fig. IV. 32 The fate of nitrates.



Fig. IV. 34 Pesticides; are they harmful to men!



Exploitation of the document

1. Why is petroleum oil a non-replaceable source of energy ?
2. Lebanon is not an industrial country, therefore we do not have any pollution. Is this true ?
3. Can we talk of a nitrate cycle ?

- The first genes transferred were antibiotic resistant genes. An agrobacterium was used to transfer a pathogen resistant gene to a tomato seedling. The resistant gene becomes a stable part of the tomato plant genome and is transmitted to the offspring.
- In the fight against insects, genetic engineering has provided solutions to reduce the use of insecticides. There are transgenic maize plants that are resistant to a devastating moth. The cultivation of this new variety relieves the farmer from using insecticides, which are an added expense and are bad for health and the environment.



Fig. IV. 21 The plant to the right is a transgenic plant resistant to insects and the one to the left is insect sensitive.



Fig. IV. 22 "Pyrale" moth.

2- Transgenic animals

By definition, a transgenic animal is an individual who has received new genetic information following the experimental introduction of foreign DNA into its genome.

Introduction of DNA is done by micro-injecting a gene, purified by cloning, in the nucleus of a fertilized animal egg. In mammals, the egg is then re-implanted in the uterus of a female.

The resulting offspring may integrate the new gene inside its chromosomes. The integrated gene becomes a stable part of the animal's genome and will be transmitted to the next generation.

In an experiment performed a few years ago, the gene of the human growth hormone was inserted into the genome of a mouse egg. After fertilization, the zygote that had integrated the human gene developed into a giant mouse (twice the normal size).

In addition to increasing the size by transgenesis, we hope to improve animal production. Thus, the quantity of meat per carcass could be increased,

and the cholesterol content of the meat or eggs could on the other hand be decreased. In addition, the average milk productivity per animal could be increased and the lactose content could be decreased in order to improve its digestibility; the quality of wool and leather could also be improved.



Fig. IV. 23 Transgenic mice.

Exploitation of the document

1. What does the expression "transgenic plants" mean ?
What are the advantages and the disadvantages of these plants ?
2. Carry out an investigation on genetically modified organisms. What are the advantages and disadvantages of using genetically modified organisms ?

Document 5 The cost of progress

Technological progress of civilization presents several drawbacks. Substances that were absent a century ago are now abundant everywhere: the air that we breathe contains several pollutants, drinking water is sometimes contaminated, and in addition there are all sorts of radiation surrounding us, and many artificial products are present in our food. What are the principal factors that threaten our lives and what is their origin ?

1- Classification of pollutants

Nature of pollutants	Atmo-sphere	Ecosystems		
		continental	fresh water	marine
1. Physical pollutants				
• Ionic radiations	+	+	+	+
• Thermal pollutants				
2. Chemical pollutants				
• Hydrocarbons and their combustion products	+	+	+	+
• Plastic material	+	+	+	+
• Pesticides		+	+	+
• Detergents			+	+
• Organic components	+	+	+	+
• Sulphur products	+	+	+	+
• Nitrates, phosphates		+	+	+
• Heavy metals		+	+	
• Fluorides		+	+	
• Mineral particle (aerosol)		+	+	
3. Biological pollutants				
• Dead organic material			+	+
• Pathogenic micro-organisms	+	+	+	+

Fig. IV. 28 Pollutants of different natures.

2- Energy production, a major source of pollution

The use of hydrocarbons in many activities of our society (factories, thermal station, cars, houses, etc.) makes them a primary source of environmental pollution. The soaring demand for energy in developed countries exhausts natural resources such as petroleum oil which cannot be renewed.

Activity	Cause of pollution	Polluted Environment	Nature of pollutants
Extraction	pit leakage	ocean	crude oil
Transport	accidents	ocean	crude oil
Refine	liquid and gas reject	atmosphere fresh water continental marine	various organic compounds sulfuric anhydride, etc.
Usage	incomplete combustion	atmosphere	sulfuric anhydride, nitrous oxide, hydrocarbon

Fig. IV. 29 Main causes of pollution associated with the use of hydrocarbons.

3- Modern chemical industry, a source of various pollutants

The extraordinary expansion of chemical industry during the last decades (metallurgy, electronics, plastics) has led to the emission in the biosphere of several organic and mineral compounds of high toxicity.

- Nitrogen oxides (NO and NO_2) as well as their derivatives, emitted from cars and power stations, cause eye irritation in humans. They are also responsible for serious damage to vegetation.
- Carbon monoxide emitted from motor exhausts causes dizziness and headaches. Its concentration in the air can be ten times more elevated during peak traffic hours.
- The increase in the concentration of carbon dioxide in the air resulted in a heightening of the "greenhouse" effect, which results in a retention of the radiation in the atmosphere causing an increase in the global temperature. This in turn may lead to the melting of glaciers and the rising of sea levels.

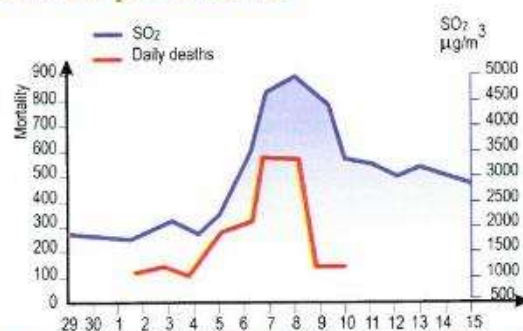
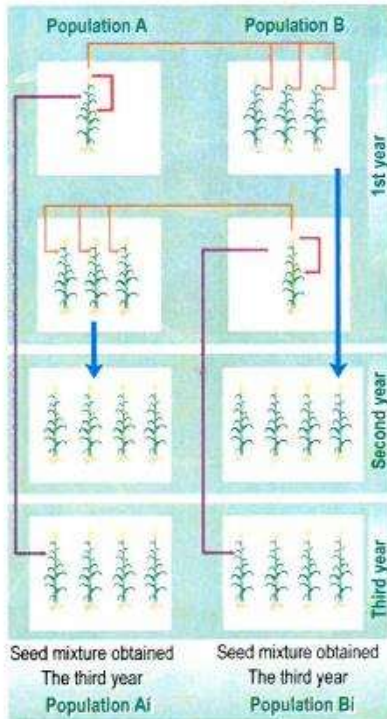


Fig. IV. 30 Toxic effect of sulphurous anhydride in the air (London, 1952).

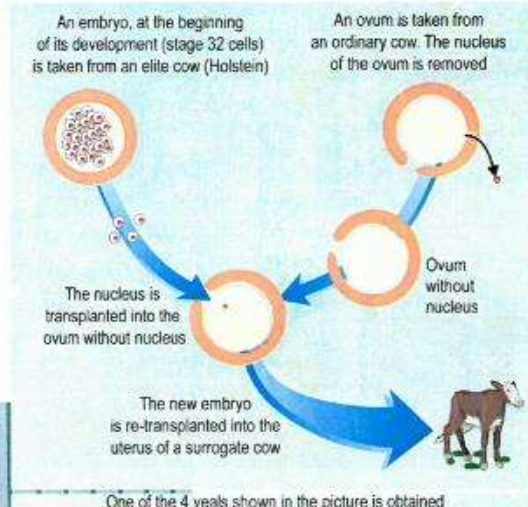


Fig. IV. 31 These products damage the ozone layer, which protects us against the harmful radiation of the sun.

EVALUATION EXERCISES



One of the 4 veals shown in the picture is obtained

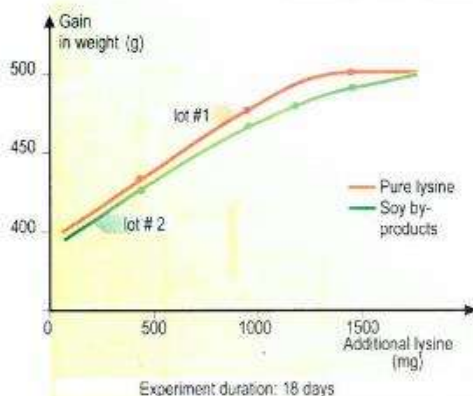


1. A method of selection between two populations A and B is considered. In the first year, ears coming from self-fertilization of each population are stored and those coming from a cross between the two populations are spread. The second year, the better hybrids are chosen and spread. The following year, the best ears stored from the first generation are sown. What is the advantage of this method?

2. New techniques concerning bovine breeding consist of dividing a zygote cell into two cells in order to obtain twins. What is the advantage of this technique?

	Average value				
	Race	Milk Production (kg/year)	Fat body (%)	Richness in proteins (%)	body weight (kg)
1973	Black "pie" (French)	4000	37	33	527
	Holstein (North america)	6474	35,3	32	557
1986	Holstein (French)	6637	38,9	32,5	600

3. Compare the milk production of these three races of cows. Which race produces most?



4. Adding proteins to poultry food can be done in two different ways. Which one is the most advantageous?

Alimentation	Chick (1 - 8 week)	Chicken (9 - 19 week)	Laying hen (20 week)
Total proteins	180	150	190
Methionine	4,5	3	4,1
Lysine	10,5	6,6	8,4
Tryptophane	11,4	10,8	1,85
Linoleic acid (lipid)	2	1,25	14
assimilated Phosphates	4,8	4,2	4,2
Calcium	10,3	11,4	40

5. What information can you deduce from this table?

For or against GMO (Genetically Modified Organisms)



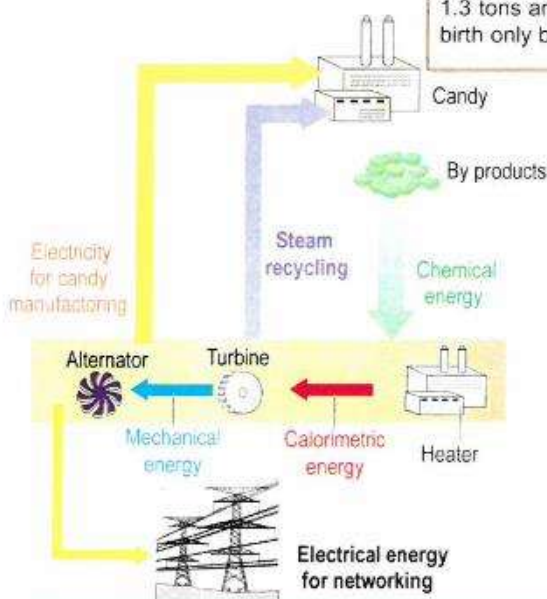
Increase production, at what price?

Nothing is lost, everything is transformed. "Animal flower", recycled from biological material taken from slaughter houses, (blood, skin, feathers) is also used as lubricants or as food for cattle. But in the latter case they end up on our plates!



A meat gene. Having a "fat bottom" gene! These animals with hypertrophied hindquarters weigh up to 1.3 tons and could be delivered at birth only by a cesarean section.

Some of these pigs will be used to produce animal flower, which will be fed to other piglets.



A way to save energy in Guadeloupe.



Air captors measuring pollution on roads and highways in Paris. Measures that have to be taken in our country too !

Additional Information

Transgenic tomatoes

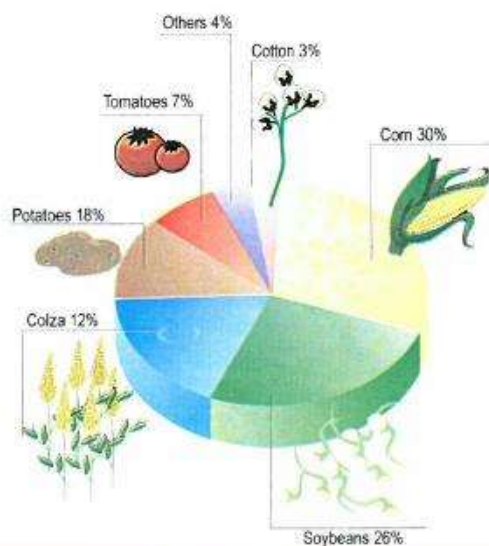
In 1994, a tomato that does not rot was developed in California. In this tomato, the gene responsible for the production of the pectinase enzyme has been removed. This enzyme monitors the aging of the tomato. It slowly dissolves the external envelope and causes the tomato to rot when it has been stored for a long time even in the refrigerator. The transgenic tomatoes will not rot even if they are stored for a few months. In 1996, British supermarkets started selling

Similar methods have been applied to fight a bacterial disease in rice. This bacterial disease has a devastating effect on rice crops in Asia and Africa. However, certain varieties of rice plants possess a gene which makes them resistant to this bacteria. Recently this gene has been cloned and used to make transgenic rice plants resistant to this bacteria.

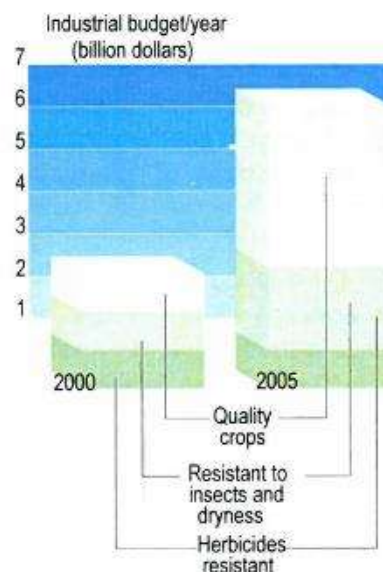
a tomato juice produced from transgenic tomatoes.



Transgenic rice for the children in Asia. In order to prevent deficiencies in vitamin A, researchers manipulated the genetic make-up of rice by introducing the gene of β -carotene in its genome in order to prevent vitamin A deficiencies in millions of children. The yellow color of transgenic rice comes from its capacity to produce β -carotene.



In the year 2005, most of the transgenic food sold in the world will be maize, followed by soy.



Industrial plants. There is a promising market for transgenic plants producing raw material for industry – such as colza for oil or potato for starch. Other transgenic plants (resistant to insects) are less profitable.

Document 6 The struggle against pollution

When the public is informed about the measures to be taken against pollution, a lot can be done. Controls and rules should be imposed in order to diminish unnecessary risks. Prevention of pollution and protection of the environment can be achieved by creating green spaces and by developing scientific research that can help avert ecological catastrophes.

1- Protection of nature

An environmental policy based on ecological planning should be undertaken at the national and the international levels in order to

- process wastes
- recycle and fight against waste of matter and energy
- create green spaces in cities
- protect landscapes
- plant forests
- benefit from solar, hydraulic and wind energy
- fight against air pollution (filters for factories and cars)
- fight against pollution of fresh water and the ocean,
- fight against noise
- encourage scientific and technical research in environmental matters
- and find biological solutions
- ...



Fig. IV. 35 Karm Schbat, a nature reserve at Khobayyat.

2- Biotechnology at the service of the environment

Bacteria eat metals



Fig. IV. 36 Iron and manganese oxides derived from industrial pollution are toxic to both plants and animals. The natural activity of bacteria which reduce iron is being stepped up in order to break up these harmful oxides.

Against pollution

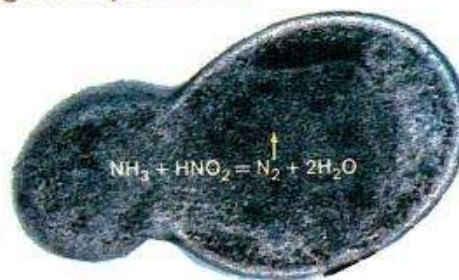


Fig. IV. 37 Ammonia present in fertilizers is one of the substances responsible for the pollution of lands suitable for cultivation. It could be transformed into nitrogen by bacteria which is at the same time capable of providing energy.

Exploitation of the document

Consider one of the most pertinent environmental problems in your town or region, make inquiries about it and discuss the measures that could be undertaken to fight it.

Essential points

Biotechnology, agriculture and environment

Amelioration of species

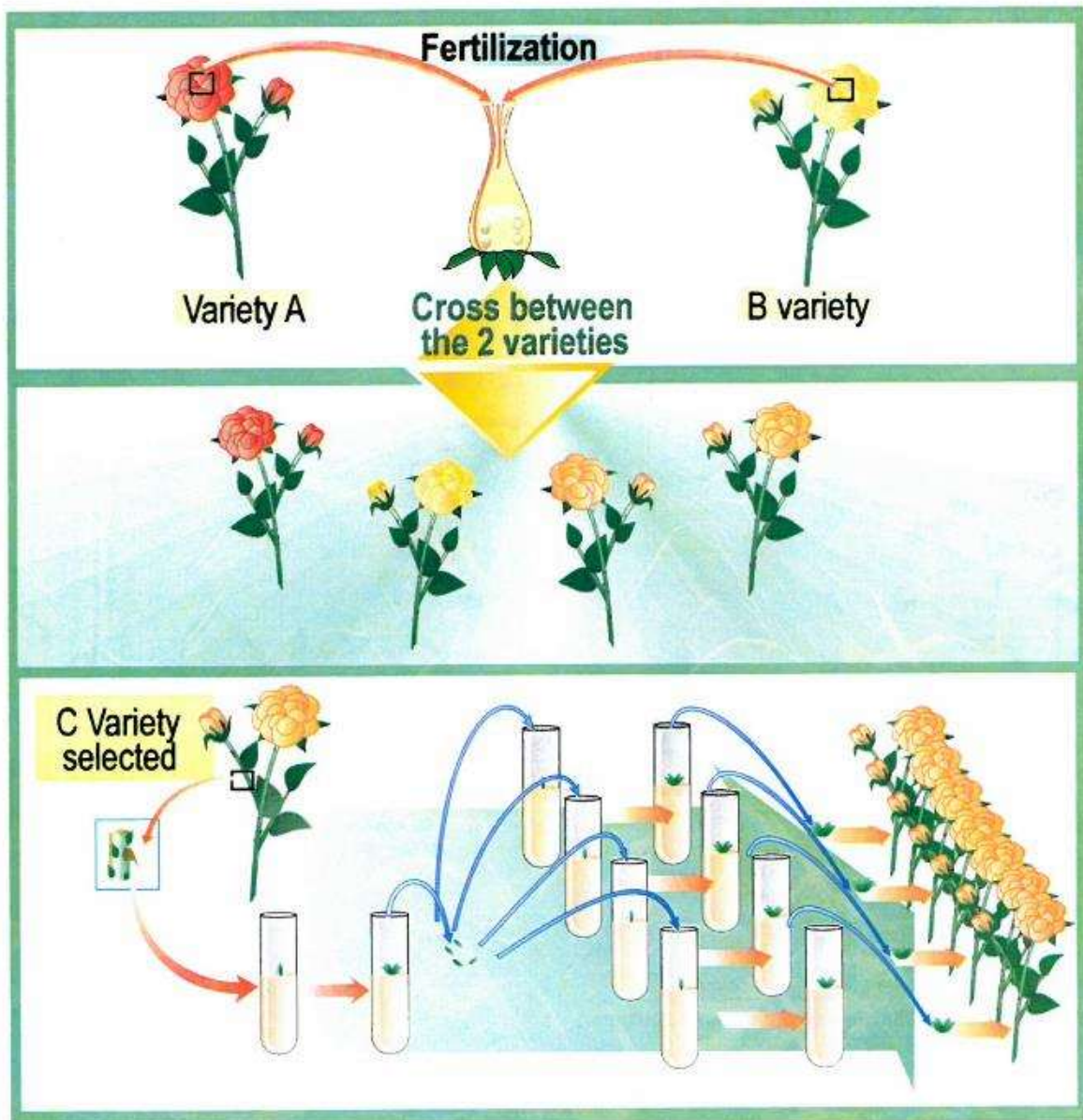
- The improvement of species exploits the variability of each species' characteristics, some of which are genetically determined while others are environmentally conditioned. However, only the genetic determinant is the one considered.
- Hybridization breeding combines certain desirable parental characteristics in an attempt to produce an offspring with the desired characteristics. It can also be used to compensate for an unfavorable characteristic. Animal hybridization breeding is only possible between closely related species.
- The principle of selection is to cross the most performing animals or plants.
- genetic engineering provides scientists with many possibilities for improving species.
- Plant transgenesis aims at improving plant species by making them resistant to disease-causing pathogens which can be deleterious to agriculture.
- Animal transgenesis relies very often on the usage of a vehicle to transfer genetic material to target cells.

Industrial breeding

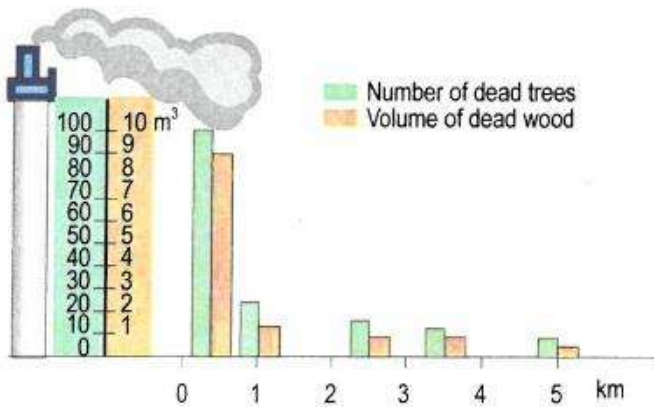
- The development of industrial breeding is linked to the increased demand of consumers for animal products.
- To breed successfully it is necessary to combine industrial and biological techniques, including an appropriate place, a well-monitored serial breeding technology, a genetic selection of productive breeds, a control of reproduction, and the use of a balanced and rationed diet at a reduced cost.
- The major concern in agriculture and food research is to improve the nutrition of animals by the addition of the amino acids necessary for growth, at a reasonable cost.

Biotechnology and environment

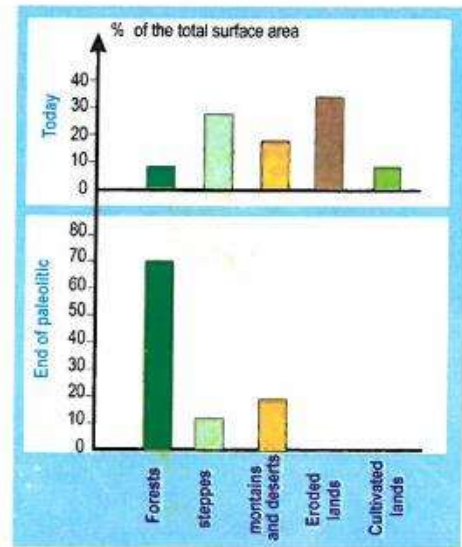
- Human aggression on the biosphere started long ago with the destruction of vegetation, erosion of the soil and extermination of fauna.
- Most of the harm done by industrial civilization comes from the disturbance of the natural flow of energy and the breakdown of the matter cycle by producing increasing amounts of non-recyclable wastes.
- We can distinguish three principal causes of biosphere contamination: production of energy, chemical industries and agricultural activity.
- Hydrocarbons, plastic substances, sulfur derivatives, nitrates, heavy metals, pesticides, aerosols, and ionizing radiation are the main pollutants.
- Pollution can affect the air that we breathe, the water that we drink, the sea, the shore and the food that we consume.
- A well-defined plan should be imposed; for efficient management of all wastes causing imbalance to the cycle of matter. It should ensure the preservation of the fauna and flora, clean water and good quality of food. It should arrange for green spaces and use biotechnology to serve mankind.
- We have no choice but to look for another civilization, one which is not based on quick profit, but on the interests of humans and the improvement of the quality of life.



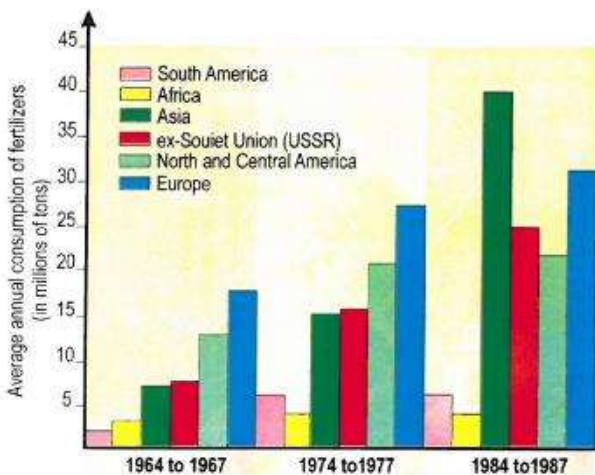
11. What is the use of this technique of roses multiplication.



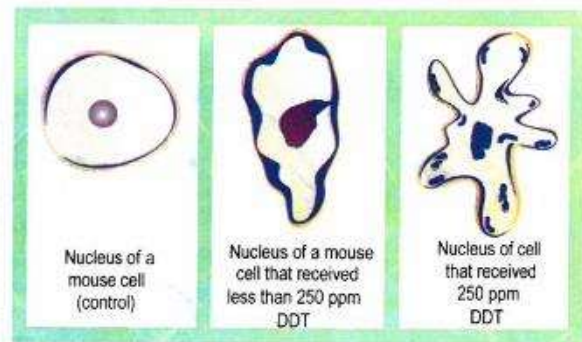
6. Comment on this graph concerning the damage observed in the forest near Carcovie (Poland) ?



7. What are the causes of the modification in the continental ecosystems in China ?



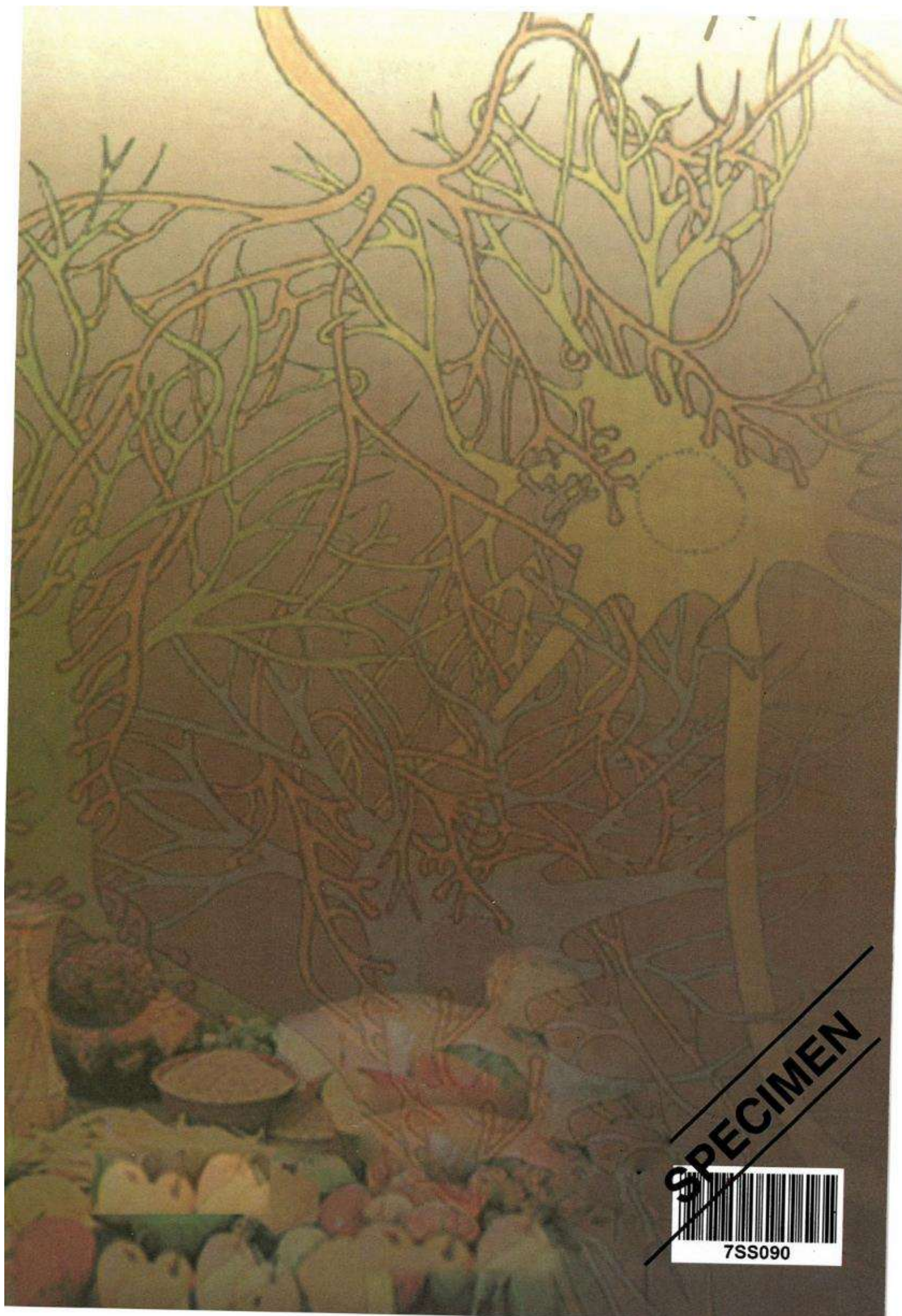
8. Comment on this graph concerning the use of fertilizers throughout the world.



9. 9- Comment on this figure !

	Finnish Foreign race ♂	+ Awassi locale race ♀	→ Finawassi hybrides ♂ and ♀
color	grey, head and legs white	off white spotted black and brown	white spotted different colors
number of lamb (prolificity)	1 - 4	1 - 2	1 - 3
Butcher's aptitude (carcas, quality of meat)	remarkable	desirable	remarkable
Growth	early	late	late
Wool	for tissue	for carpet	carpet and tissue
Tail	small	fatty	different form
Adaptation	weak	perfect	good

10. Do you find this cross advantageous ?



SPECIMEN



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